D365 Design Studio

Functional guide: the complete feature set, end to end

A browser-based design tool for Dynamics 365 functional consultants and business analysts. It shows the design as a Dynamics 365 app, and reads and writes the Excel design workbook that the deployment engine uses to build the solution.

Item	Detail
Audience	Business analysts, functional leads, and anyone reviewing or signing off a design
Version covered	Studio release built 1 November 2026 (single file: D365-Design-Studio.html)
Screenshots	Taken from the built-in sample workbook ("Sample Permit CRM"), which contains no client data
Companion	CRMMetadataDeploy engine (reads the same workbook and builds it in Dataverse)

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1. What the Studio is

The Studio lets a business analyst design a Dynamics 365 application visually, with the look and behaviour of the Unified Interface, instead of filling an Excel template cell by cell. The analyst works on screen. The Studio writes the result back into the same Excel workbook. The deployment engine then builds exactly what the workbook says.

1.1 How it fits together

Step	Who / what	What happens
1. Design	Business analyst in the Studio	Opens the workbook, designs tables, fields, forms, views, flows, roles, organisation, scripts and documents. Checks the design and records notes.
2. Save	Studio	Save to Excel downloads the updated workbook. Only the sheets that were changed are altered; every other sheet is kept as it was.
3. Validate	Engine (CRMMetadataDeploy)	The validate action reads the workbook and reports errors and warnings before anything is built.
4. Build	Engine	Deploy, views and org actions create tables, columns, choices, views, business units and teams in Dataverse. A dry run is available first.
5. Review	BA and consultant	Compare versions shows what changed between two copies of the workbook, with a Word change list for the client.

1.2 Principles

- **The workbook is the single source of truth.** Nothing is stored on a server. The engine never writes the workbook; only the Studio does.
- **Everything stays in your browser.** Opening a workbook does not upload it. The only optional network call is the AI feature, and it has a no-key copy-and-paste mode.
- **Every change can be undone.** Undo and Redo (Ctrl+Z / Ctrl+Y) cover every edit, including bulk changes from AI design.
- **Design first, build later.** Several features record intent for the developer (default values, validation rules, scripts, documents) without building it yet.
- **The Studio is honest about what it checks.** Check design lists what would break the build and what is worth a look, and each item jumps to the place to fix it.

2. Getting started

2.1 Open the tool

Double-click D365-Design-Studio.html in Chrome or Edge. It is one self-contained file and needs no installation. The start screen offers two ways in.

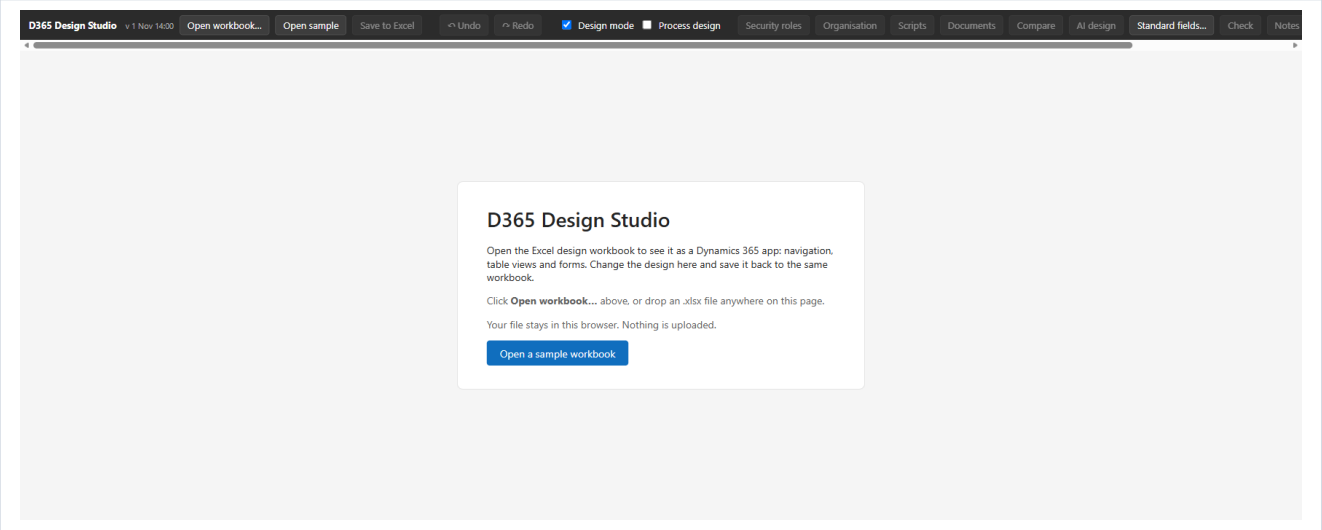


Figure 1. *Start screen. Open workbook… or drop an .xlsx on the page; Open sample loads the built-in example.*

1. Click Open workbook… (or drag the .xlsx onto the page).
2. Or click Open a sample workbook to explore with invented data.
3. Work in Design mode (on by default). Untick it to see the app exactly as a user would.
4. Click Save to Excel when done. A dot on the button shows there are unsaved changes.

2.2 The top bar

The dark bar at the top holds every tool. The blue bar below it is the app itself, as users will see it.

Button	What it does
Open workbook… / Open sample	Loads an Excel design workbook, or the built-in sample.
Save to Excel	Downloads the updated workbook. Only changed sheets are altered.
Undo / Redo	Steps back and forward through every edit.
Design mode	On: all design controls visible. Off: preview, looks like the live app.
Process design	A tick box. Turns on the process designer tools: the process bar on the Ordinary form, the process canvas, Fields per step and the Tools menu (section 16). Off by default. It does not decide what a record shows: a record whose process has step fields opens on the process screen with the tick on or off (section 16.17).
Security roles	Privilege grid, field security, navigation and AI draft for each role (section 9).
Organisation	Business units and teams, and the roles each team holds (section 10).
Scripts	Web resources and form scripts (section 11).
Documents	Word / PDF templates produced from a record (section 12).
Compare	What changed between two versions of the workbook (section 13).
AI design	Describe a process; the AI proposes tables, fields and a process flow (section 14).

Button	What it does
Standard fields…	Loads a list of standard Dynamics fields, so standard columns are reused rather than recreated (section 4.4).
Check / Notes / Migration	Review panel on the right (section 8).
View as	Redraws the app as a chosen role would see it (section 7.2).

3. Browsing the app and designing views

3.1 The app shell

Once a workbook is open, the Studio draws a Dynamics 365 app: app name, search bar, left navigation with groups and pages, a command bar and a list of records. The list shows one sample record built from each field's Example column, so the analyst sees realistic data without any database.

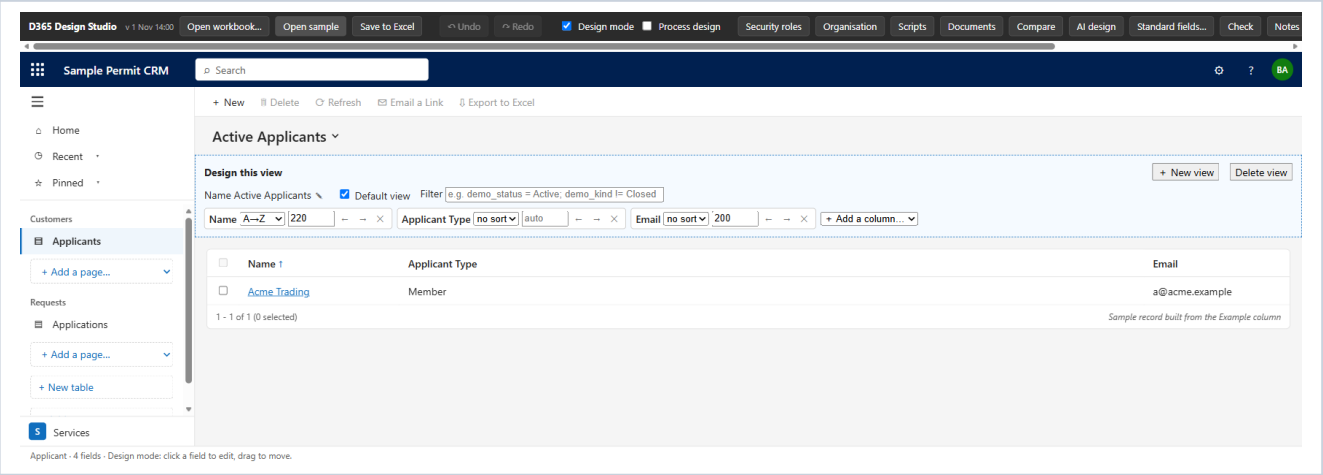


Figure 2. *The app shell. Navigation on the left, command bar on top, view designer (dotted blue box) above the list.*

Navigation (sitemap) design

- Rename a group or page with the pencil icon; reorder with the ↑ / ↓ arrows; remove with ✕.
- Add a page with + Add a page… (choose any table), and a group with + Add a group.
- Each change is saved to the Sitemap sheet of the workbook.

3.2 Views

Every table has one or more views, chosen from the view picker beside the title. In design mode a dotted panel shows what the BA can change.

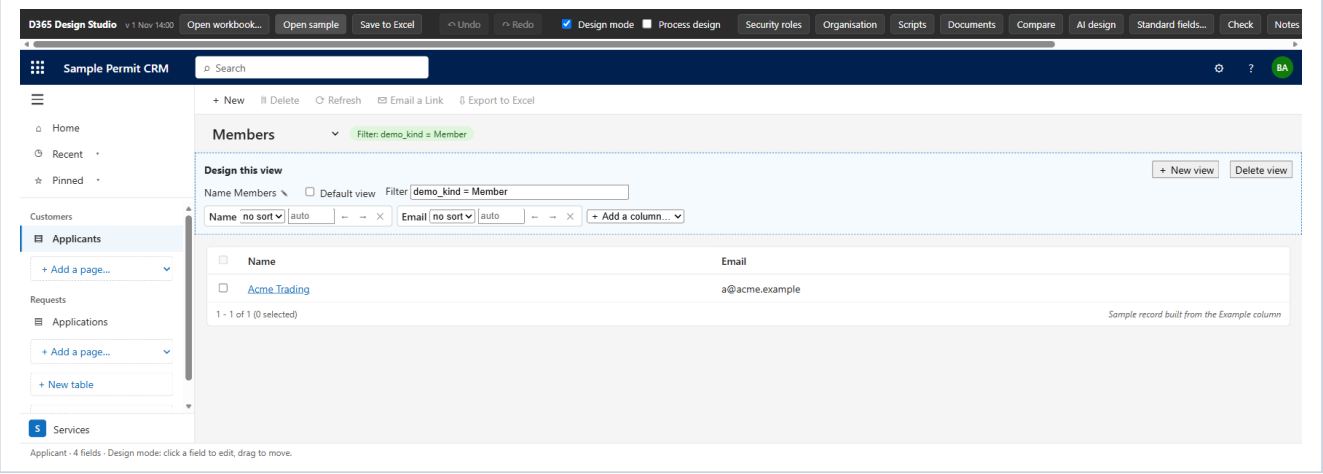


Figure 3. *A second view (Members) with its filter shown as a green chip beside the title.*

Control	Purpose
Name	Rename the view.
Default view	Tick to make this the view users land on.

Control	Purpose
Filter	A short rule such as demo_status = Active; demo_kind != Closed. The list and the Excel sheet use it.
Column boxes	Per column: sort (none, A→Z, Z→A), width, and ← → arrows to reorder; ✕ removes.
+ Add a column…	Adds any column of the table, including system columns: Created On, Created By, Owner, Modified On, Modified By.
+ New view / Delete view	Create a named view or remove one.

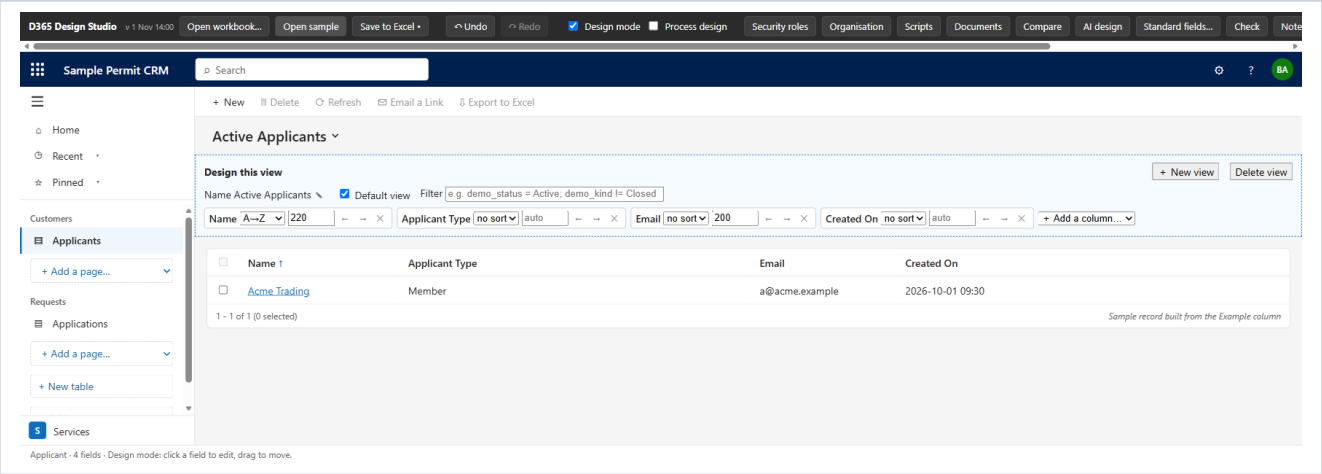


Figure 4. *Adding Created On to the default view. It appears at once in the list.*

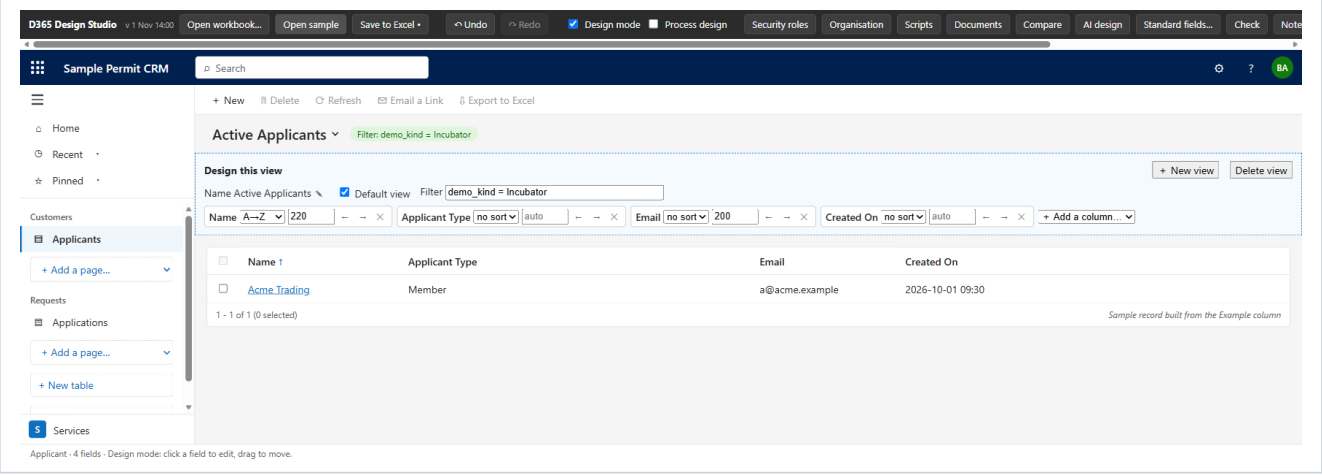


Figure 5. *Typing a filter in the view designer. Views are saved to the Views sheet.*

3.3 Adding a new table

A business analyst can add a table without any Excel work. In design mode, click + New table under the navigation. It sits above + Add a group.

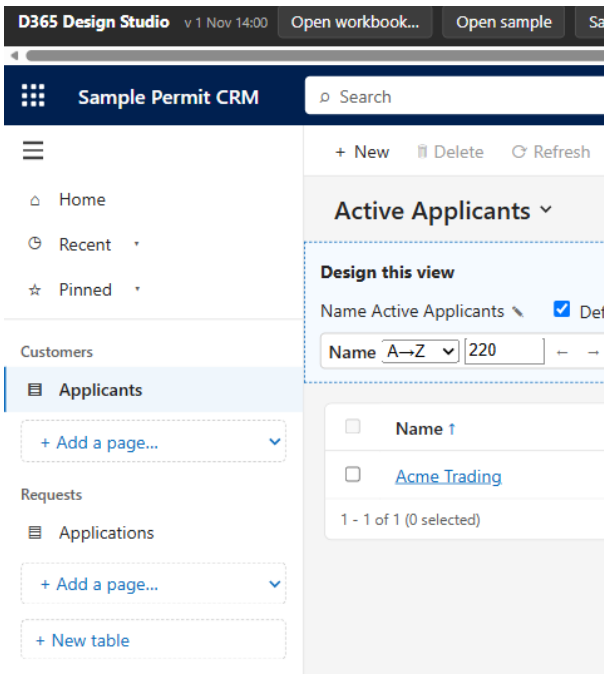


Figure 6. *The + New table button in the navigation (design mode).*

A small form opens. Only the display name is required.

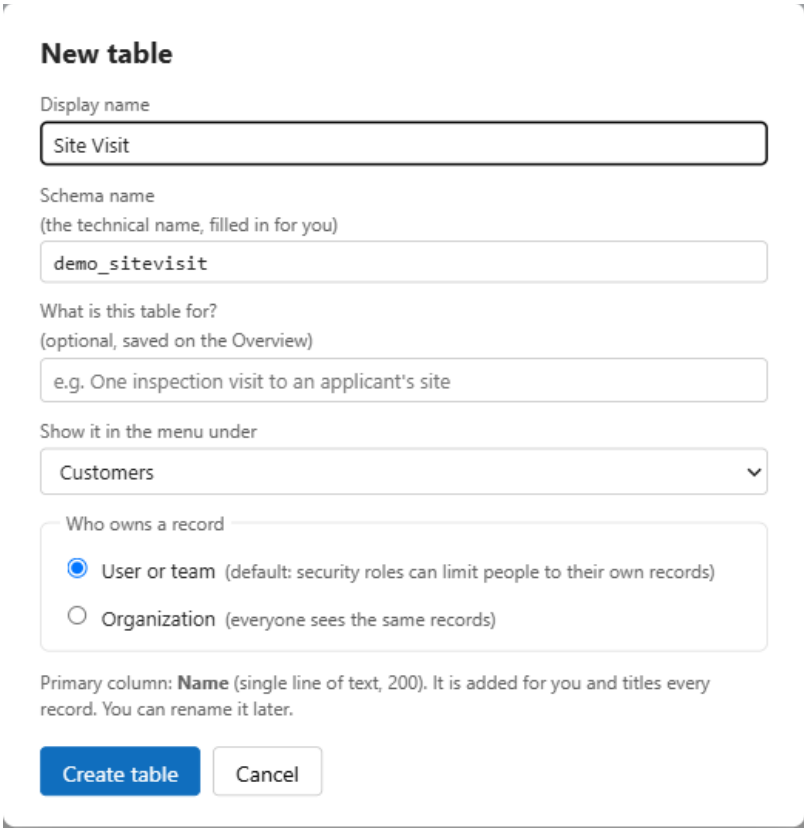


Figure 7. *New table dialog. The schema name (demo_sitevisit) filled itself in from the display name.*

Field	What it does
Display name	The name users see, for example Site Visit. It must not match an existing table.
Schema name	Filled in for you: the publisher prefix plus the name in lower case without spaces (demo_sitevisit). If it is already taken, a number is added. Type in the box to change it; Reset returns to the suggestion. Maximum 50 characters including the prefix.

Field	What it does
What is this table for?	Optional. Saved as the purpose on the Overview sheet and in the table sheet title.
Show it in the menu under	Pick one of the current group names, or choose A new group… and type a name. This is also written to the Block column of the Overview.
Who owns a record	User or team is selected by default. Organization is the other choice.
Primary column	Always Name (single line of text, 200 characters, marked as the record title). It is added for you and can be renamed later.

Create table adds everything in one step, and the new table opens on its form so fields can be added straight away.

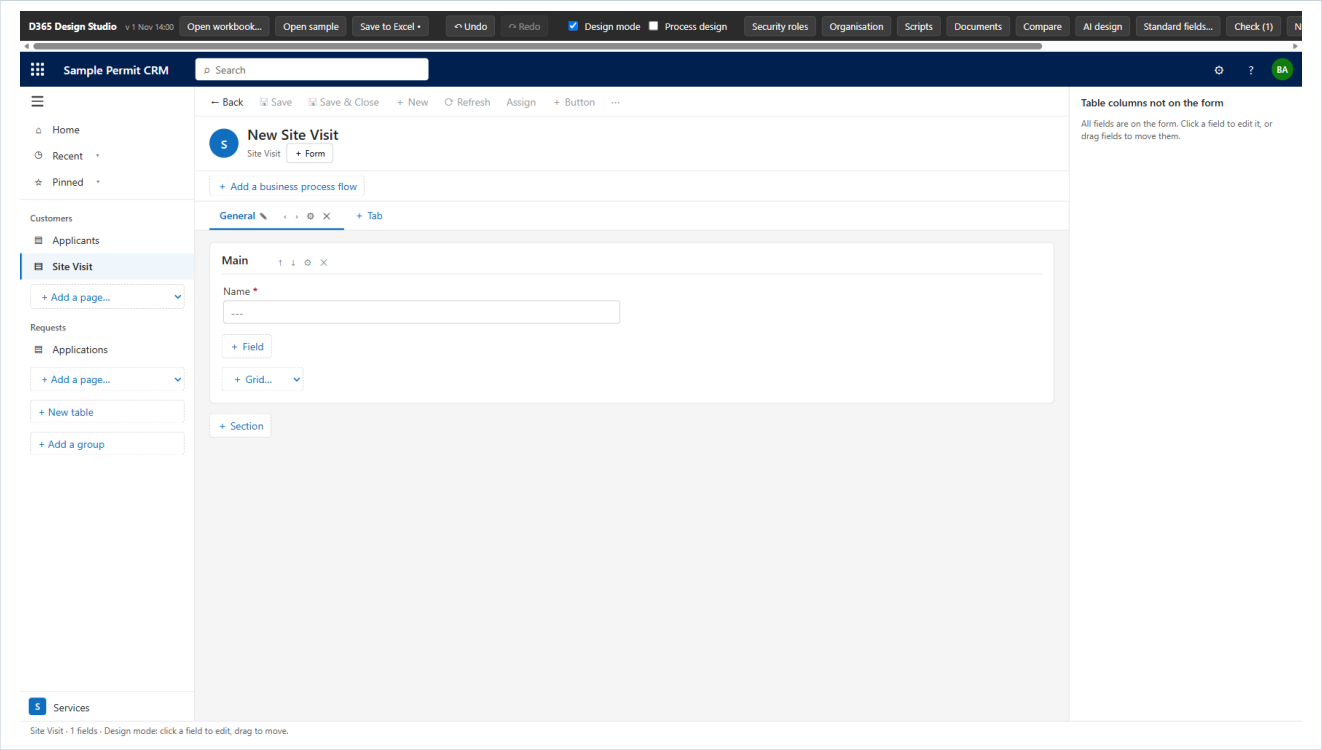


Figure 8. *The new table opens on its form with the Name primary column, and appears in the navigation.*

- **What is created:** a table sheet (title, Type, Block, Grain, Schema name and Ownership lines, then the field table with the Name column), a row on the Overview sheet, and a menu item. The worksheet itself is written when you press Save to Excel.
- **Undo:** one Undo removes the table, its Overview row and its menu item together.
- **Checks:** the dialog refuses a duplicate display name, a duplicate or invalid schema name, and a schema name without the publisher prefix, and says why in plain words.
- **Check design:** a new table is flagged "No role can read Site Visit" until you give a role Read access on the Security roles screen. This is expected; it stops a table nobody can open from reaching the build.

Ownership: User or team, or Organization

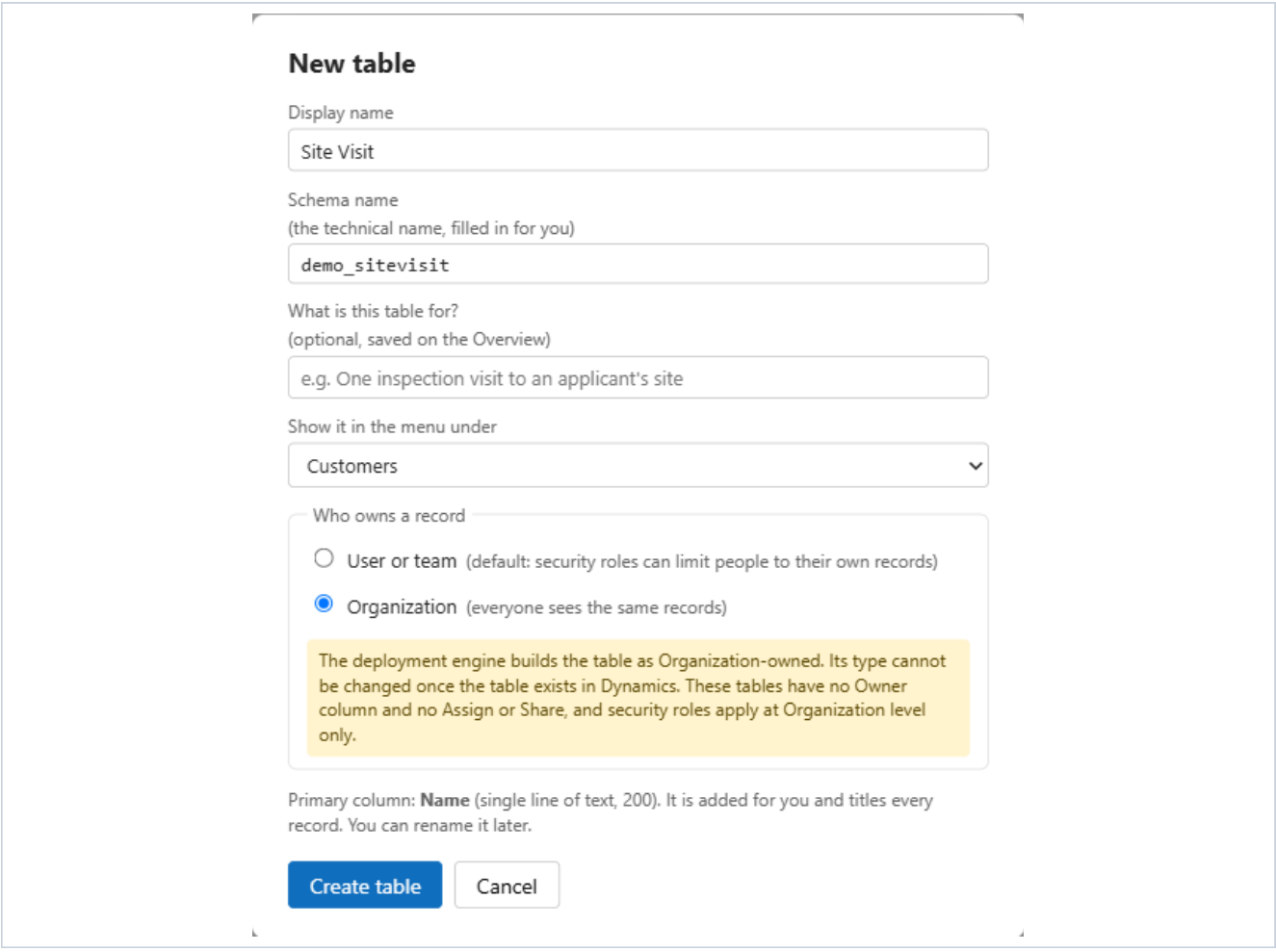


Figure 9. *Choosing Organization shows what the engine does and the limits of an organization-owned table.*

User or team is the Dataverse default and lets security roles limit people to their own records. Organization means every record is shared by everyone. The choice is saved on the Ownership line of the table sheet. A sheet without that line is read as User or team.

> **Engine** The deployment engine reads the Ownership line and builds an Organization table as Organization-owned. The type cannot be changed once the table exists in Dynamics, so decide before the first deployment. These tables have no Owner column, Assign or Share, and security roles apply at Organization level only. The engine warns about an owner field, owner-based views and privileges below Organization level.

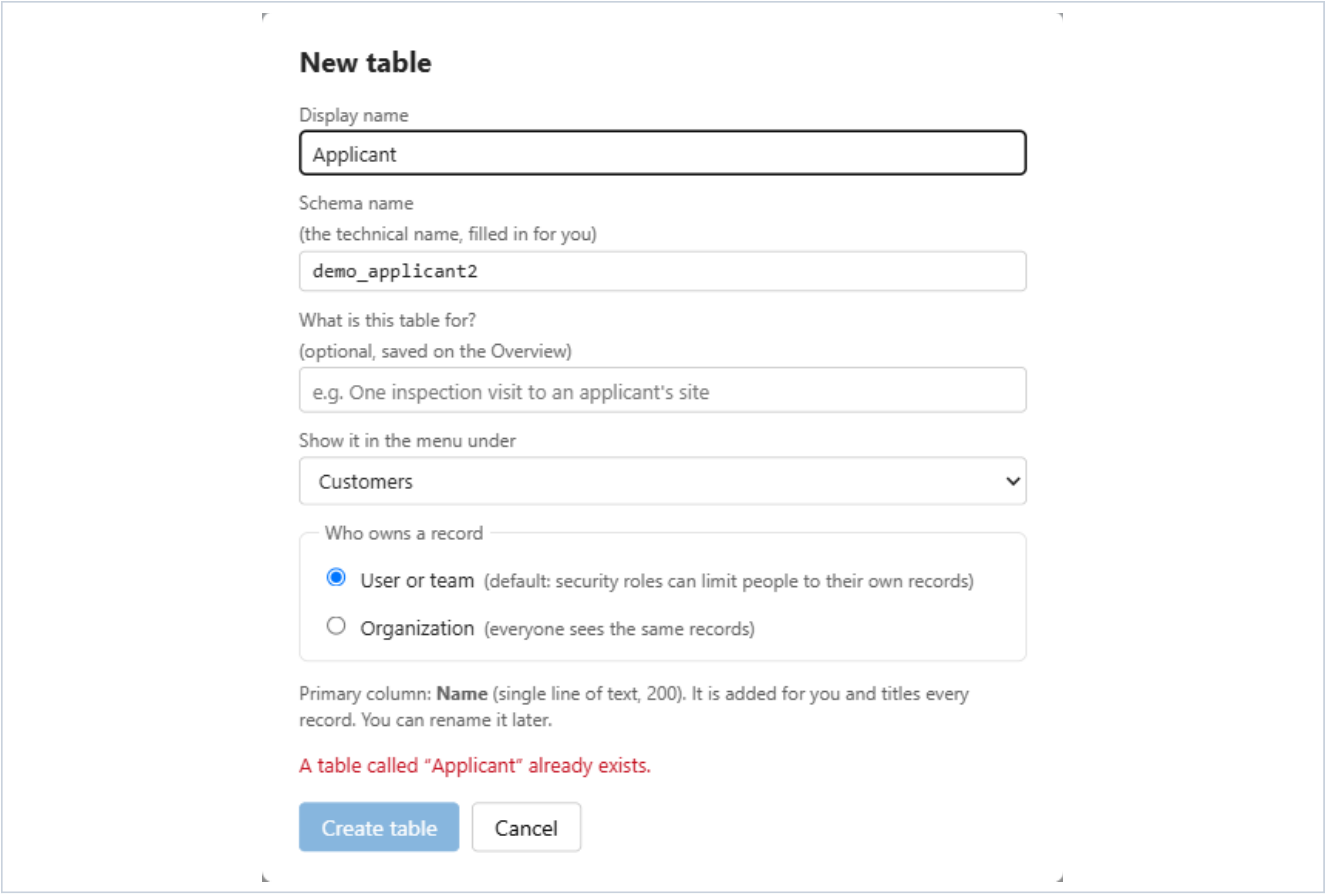


Figure 10. *Entering a name that already exists: the message appears and Create table stays disabled.*

4. The form designer

Click a record in the list to open its main form. The form is built from tabs, sections and fields, exactly as in Dynamics 365. In design mode every part is editable. If the record opens on the process screen, because its process has step fields, as both tables in the sample do, click Ordinary form at the top to design the form (section 16.17).

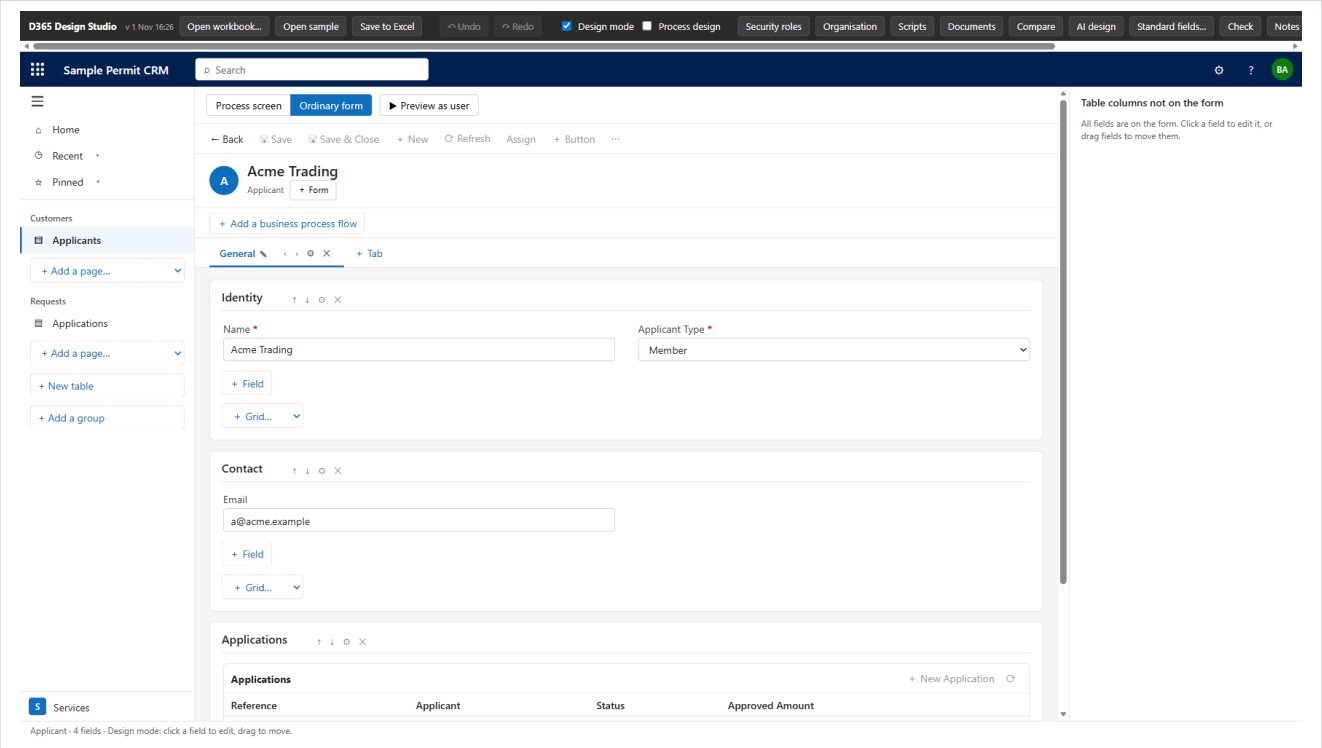


Figure 11. *A form in design mode: tabs (General), sections (Identity, Contact, Applications) and fields. The right panel lists table columns not yet on the form.*

4.1 Tabs, sections and fields

- **Tabs:** click the pencil to rename, ‹ › to reorder, ✕ to remove, + Tab to add.
- **Sections:** rename by clicking the title; drag by the title or use ↑ ↓ to reorder (drag onto a tab to move it to that tab); ✕ removes; + Section adds one.
- **Fields:** drag a field to another section or position. + Field adds a new field to that section.
- **Table columns not on the form:** the right-hand panel lists columns that exist but are not placed. Drag one onto the form, or click it to edit.
- **Grids:** use + Grid… in a section to add a related-records grid for any table (shown as Applications above).

4.2 Field properties

Click any field to open its properties on the right.

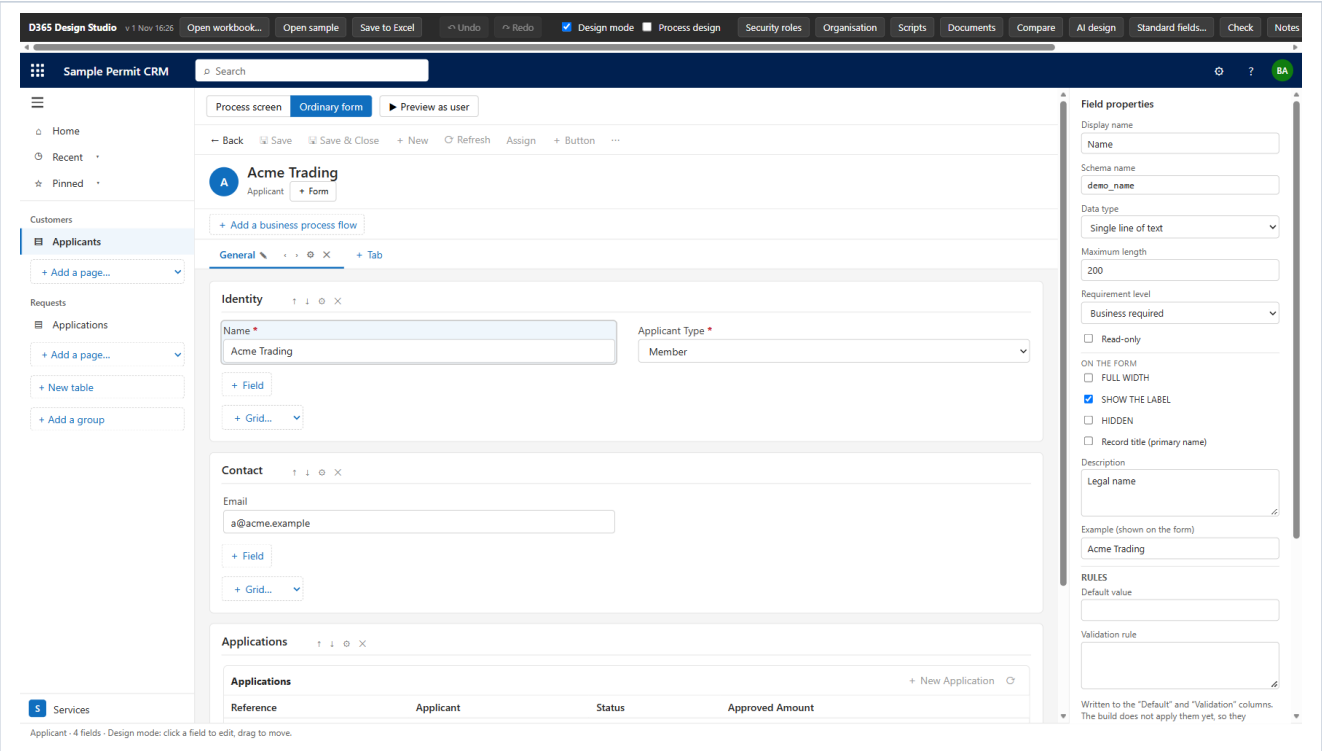


Figure 12. *Field properties for Name: type, length, requirement level, record title, description, example value, and the migration fields at the bottom.*

Property	What it means
Display name / Schema name	The label users see, and the technical name. A new field fills its schema name itself from the display name (publisher prefix + name).
Data type	Single line of text, multiple lines, whole number, decimal, currency, date only, date and time, Yes/No, Choice, Choices (multi-select), Lookup, Subgrid, Autonumber, Email, Phone, Owner.
Maximum length	For text types.
Requirement level	Optional, Business required, or System required. Required fields show a red star.
Read-only	Users cannot edit the field.
Record title (primary name)	Tick on the one text or autonumber column that names each record. Only one per table.
Description / Example	Description is saved for the developer. Example is the value shown in the sample record.
Default value / Validation rule	Recorded for the developer. The build does not apply them yet.
Legacy field / Migration notes	Where the data comes from in the old CRM (section 8.3).
Remove from form / Delete field	Remove keeps the column in the table (OnForm = Removed). Delete removes its row from the workbook.

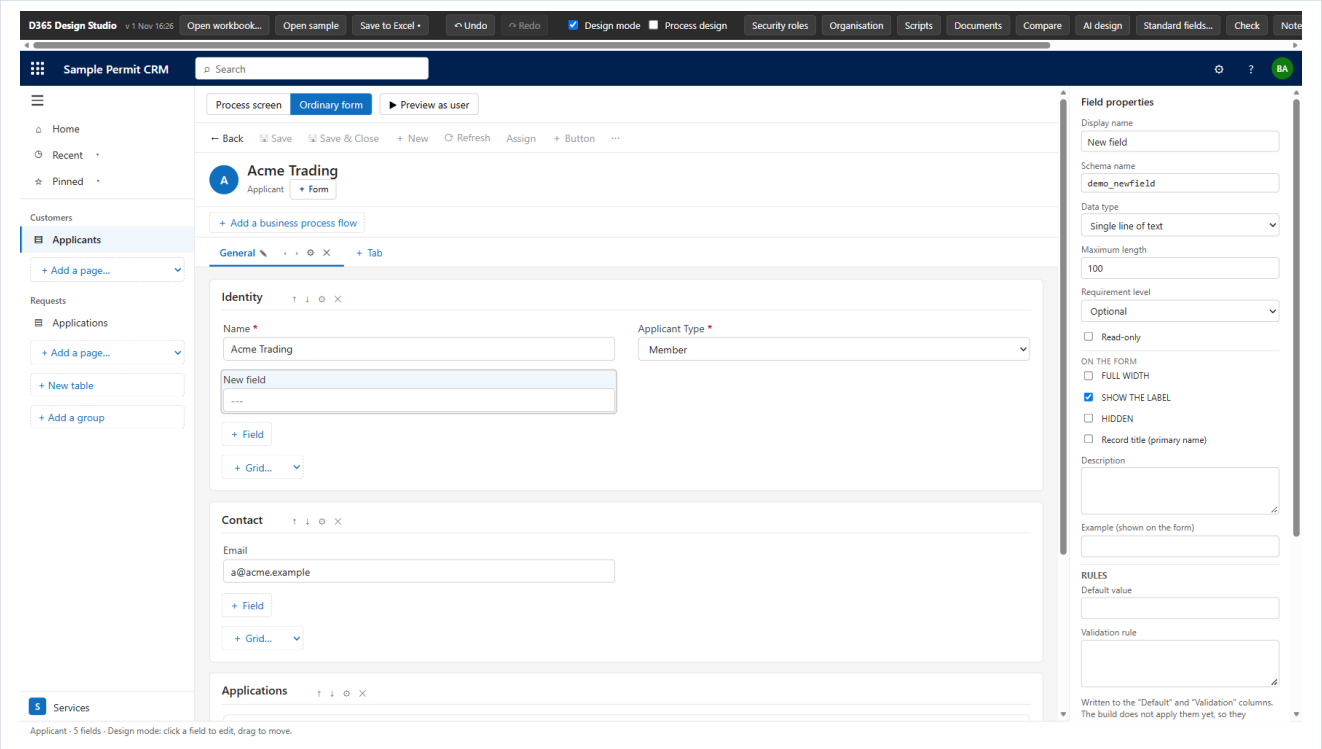


Figure 13. *A new field added to the Identity section. Its schema name (demo_newfield) was generated automatically.*

4.3 Several forms per table

Under the record title, + Form adds another main form that starts as a copy of the current one. Name it, press Enter, and switch between forms from the same place. Each form has its own tabs, sections and fields, saved in the Form column of the workbook.

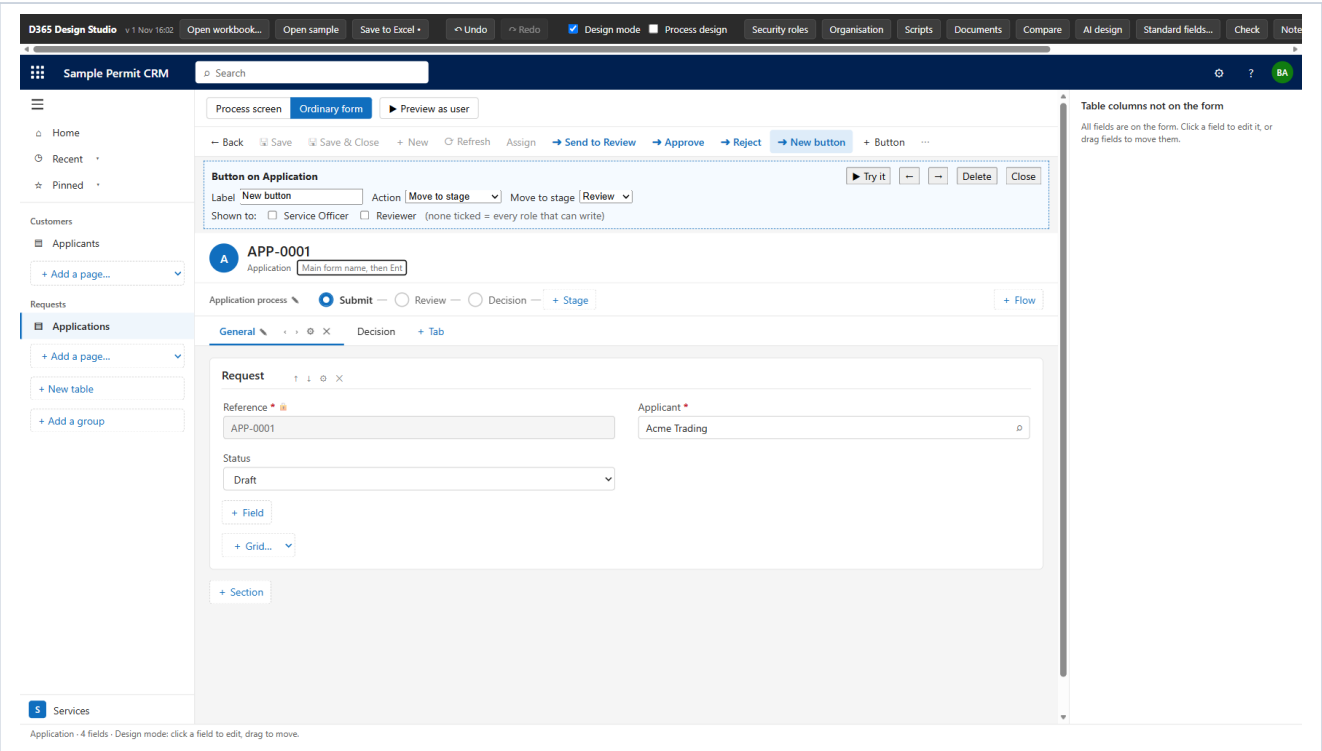


Figure 14. *Naming a new form (the box under the record title). The command-bar button editor is open above.*

4.4 Standard fields first

On a standard Dynamics table (Account, Contact and so on), Standard fields… offers the columns Dynamics already has, with the ones on the standard form listed first. This avoids creating a duplicate of something that exists. The list comes from a small file exported from your environment by the ExportStandardFields tool (see docs/STANDARD_FIELDS.md). The custom tables in the sample workbook do not need it, so it is not illustrated here.

4.5 Controlling the layout: tabs, sections and fields

In design mode, the gear icon next to a tab or a section opens its layout settings. A field has the same settings in the right-hand panel, under On the form.

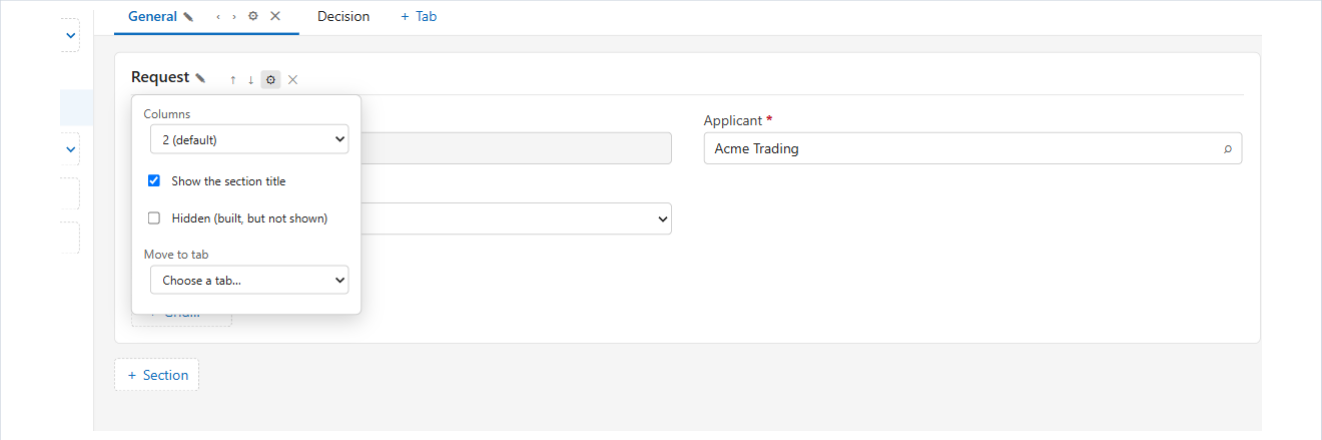


Figure 15. *The gear menu of a section: columns, section title, hidden, and Move to tab.*

Setting	What it does
Move a section to another tab	Drag the section title onto a tab, or use Move to tab in its gear menu. If the other tab already has a section with the same name, the fields join it. The section's settings move with it.
Move a field to another tab	Drag the field onto a tab. It lands in the first section of that tab. Drag it onto a field to place it before that field.
Merge tabs	In the tab's gear menu, Merge this tab into… moves every section across. One Undo reverses it.
Columns (section)	1, 2, 3 or 4 columns. Two is the default and is not written to the workbook.
Show the section title (section)	Untick to hide the title. In design mode the title stays visible but struck through, so you can still select it.
Hidden (tab, section, field)	The item is built on the form but not shown. While designing it appears dimmed with a "hidden" badge. In Preview it disappears.
Full width (field)	The field spans the whole width of its section, whatever the column count. Multi-line fields and grids are already full width.
Show the label (field)	Untick to show the field without its label.

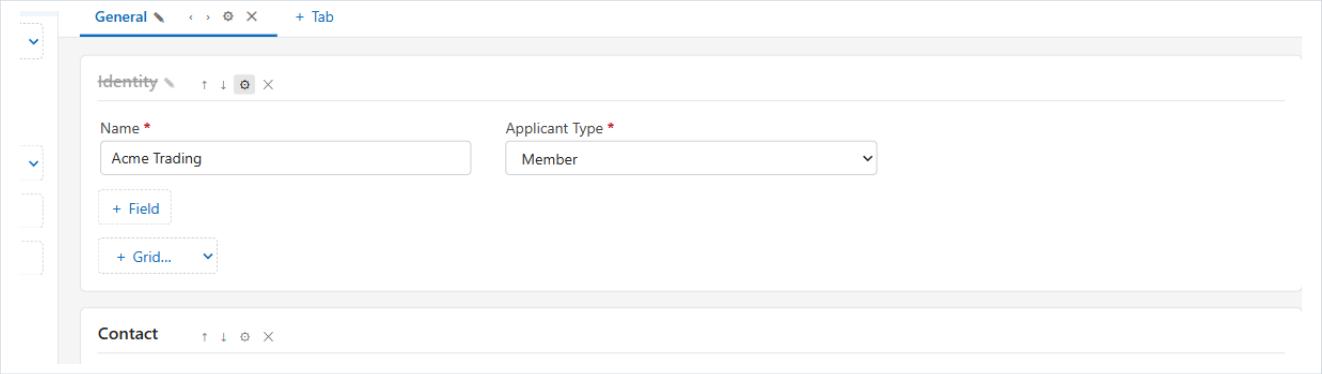


Figure 16. *A section set to three columns with its title hidden (struck through while designing). Preview shows no title.*

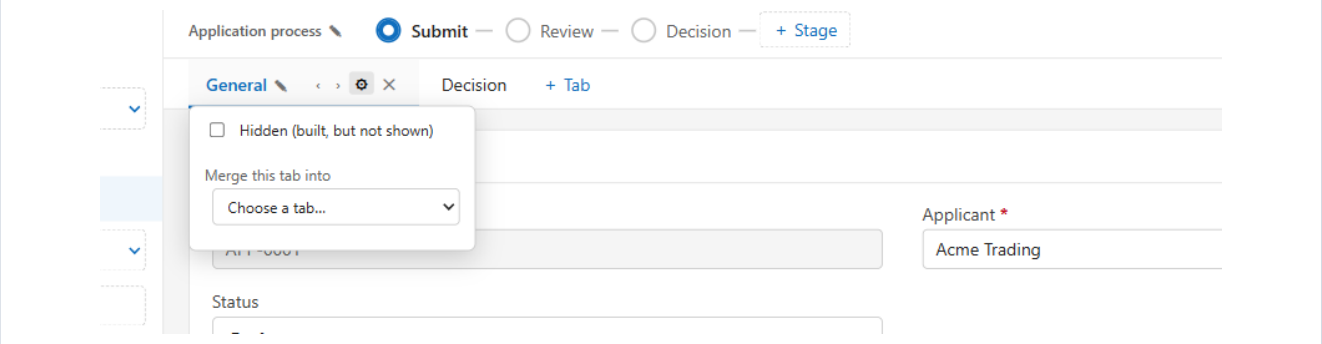


Figure 17. *The gear menu of a tab: Hidden, and Merge this tab into another tab.*

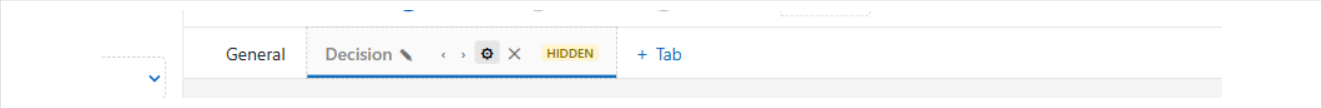


Figure 18. *A hidden tab, shown dimmed with a "hidden" badge. It does not appear in Preview.*



Figure 19. *Field properties: Full width, Show the label and Hidden, under On the form.*

Where it is saved Which tab and section a field is in, and the order of tabs, sections and fields, are saved in the table sheet, as before. Columns, hidden, label and full-width settings are saved in a new optional Form Layout sheet (Excel format, section 6.16). The engine reads that sheet when it builds the form, and a workbook without the sheet builds exactly as before. This has been checked offline on a real form, not yet in a live deployment. One limit: a field that is already on a live form is not moved when its tab or section changes in the workbook. Details are in docs/ENGINE_FORM_LAYOUT.md.

5. Business process flows

A business process flow is the stage bar across the top of a form. Click a stage to see its steps. In design mode the BA builds the flow directly on the form.

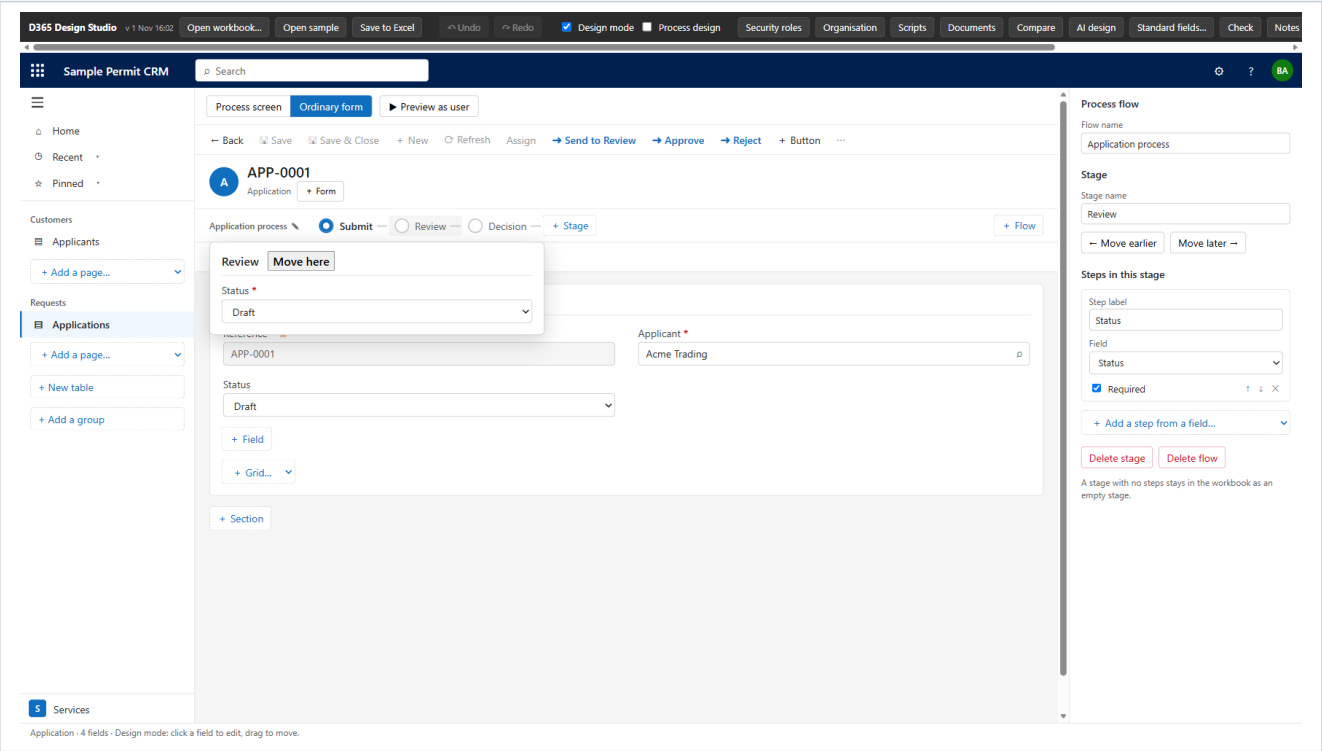


Figure 20. *A flow with two stages. + Stage adds another, + Flow adds a second flow, and the right panel edits the selected stage.*

- Click + Add a business process flow to create one, then + Stage to add stages.
- Select a stage to rename it, move it earlier or later, or delete it. Delete flow removes the whole flow.
- Add steps from a field: each step is a field the user fills in before leaving the stage. Tick Required where the process cannot move on without it.
- A stage with no steps is kept in the workbook as an empty stage.
- Flows are saved to the Process Flows sheet.

Need more? For requests that move between people, with assignment rules, time allowed, reminders, rejection and a preview of what users see, use the optional process designer in section 16.

6. Command-bar buttons

The command bar above a form holds the standard buttons (Save, Save & Close, New, Refresh, Assign, Share). In design mode the BA can add stage buttons that move the record to a chosen stage.

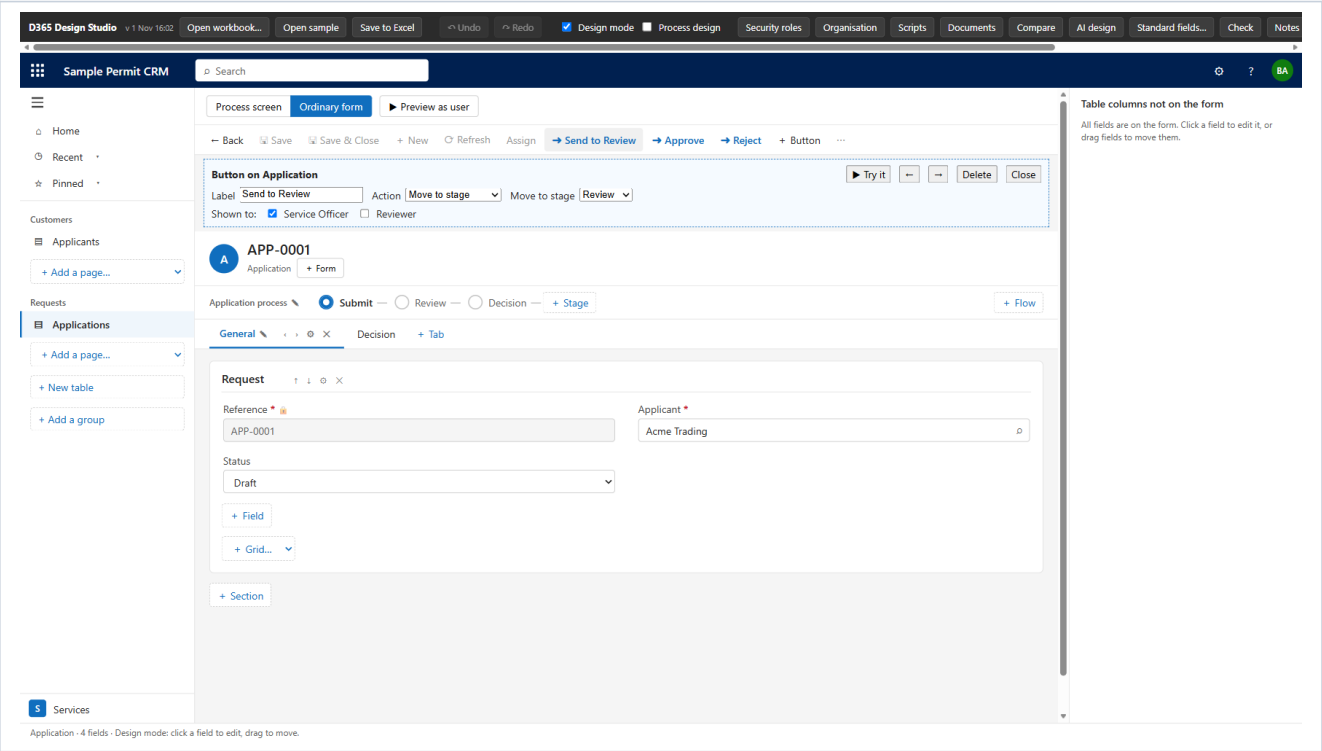


Figure 21. *Editing a stage button: label, action (Move to stage), target stage and the roles who see it.*

Setting	Meaning
Label	The text on the button.
Action / Move to stage	Moves the record to the chosen stage, optionally setting a field first.
Shown to	Tick roles to limit who sees it. None ticked means every role that can write.
Try it	Test the button on the sample record. A banner explains what happened.

Buttons are saved to the Command Bar sheet. In the stage flyout of a flow, Move here does the same by hand.

7. Preview and View as

7.1 Preview

Untick Design mode to hide every design control. The app looks and behaves like the final product, which is the right view for a client walkthrough. To walk through a process, use the process screen a record opens on when its process has step fields (section 16.17).

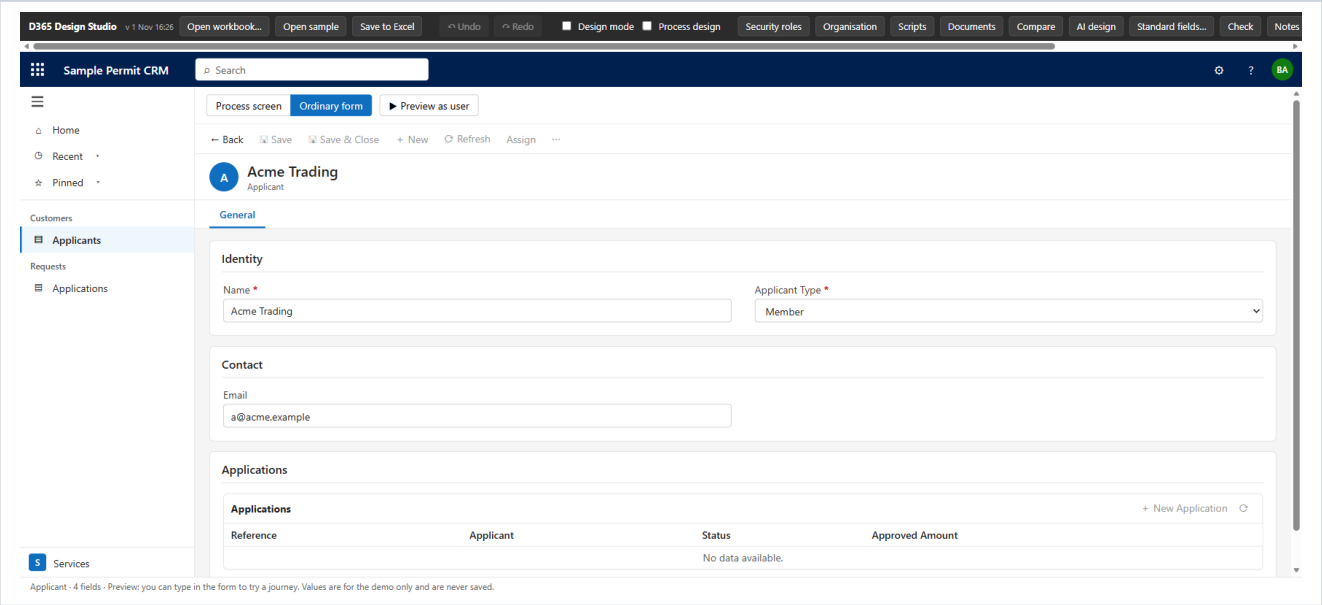


Figure 22. *Preview: no design controls, just the app.*

In Preview the BA can type or pick values on a form to walk a client through a journey. These sample values are held only while the page is open: they are never written to the workbook and the engine never sees them. They also appear in the list (the footer says so), and the form title follows the record title. The Reset sample data button on the top bar forgets them, and so does opening another workbook. Refreshing the page also clears them.

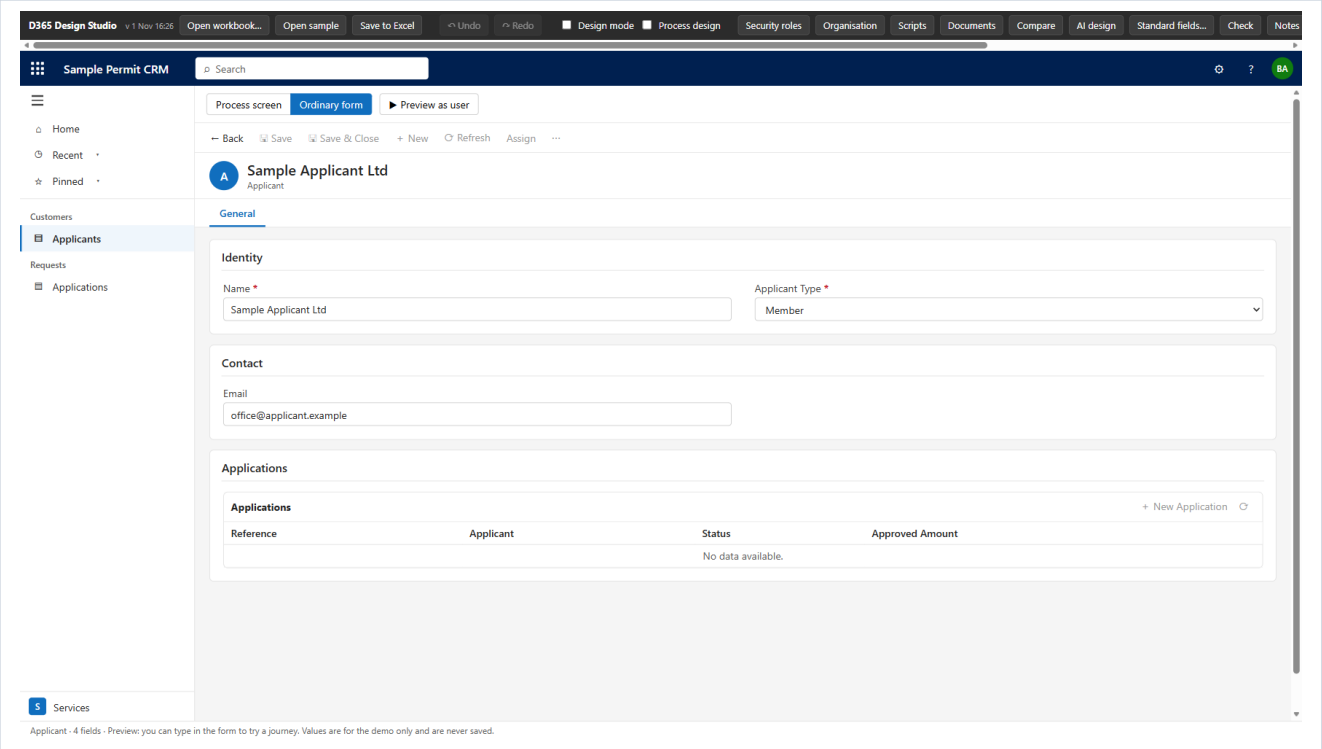


Figure 23. *Preview with typed sample values: the title follows the Name field and Reset sample data (2) appears on the top bar.*

7.2 View as a role

Pick a role in View as on the top bar. The Studio redraws the whole app as that user would see it, from the role's privileges.

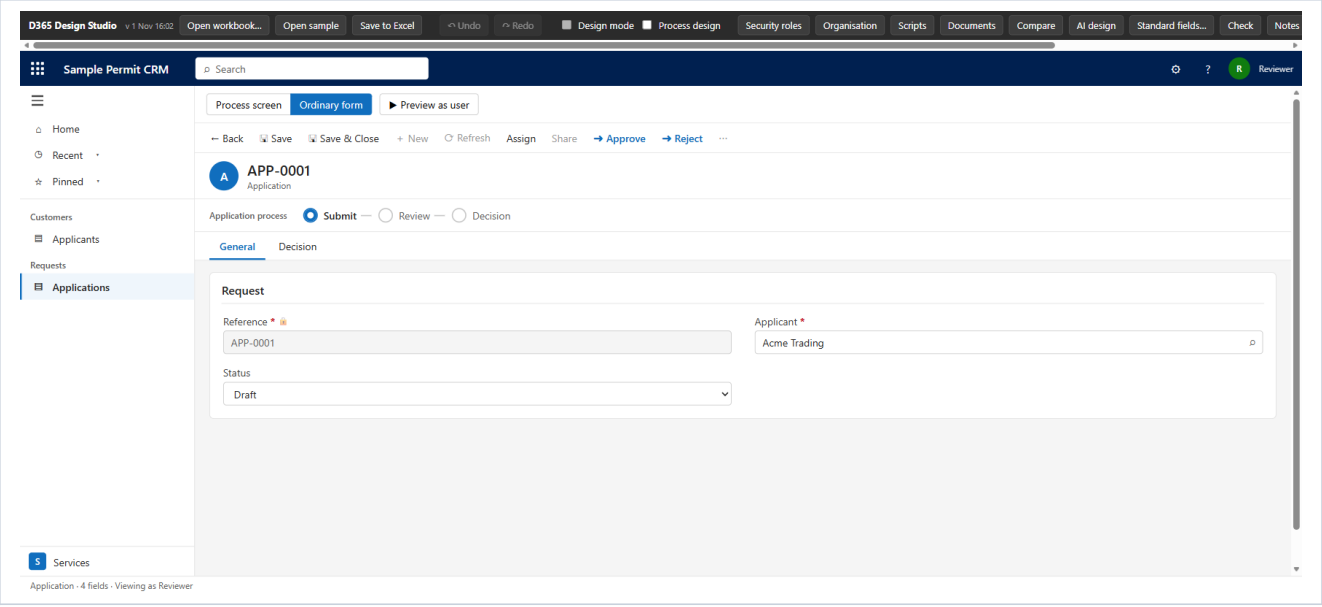


Figure 24. *View as "Reviewer": read-only fields show a lock, buttons the role lacks are hidden, navigation is limited.*

- Navigation shows only the areas the role is allowed to open.
- Buttons appear only if the role has the matching privilege: New (Create), Delete, Assign, Share, Save (Write).
- Fields restricted by field security show as hidden or read-only.
- The list shows only the records the role's Read scope allows.
- Pick Designer (all access) to return to the full view.

8. Check design, Notes and Migration

Three tabs in the right-hand review panel help the BA keep the design clean and the project organised.

8.1 Check design

One click lists what would break the build and what is worth a look. Each item can be clicked to jump to it.

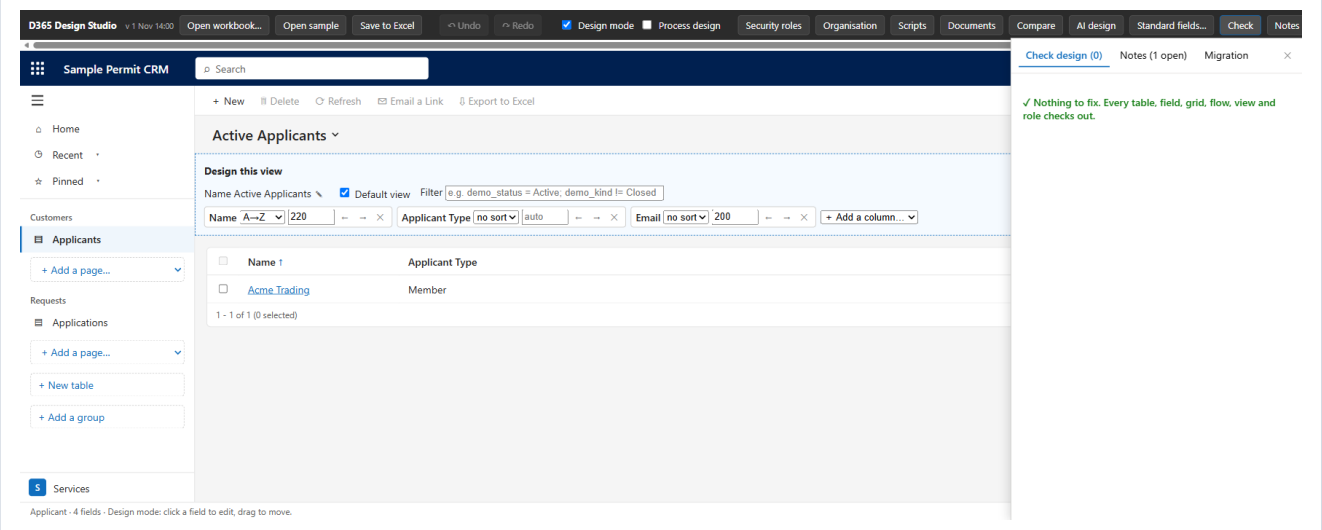


Figure 25. *Check design: green when nothing needs fixing.*

Kind	Examples
Errors (would break the build)	A choice field with no options, a grid that points to a missing table, duplicate or invalid schema names, a lookup that does not say what it looks up, a navigation page or flow that points at a missing table, a stage button aimed at a missing stage, two columns marked as the record title.
Process area (when Process design is on)	A human step with no actions, a rule with no Otherwise row, a rule that looks at a missing field, a reminder with nothing to count from, an AI step with no agent (section 16.13).
Warnings (worth a look)	A table nobody can read, a table missing from the menu, a view column that does not exist, a role that names a missing table or field, a lookup to a table with no sheet (fine if it is a standard Dynamics table).

8.2 Notes

Log questions, decisions and actions during a session, attached to a table or a field, and tick them off. Notes are saved to a Notes sheet that the engine ignores, so the workshop record travels with the workbook.

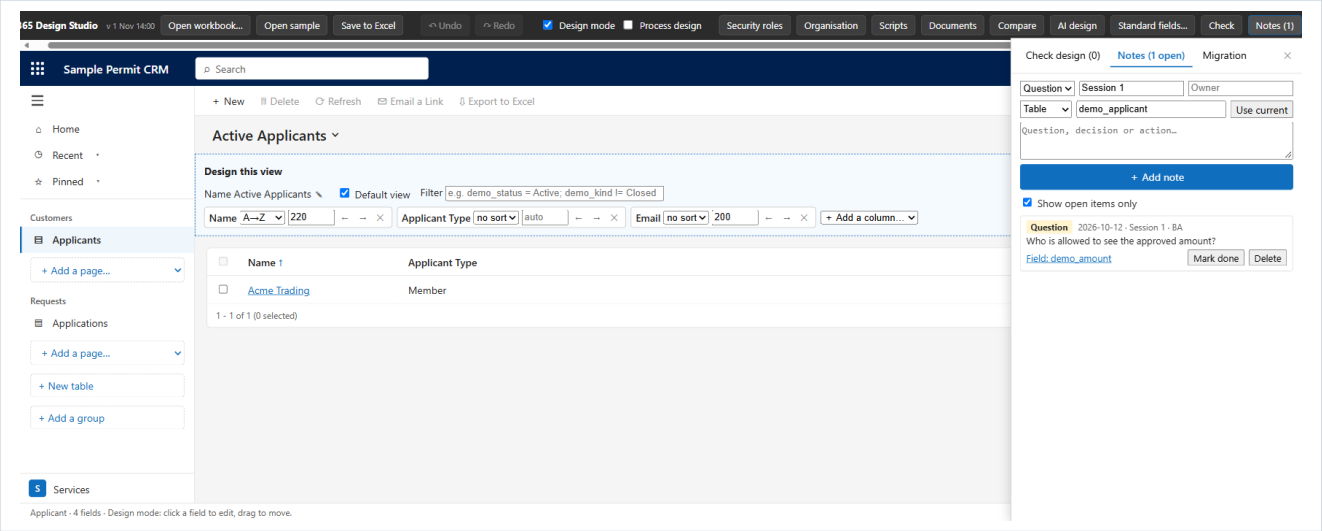


Figure 26. *Notes panel: type (Question, Decision, Action), session, owner, target table or field, and the open-items filter.*

8.3 Data migration mapping

Each field has a Legacy field and Migration notes box. The Migration tab shows how many fields are mapped per table, so migration progress is visible. They are saved as two extra columns on the table sheet, which the engine ignores.

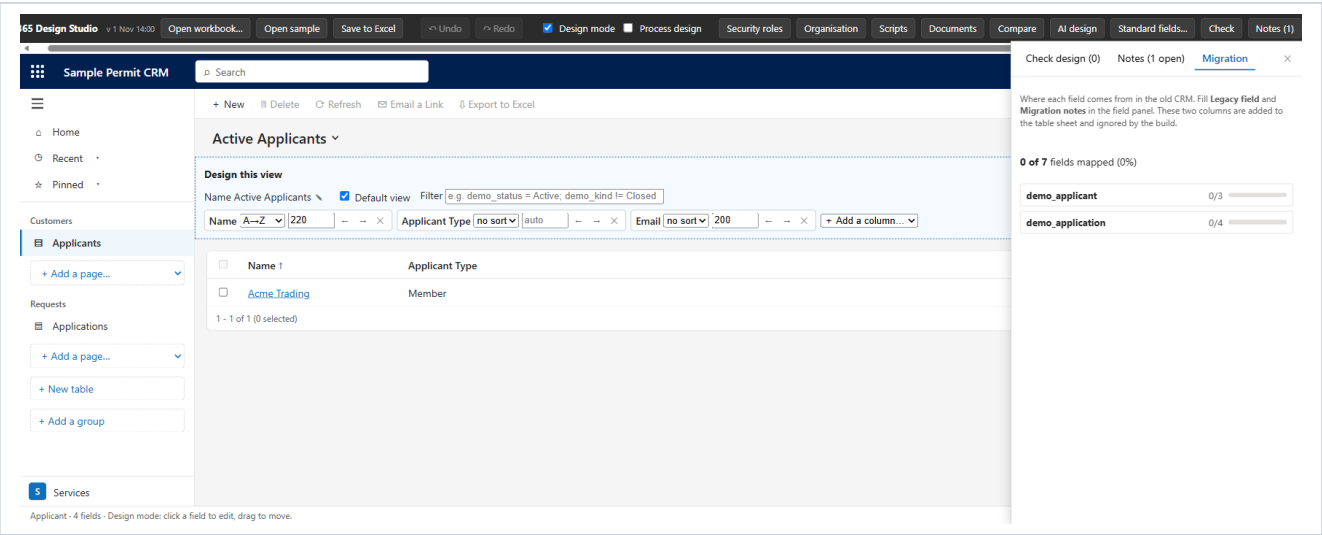


Figure 27. *Migration tab: fields mapped per table and overall percentage.*

9. Security roles

The Security roles screen defines what each role can do, see and open. It uses the same circles Dynamics 365 uses: none, own records (User), Business Unit, Parent-Child BU, and Organization.

9.1 Table privileges

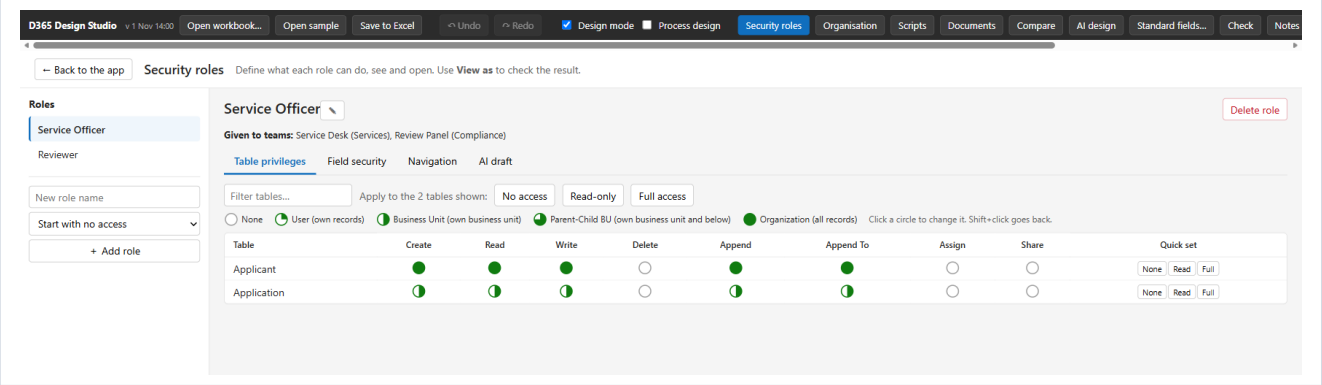


Figure 28. *Table privileges grid for Service Officer. Click a circle to change the scope; Shift+click goes back.*

- Rows are tables; columns are Create, Read, Write, Delete, Append, Append To, Assign, Share.
- Quick set per table: None, Read, Full. Apply to the shown tables: No access, Read-only, Full access.
- Create a role with + Add role and choose how it starts: no access, read-only on all tables, or full access.
- "Given to teams" under the title shows which teams hold the role.

9.2 Field security

Choose a table, then set each field to Editable, Read-only or Hidden for the role. Fields not listed stay editable. Dynamics secures a whole column and uses profiles, so the build gives each role in the workbook its own profile.

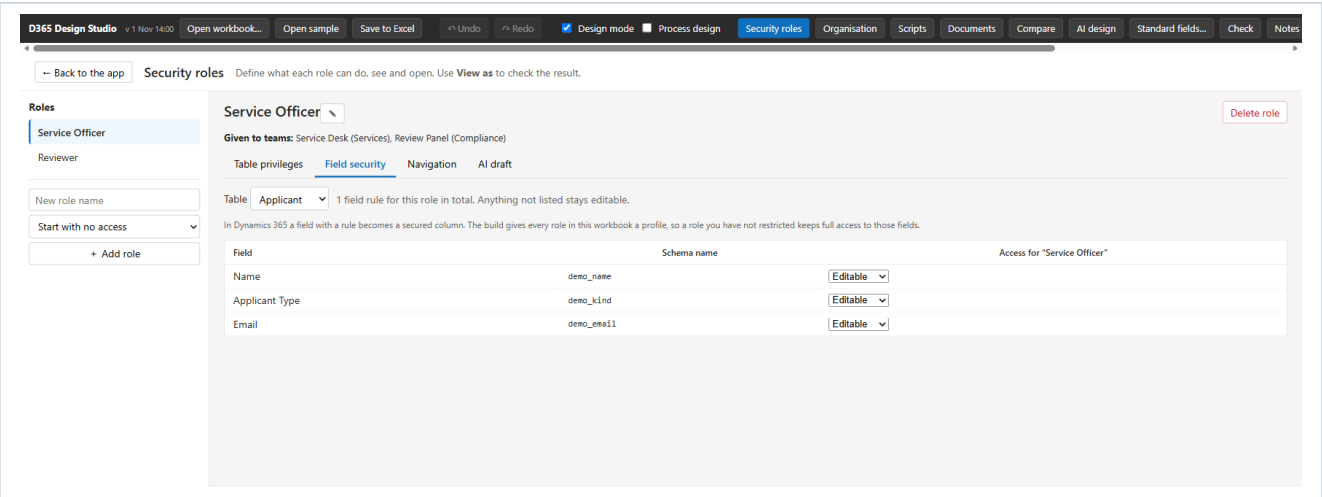


Figure 29. *Field security for the Applicant table.*

9.3 Navigation

Tick Limit this role to the areas ticked below, then tick the areas the role may open. These limits shape View as and are saved to the Role Navigation sheet.

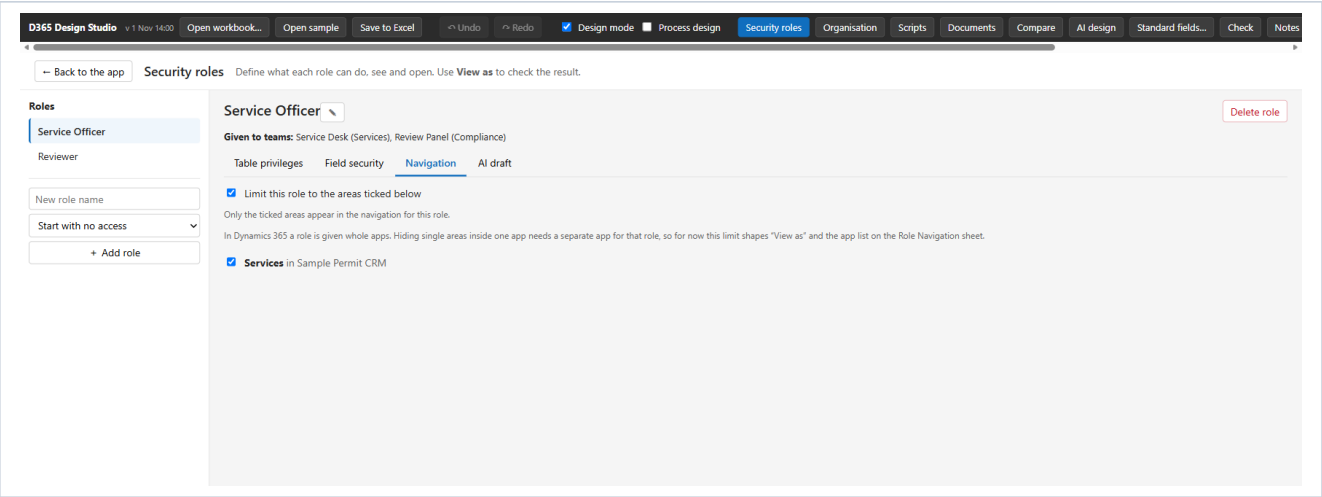


Figure 30. *Navigation limits per role.*

9.4 AI draft for roles (optional)

Describe the department in plain words and the AI proposes table privileges and field restrictions. The BA reviews, unticks anything they disagree with, and presses Apply. Undo reverses it. It needs the BA's own Anthropic API key, which is kept only in the browser. By default only the description and table names are sent; table purposes and field names are opt-in, and a "What is sent" box shows the exact text.

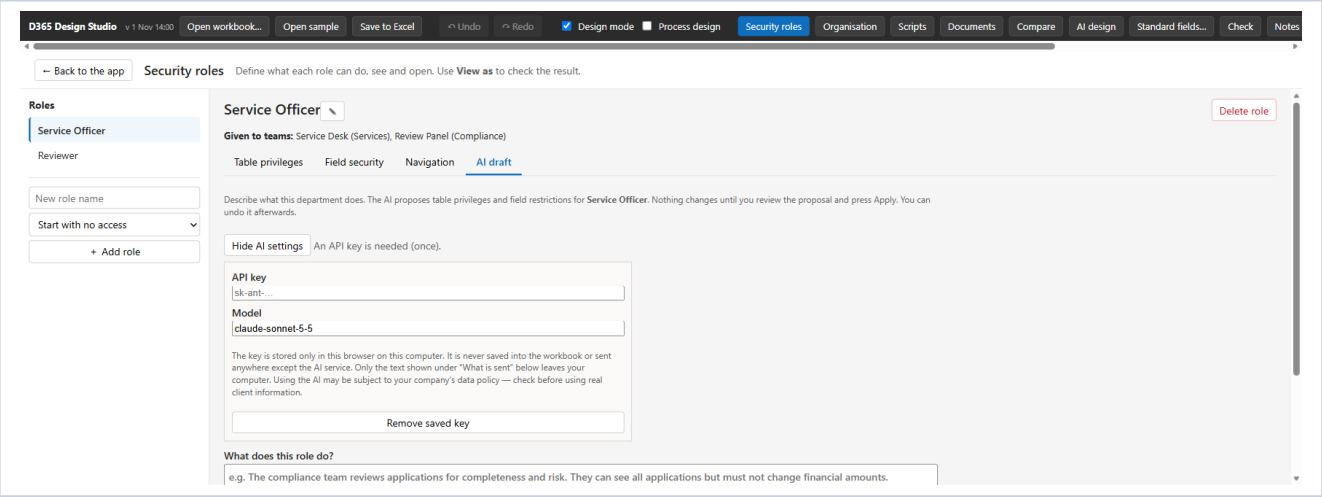


Figure 31. *AI draft tab: key settings, description box and the privacy note.*

> **Note** The role sheets written are Security Roles, Role Field Rules and Role Navigation. The Studio works fully without the AI feature.

10. Organisation: business units and teams

Business units decide what "Business Unit" and "Parent-Child" mean in a security role. Teams hold the roles. The Organisation screen shows both as a tree.

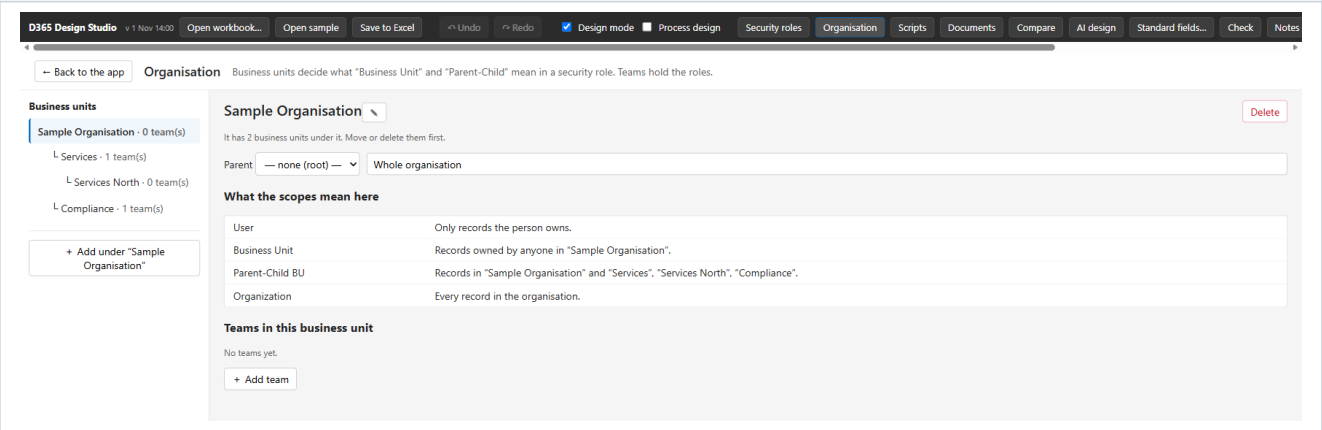


Figure 32. *Business unit tree with the plain-words table "What the scopes mean here".*

- Add a unit under any unit; rename, move (parent) or delete it. Deleting needs the unit to be empty first.
- The "What the scopes mean here" table updates as the tree changes, so the BA can explain each scope to the client in concrete terms.
- Add teams to a unit with + Add team. Each team has a name, a type (Owner or Access) and a business unit.
- Tick the roles each team holds. The Roles screen then shows which teams hold each role.

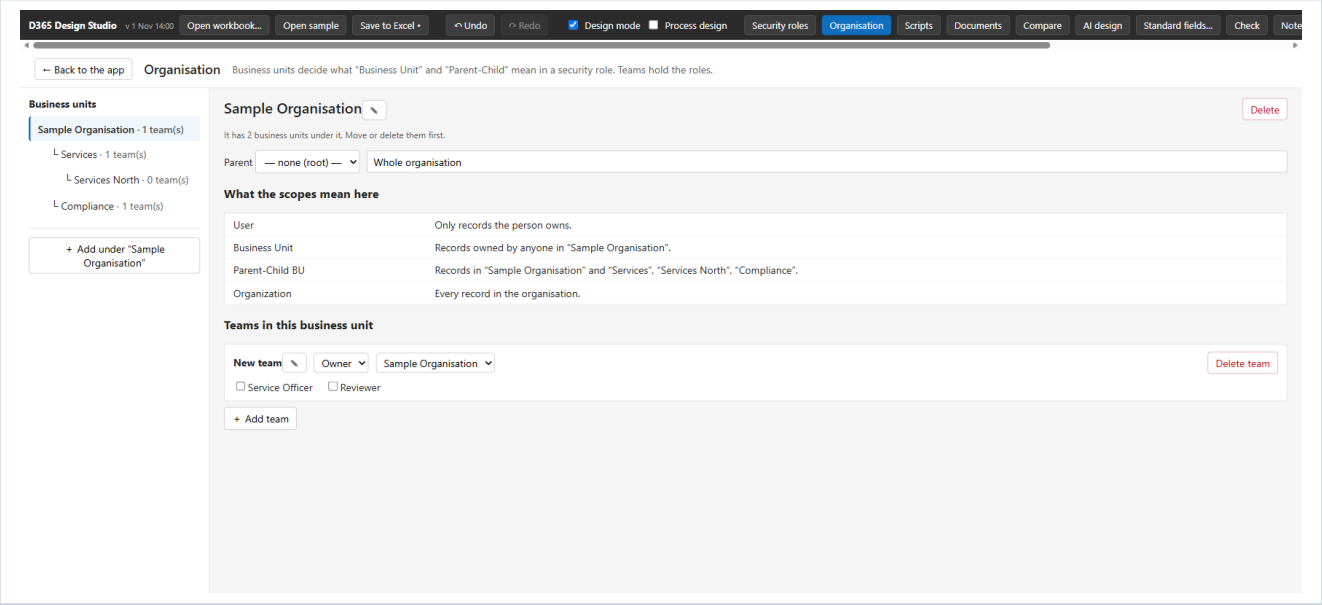


Figure 33. *A new team with its type, business unit and role tick boxes.*

> **Rule** Access teams share individual records and cannot hold security roles in Dataverse. The Studio shows a note instead of the role boxes, and the engine skips roles on Access teams. Choose Owner if the team needs roles.

Saved to the Business Units and Teams sheets, which the engine uses to create them in Dataverse (org action, dry run available).

11. Scripts

Scripts record which JavaScript files the solution publishes and which function runs on which form event. They edit the Web Resources and Form Scripts sheets.

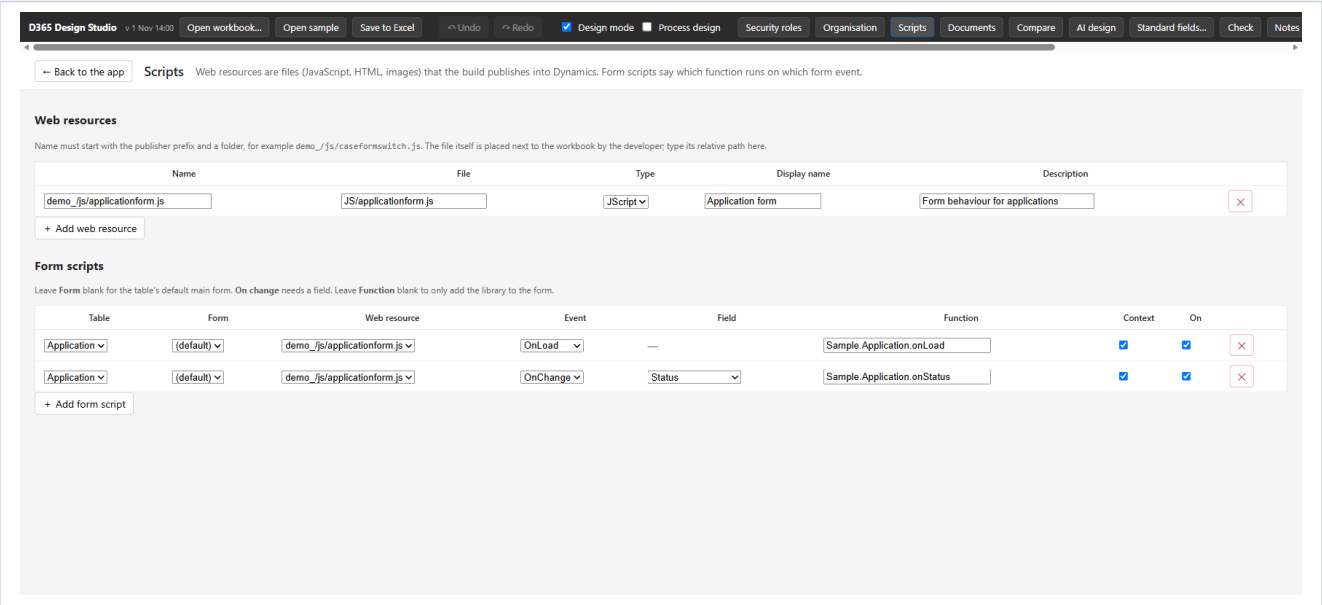


Figure 34. *Web resources (top) and form scripts (bottom).*

- **Web resources:** name (publisher prefix and folder), file path next to the workbook, type, display name and description.
- **Form scripts:** table, form (blank = default main form), web resource, event (OnLoad, OnChange…), field (for OnChange), function name, pass execution context, enabled.
- Leave Function blank to add only the library to the form.

12. Documents

Documents are the Word or PDF letters, certificates and summaries Dynamics will produce from a record. The BA designs the content; each section is built from lines.

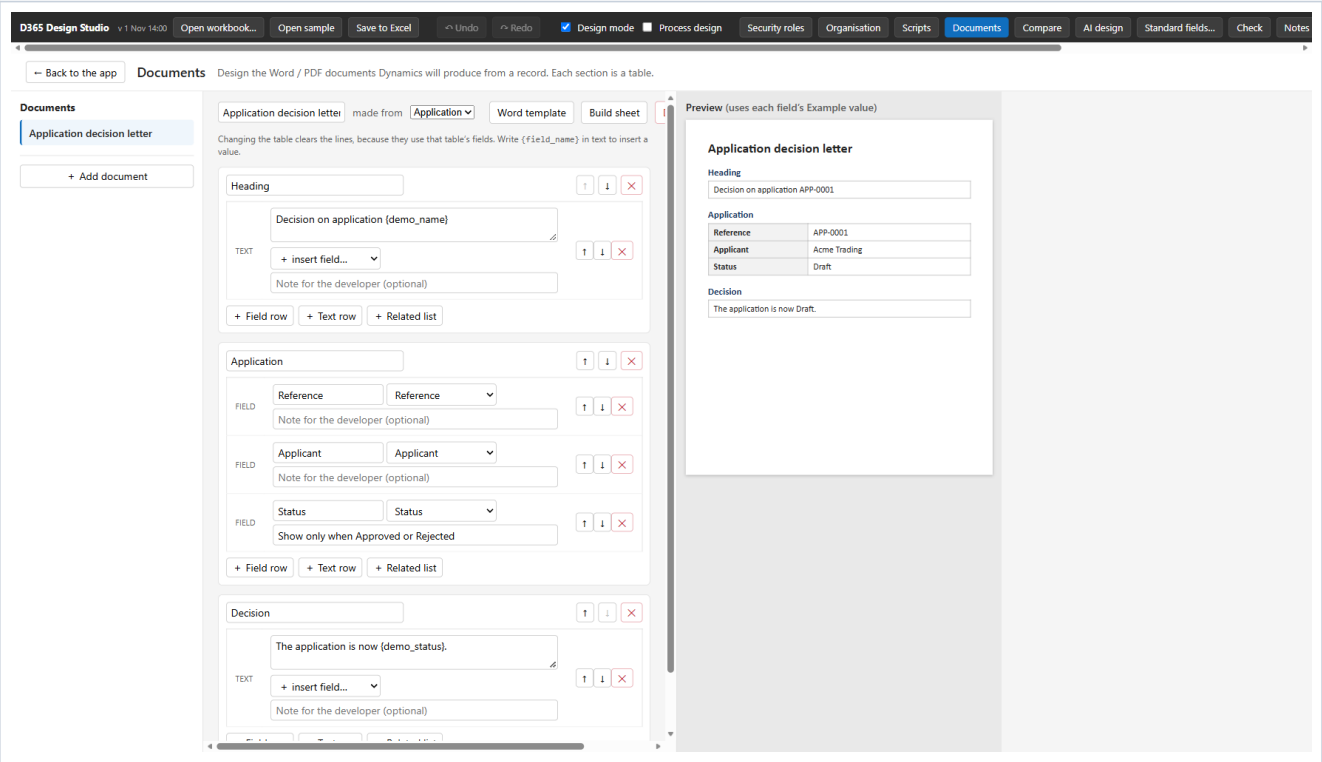


Figure 35. *A document design with live preview on the right (uses each field's Example value).*

Line type	Use
Text row	Free text. Write {field_name} to insert a value, or use the insert field… list.
Field row	A label plus a field from the chosen table.
Related list	A small table of child records, with chosen columns.

- made from selects the table. Changing it clears the lines, because they use that table's fields.
- Note for the developer is saved with each line (for example, "Show only when Approved or Rejected").
- Word template exports a .docx with a content control for every field. Build sheet exports a short Word sheet for the developer.

13. Compare versions

Keep a copy of the workbook before each working session, then compare to show the client exactly what changed. Nothing is edited.

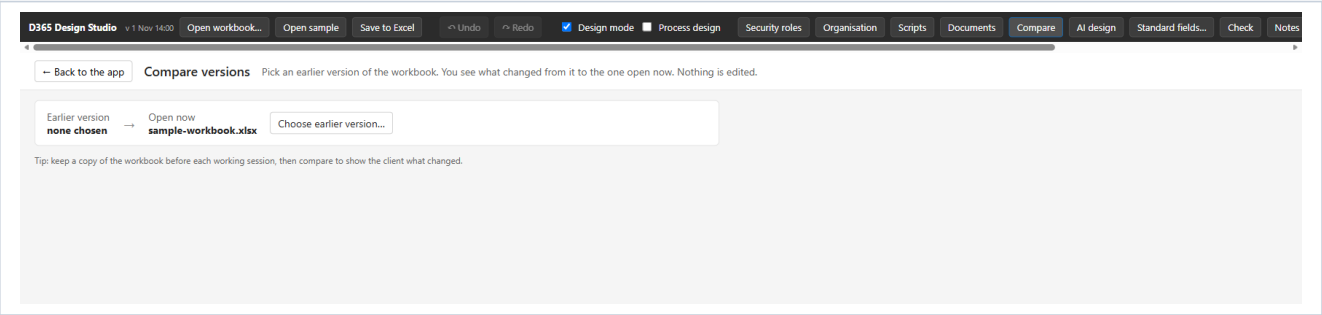


Figure 36. *Compare starts with the open workbook and asks for an earlier version.*

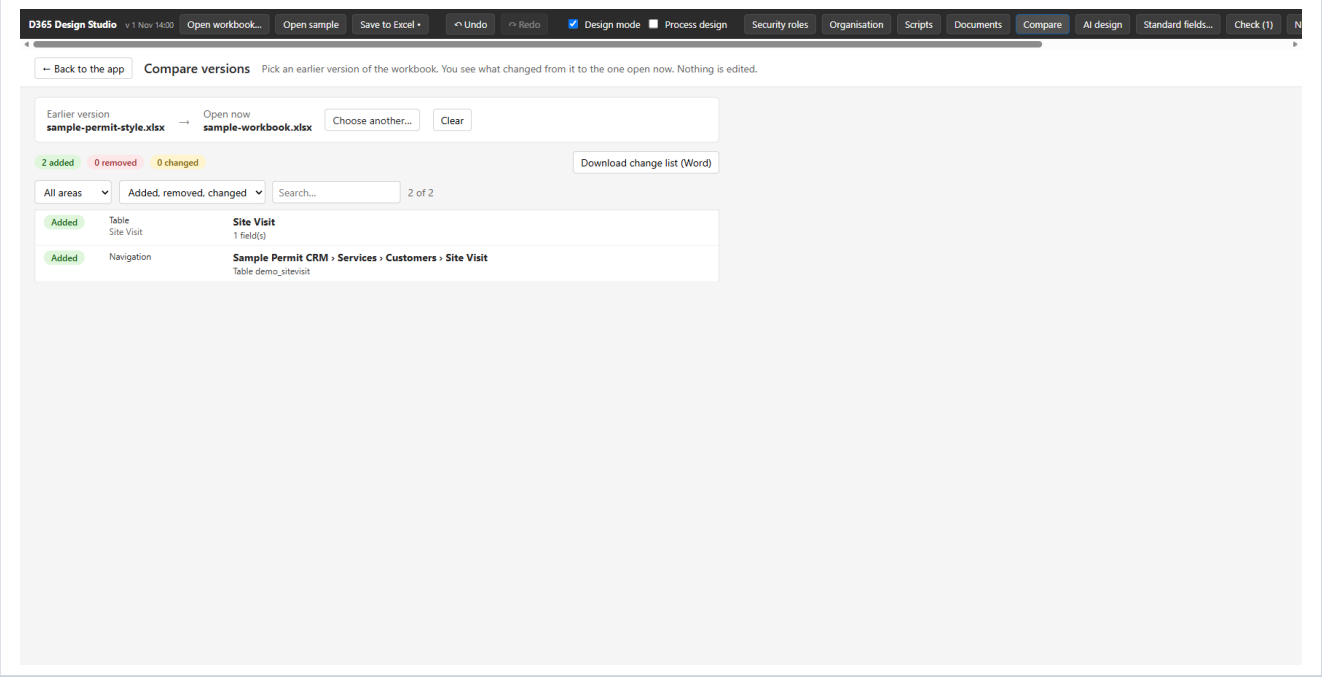


Figure 37. *Result after changes: 4 added, 0 removed, 1 changed. Filter by area or search.*

- Click Choose earlier version… and pick the older .xlsx.
- The summary counts added, removed and changed items across tables, fields, views, forms, flows, roles, organisation and buttons.
- Filter by area and by Added, Removed or Changed; search by name.
- Download change list (Word) produces a client-ready .docx of all changes.
- Details such as old to new filters, column lists and widths are shown on the changed rows.

14. AI design: from a process description to a design

The BA pastes a workshop note or requirement, and the AI proposes a data model and then a process flow. The AI only adds; it never changes or removes anything already in the design. The BA ticks what to keep and applies it as one undoable step.

14.1 Two ways to use it

Mode	How it works
Copy and paste (default, no key)	The Studio builds a prompt. The BA copies it into Claude, Copilot or any chat they already have, then pastes the reply back. The reply goes through the same checks as the direct mode.
Direct (needs an API key)	The Studio calls the Anthropic API from the browser. The key is kept only in the browser.

14.2 Step by step

1. Open AI design and write or paste the process description. It is kept in the browser, so the BA can come back to it.

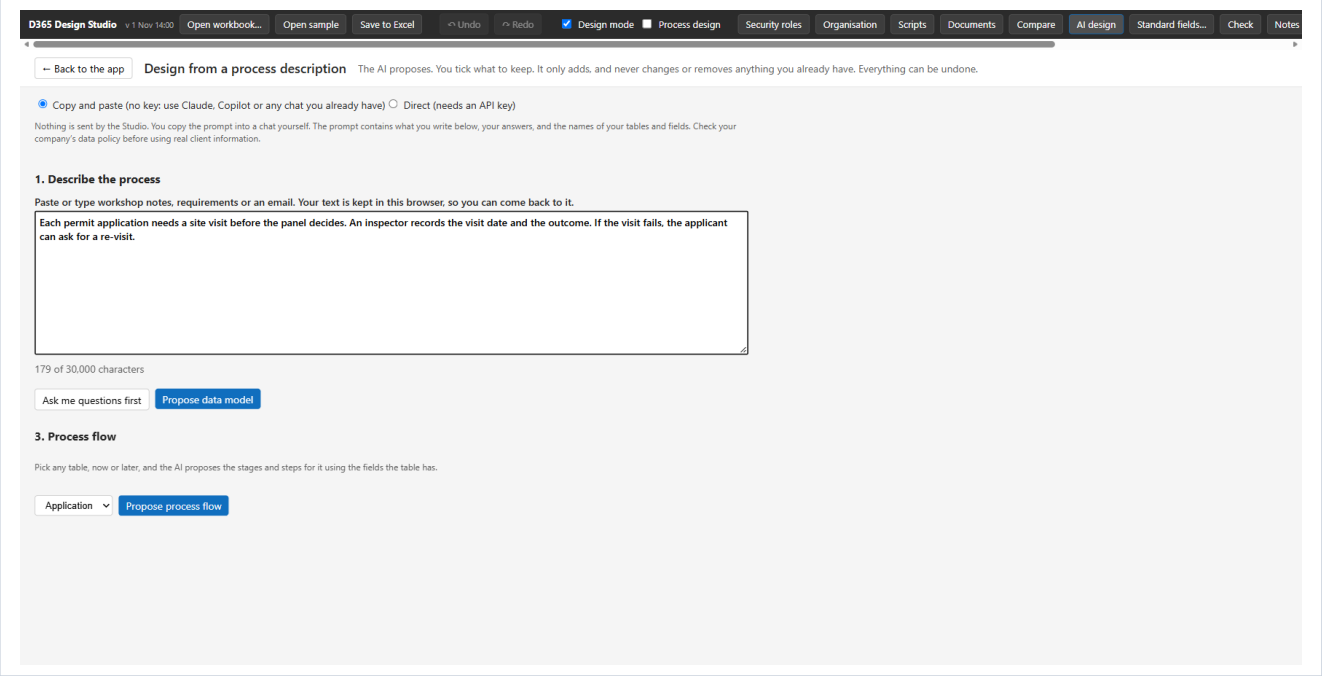


Figure 38. *Step 1: describe the process. The character counter allows up to 30,000.*

2. Optional: Ask me questions first. The AI asks up to six short questions about what the description leaves open. Answers are included in the next prompt.
3. Click Propose data model. In copy-and-paste mode a panel gives the prompt and a box for the reply.

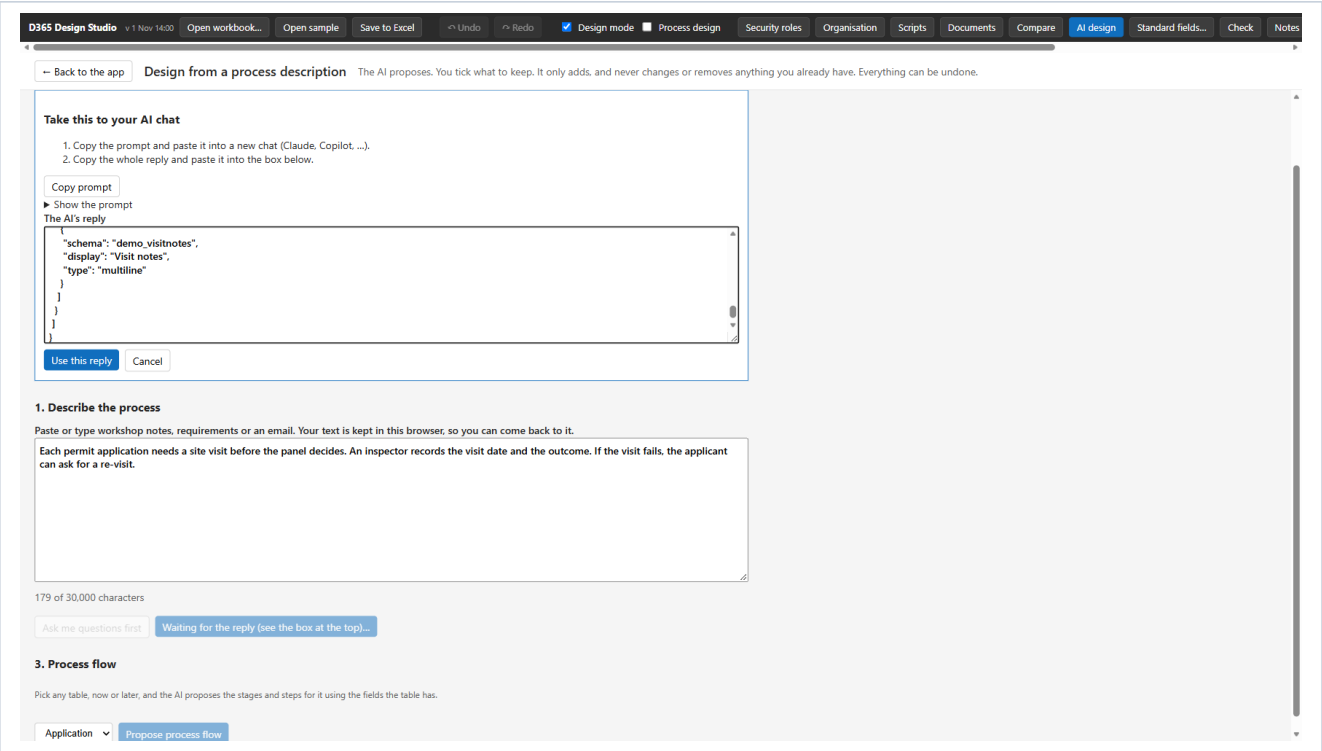


Figure 39. *Step 3: Copy prompt, run it in your AI chat, paste the whole reply and press Use this reply.*

4. Review the proposal. New tables and the new fields of existing tables are listed with the reason for each. Untick anything you disagree with.

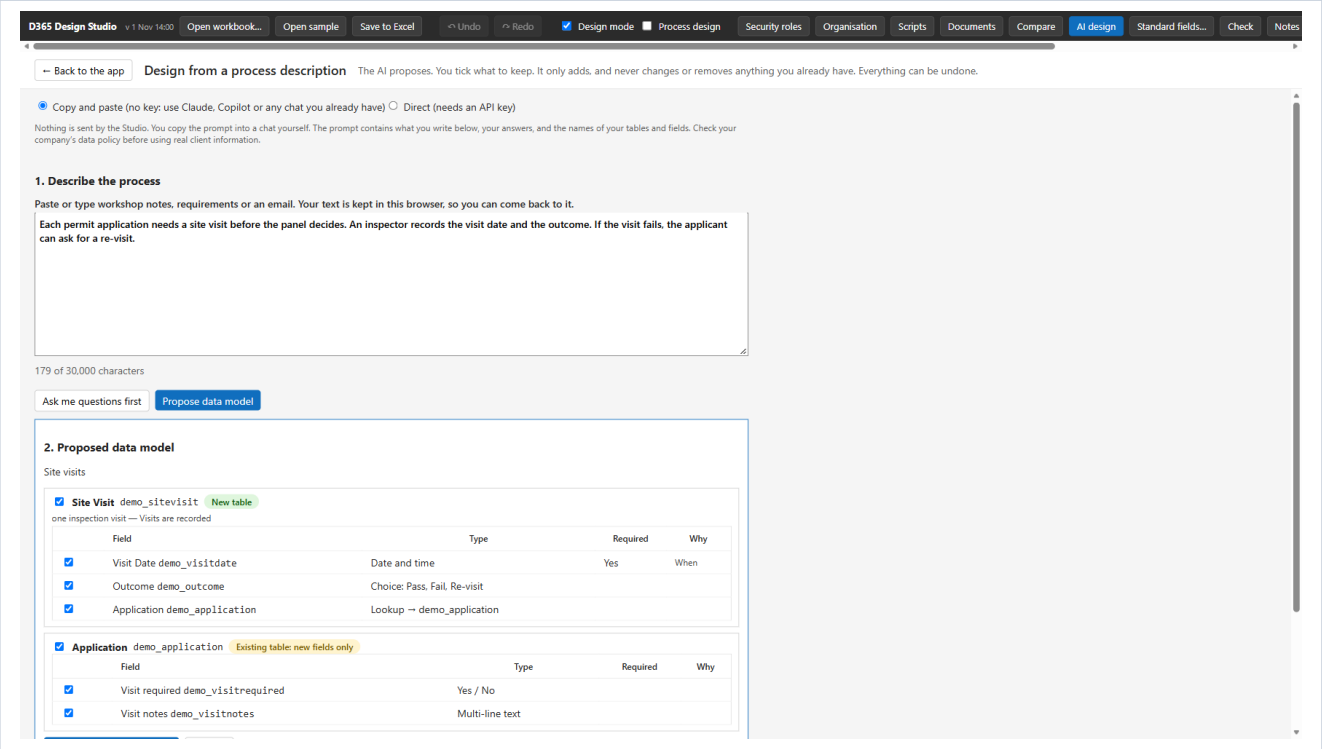


Figure 40. *Review: a new table (Site Visit) and new fields on an existing table (Application, "new fields only").*

5. Click Add selected to the design. The tables are created, added to the Overview and the navigation, and the result is checked automatically.

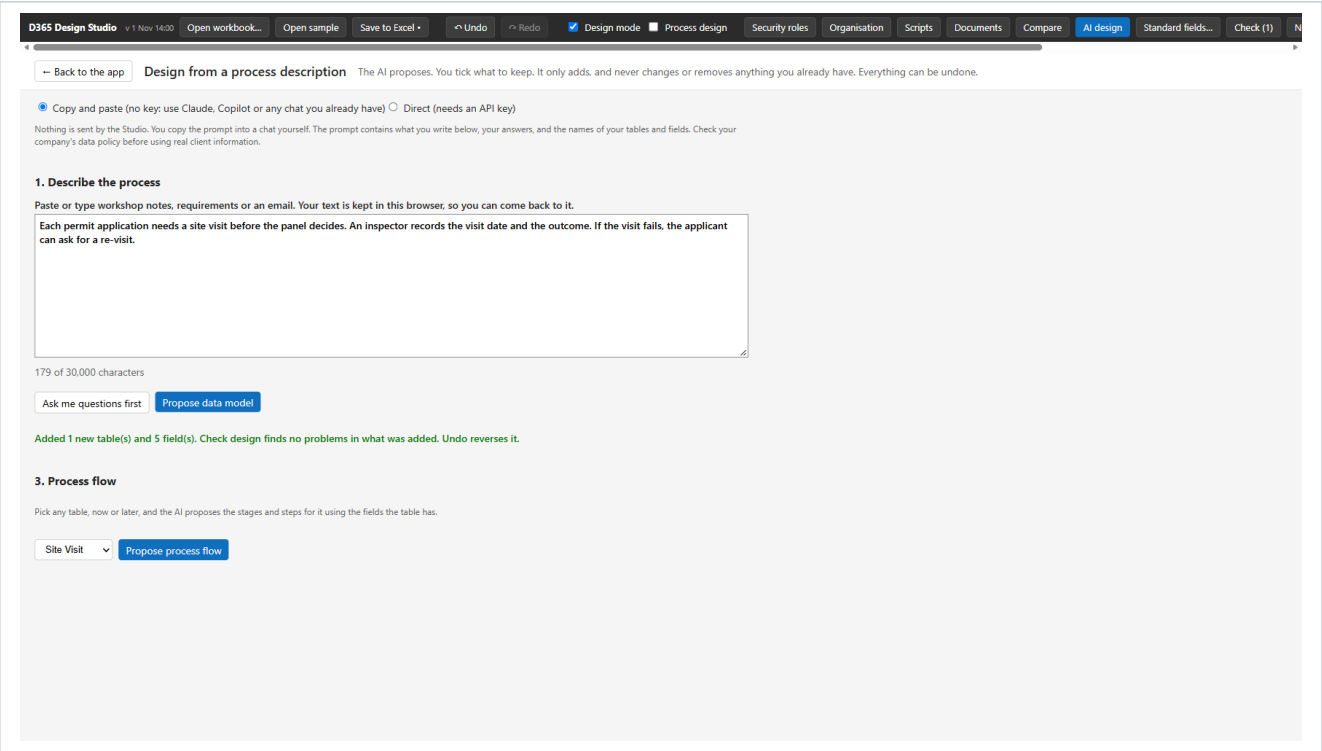


Figure 41. *Applied: "Added 1 new table(s) and 5 field(s). Check design finds no problems." Undo reverses it.*

6. Process flow: pick any table and click Propose process flow. Review the stages and steps and add the flow.

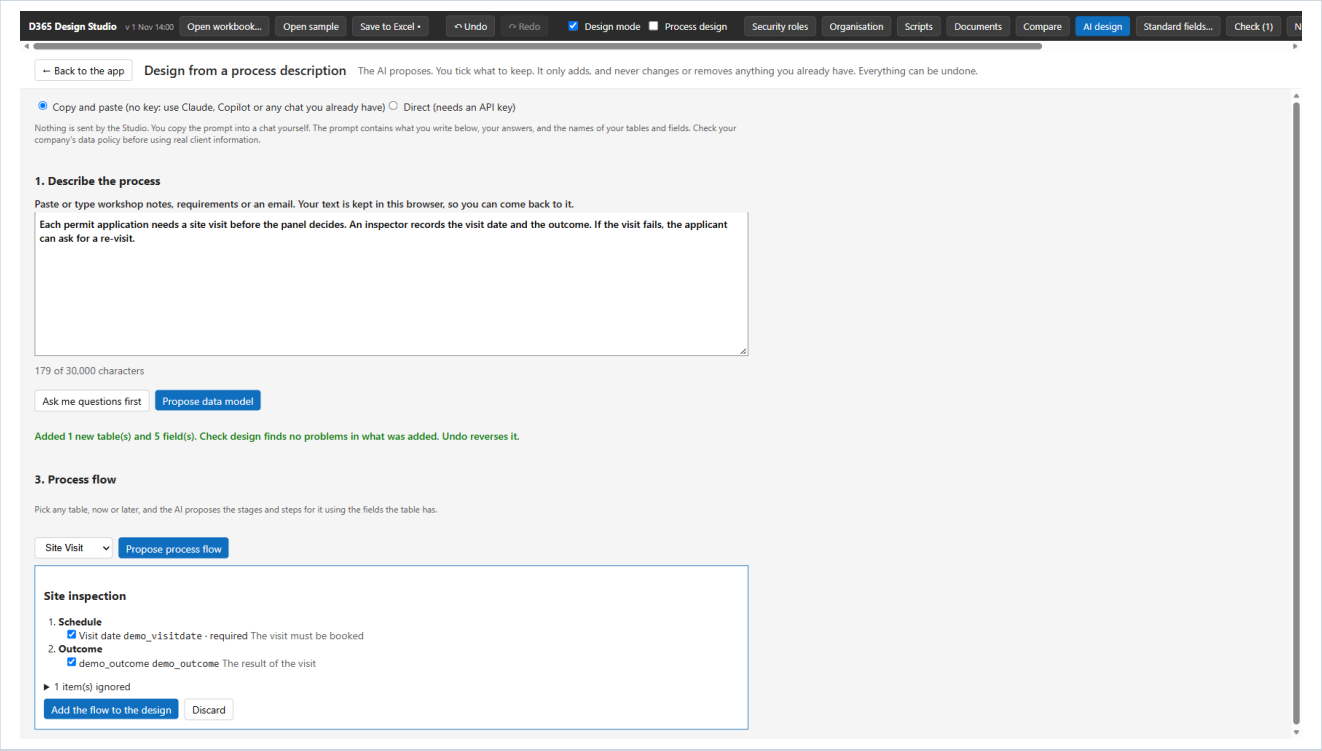


Figure 42. *Flow review: stages, steps, required flags and reasons. Items that name a field not on the table are listed as ignored.*

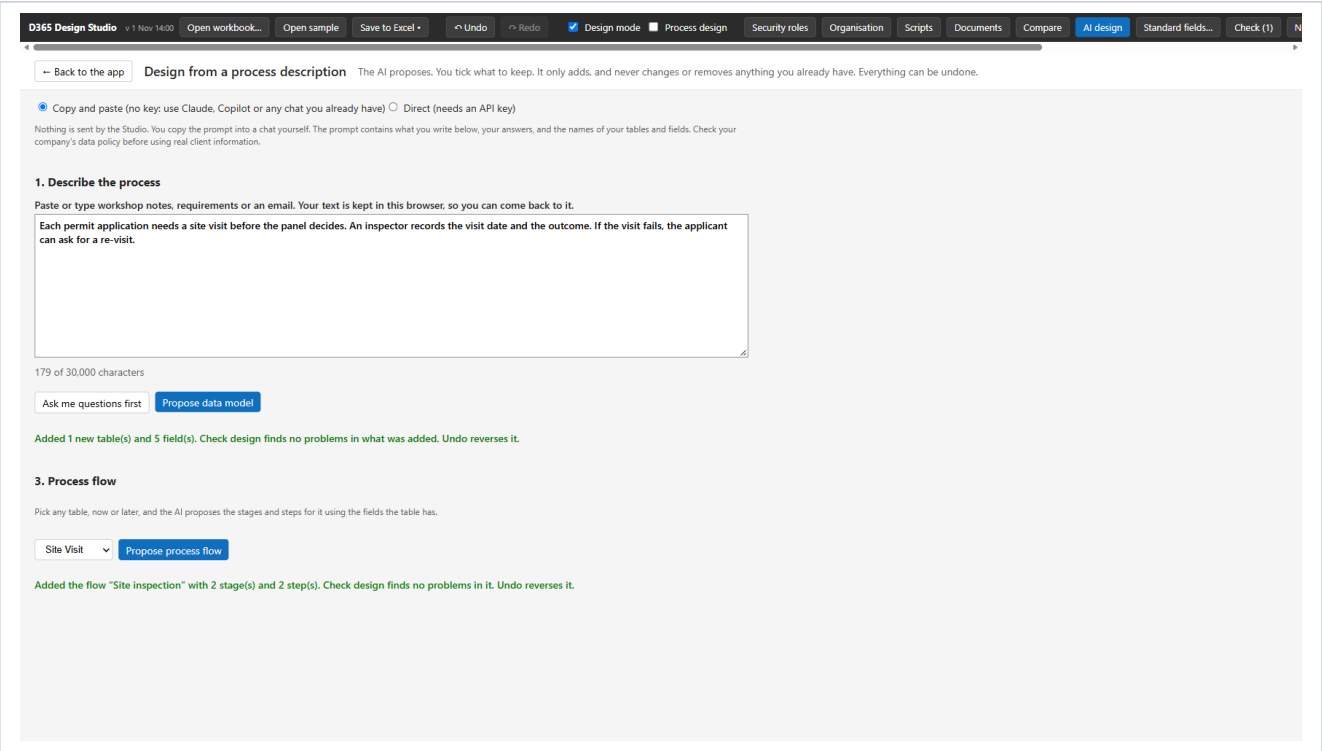


Figure 43. *The flow is added to the design as one undoable step.*

14.3 Safeguards

- The reply must be valid JSON that matches the Studio's format. Unknown items are dropped and listed under "item(s) ignored".
- Schema names are lower case and carry the publisher prefix. Lookup targets must exist, be in the proposal, or be a standard table.
- Limits per proposal: 12 new tables, 40 fields per table, 8 stages, 8 steps per stage.
- Only your description, answers and the table (and optionally field) names are sent; nothing is uploaded unless you copy or send it yourself.

> **Data policy** Check your company's policy before pasting real client information into any AI chat.

15. Saving, undo and the workbook

- **Save to Excel** downloads the workbook. It writes only the cells that changed and leaves every other sheet and style as it was, so the workbook remains safe to use with the engine. When a process row is deleted, the rows under it move up, so no blank row is left, and cells typed by hand beside a row on any process sheet go with that row, even in a column with no header. A cell's colour stays where it was.
- **New tables** created in the Studio get a worksheet (header block, field header row, a primary name field), an Overview row and a sitemap entry when saved.
- **Choice fields** add rows to an Option Sets sheet, which is created if missing.
- **Undo and Redo** work across the whole session, including AI changes.

15.1 Workbook sheets the Studio reads and writes

Sheet	Content
Overview	One row per table: display name, schema name, block, purpose.
One sheet per table	Header block (title, Type, Block, Grain, Schema name, Ownership), then one row per field: names, type, length, requirement, form placement (tab, section, order, form), description, example, phase, populated by, default, validation, legacy field, migration notes.
Option Sets	Choice values for choice fields.
Sitemap	Navigation groups and pages.
Views	Named views, columns, widths, sort, filter, default flag.
Process Flows	Flows, stages and steps.
Command Bar	Form buttons.
Security Roles / Role Field Rules / Role Navigation	Table privileges, field restrictions and navigation limits per role.
Business Units / Teams	Organisation tree, teams, team type and roles.
Web Resources / Form Scripts	Scripts and the events that run them.
Documents	Document templates and their lines.
Notes	Workshop questions, decisions and actions.
Process Design … Process Step Documents	Fifteen optional sheets for the process designer (section 16.12). Created only when used.
Form Layout	Optional. Section columns, hidden tabs, sections and fields, label visibility and full-width fields.

The full layout is defined in docs/EXCEL_FORMAT.md in the Studio repository. That document is the contract for both the Studio and the engine.

16. Process design (optional)

Section 5 describes the business process flow, the simple stage bar on a form. This section describes a much fuller option: a process designer for requests that move between people, with who does each step, how long it may take, what happens when someone rejects, and what the user sees on the record. It is optional and off by default. A designer who does not need it never sees it.

> **Why it exists** It is the design tool, and the written specification, for a request process engine that is built separately. The Studio records what the process must do in the workbook. The process engine reads these sheets, apart from the newest settings listed in section 16.14.

16.1 Turn it on and find the process

1. Tick Process design in the top bar. It is remembered in this browser. The tick only controls the designer tools: what a record shows does not depend on it.
2. Open a record of the table the process runs on. If its process has no step fields, a process bar appears under the title, with one chevron per stage and the number of steps in each. If it has step fields, the record opens on the process screen (section 16.17): click Show full process there, or click Ordinary form to see the bar.
3. In design mode, a table with no process shows + Add a process. Click it, then Show full process to open the designer.

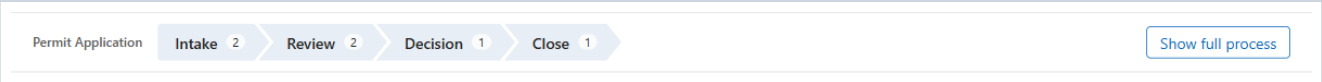


Figure 44. *The process bar under the record title. Each chevron is a stage; Show full process opens the designer. When a table has a process, the older process-flow bar from section 5 is hidden, so there is only one bar.*

16.2 The process canvas

Show full process opens the whole process as a diagram. Stages are columns, steps are cards inside them, and the lines show where each action leads. The designer reads left to right.

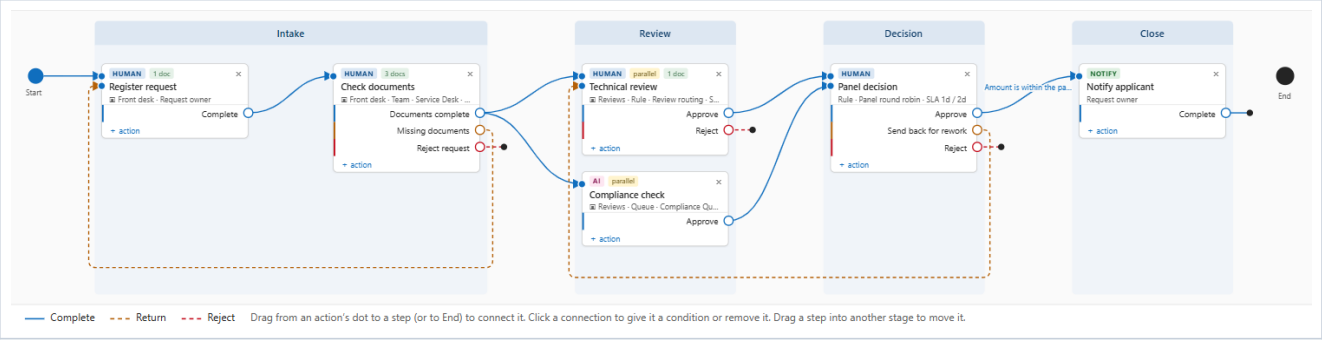


Figure 45. *The process canvas. Two reviews sit one above the other because they open together; the dashed lines are Return and Reject routes.*

On the canvas	What it means
Step card	One piece of work. The coloured tag shows the type (Human, System, Notify, AI). Below the name: who it goes to and the time allowed.
Dot on the right of an action	One dot per action (button) on the step. Drag from the dot to another step, or to End, to say where the action leads.
Dot on the left of a step	One for each way into the step.

On the canvas	What it means
Blue line	A Complete route. Orange dashed is a Return (goes back). Red dashed is a Reject.
Steps stacked in one column	They open at the same time (parallel). This happens when one action has several routes with no condition.
Route with a label	The route has a condition in plain words. Two or more conditional routes from one action are a choice: the process takes one of them.
+ action	Adds another button to the step. + Step and + Stage in the header add steps and stages.

- Click a step, a stage, a connection or the empty canvas to see its settings in the right-hand panel.
- Drag a step into another stage band to move it. The layout is automatic, so there is no need to place anything by hand.
- Click a connection to give it a condition or remove it. The × in a step card's corner deletes the step, after asking (see Deleting a step, under section 16.3).
- Problems appear in a strip at the bottom (for example a step that leads nowhere), each one clickable.

16.3 The step panel

Click a step to describe it. Everything about the step is on one panel, from top to bottom: name and stage, type, who it is assigned to, weight, time allowed, documents, notifications and actions.

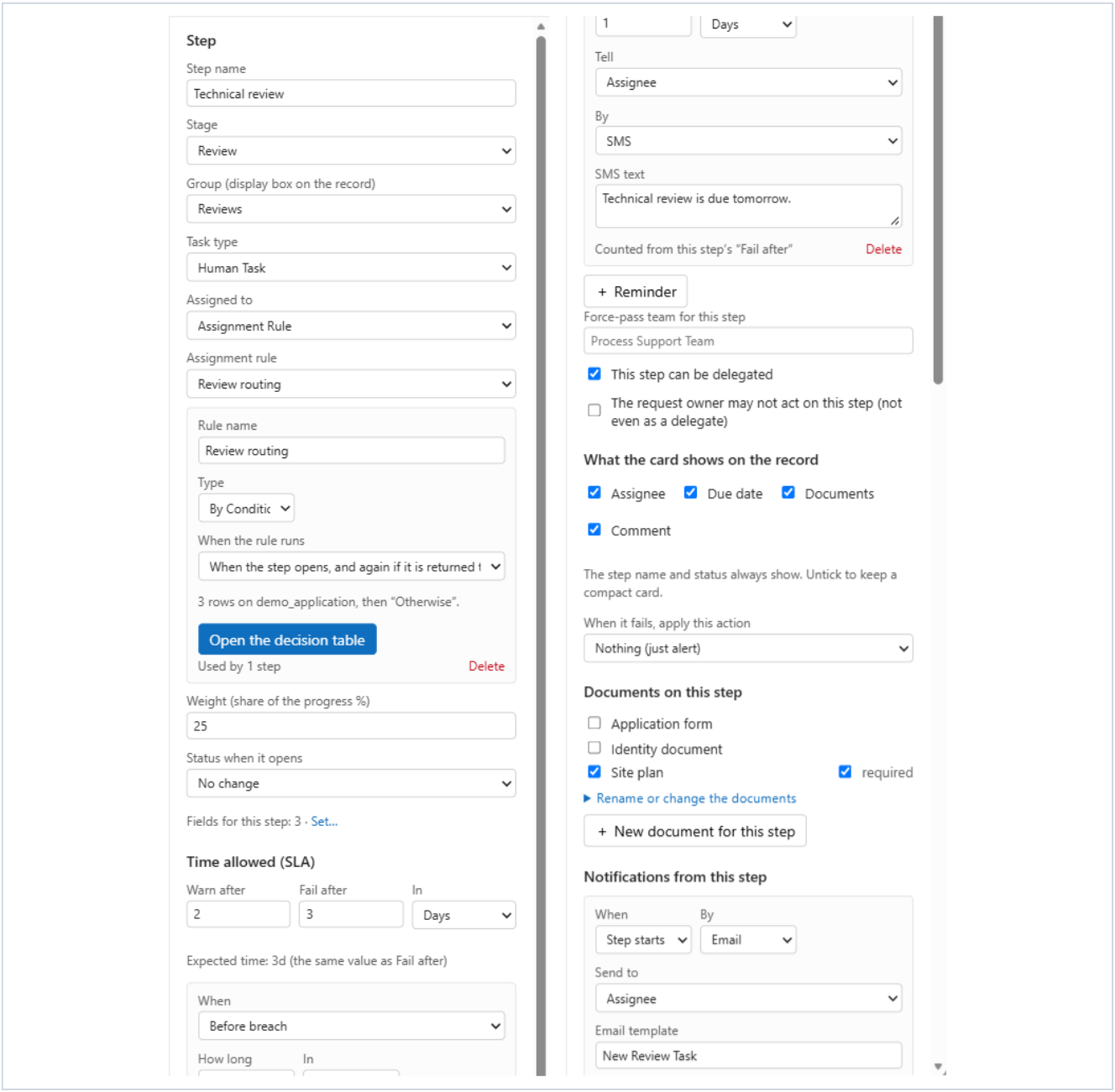


Figure 46. *A step's panel. Left: type, assignment (here by a rule), weight and time allowed, with a reminder. Right: lower down, documents, notifications and actions.*

Setting	What it does
Task type	Human Task (a person acts), System Task (runs by itself, see "System tasks, popups and Super user" in section 16.17), Notification (sends a message and moves on), AI Agent Task (an agent screens, section 16.6).
Assigned to	User, Team, Queue, Request Owner, Manager of Request Owner, or Assignment Rule (a rule decides, section 16.4).
Weight	This step's share of the request's progress %. Weights are added up across the process.
Time allowed	Warn after and Fail after, in hours or days. A reminder can be tied to each (section 16.5). When a step fails, one of its actions can be applied automatically.
Force-pass team	Optional team for this step only; the process setting is used when blank (section 16.7).
Delegation flags	Can this step be delegated, and may the request owner act on it (section 16.10).
Documents	Tick the documents the step needs and whether each is required. A new document can be created right here.

Setting	What it does
Notifications	Email or SMS when the step starts, completes or an action is taken, to the assignee, the request owner, the owner's manager, a team or a user. Email uses the name of an ordinary Dynamics email template; SMS carries its own text.
Actions	The buttons a person sees: Complete, Return or Reject, whether a comment is required, and the status set on the request. Where each action leads is set by its routes.

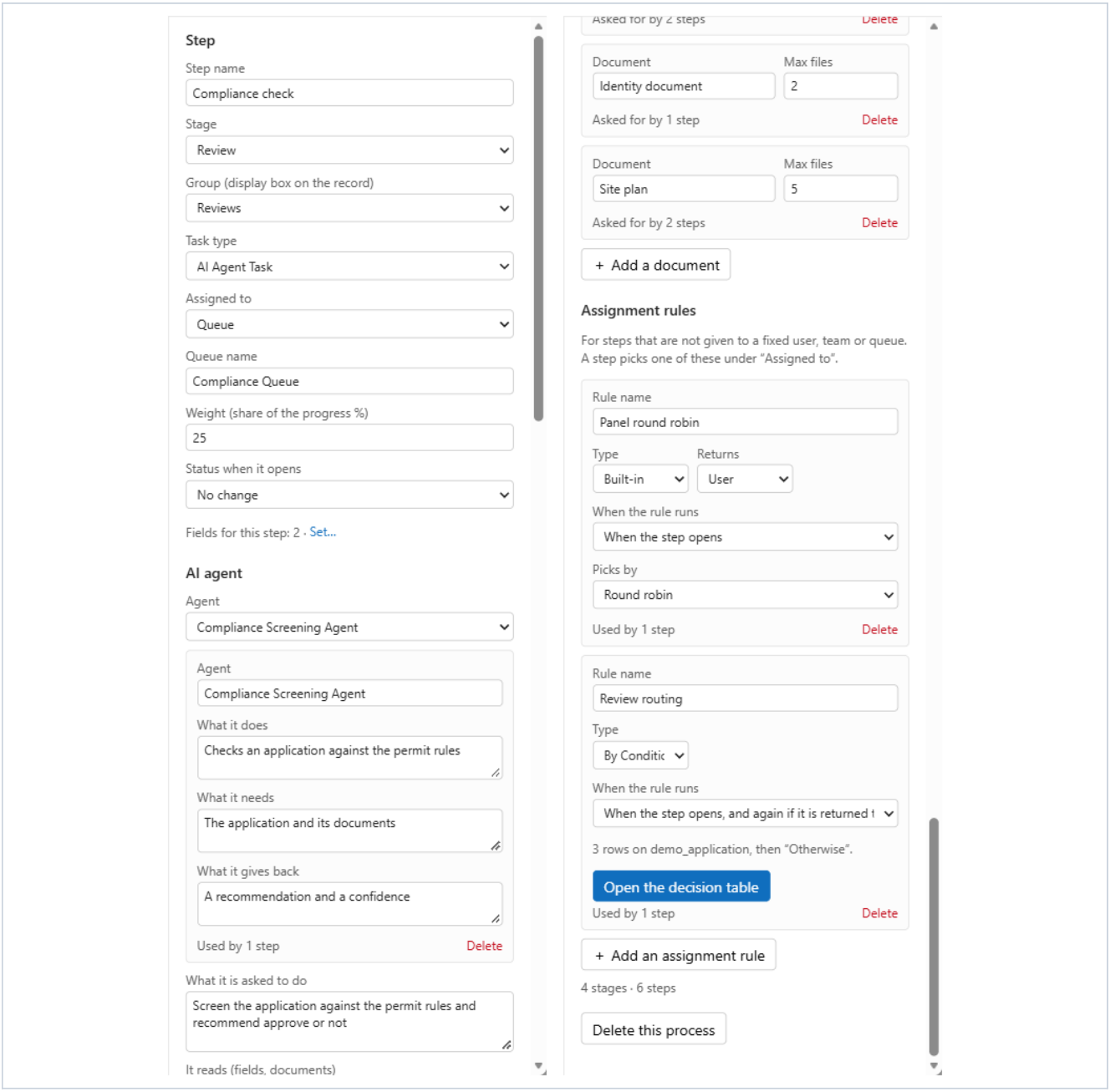


Figure 47. *Left: an AI Agent step (the agent, what it is asked to do, what it reads and fills in). Right: the assignment rules list in the process settings.*

Deleting a step

In Design mode, with Process design on, each step card on the diagram has a × in its corner and the step panel has a Delete step button. Either one asks first, and lists in plain words what goes with the step: its actions, step fields, notifications, SLA reminders, pause rules, document requests and agent step set-up, and every connection that starts at it or points to it, with counts and names.

A connection pointing to the step is removed, not moved to another step, so the action it left is connected to nothing until you connect it again. An assignment rule that only this step used goes with it; one that other steps use stays, and the confirmation says so. A document always stays, as only its link to the step goes. If no step asks for it any more, the confirmation says so and Check design lists it as a warning, so you can ask for it on another step or delete it. The shared list of AI agents never changes.

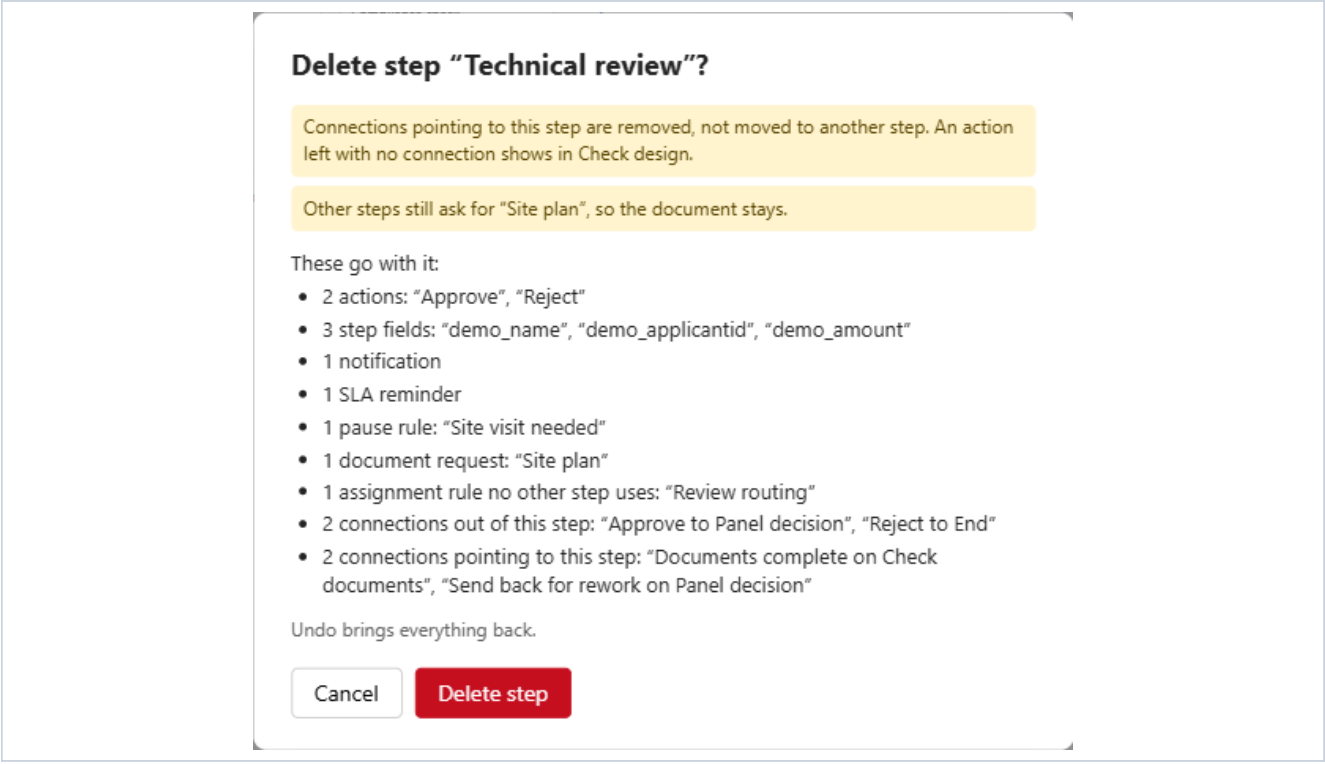


Figure 48. *The confirmation names what goes, and what stays because other steps use it.*

The confirmation also says when the step is the first or the only one of the process; the decision stays with you. One Undo brings everything back, and Redo removes it again. On saving, the rows leave every sheet they were on, so reopening the workbook shows the step gone.

Check design then says what is missing, and each line opens the process, on the step when it has one. An action connected to nothing is a warning. A Return with nowhere to go is an error, as it cannot run. A deleted first step is explained until another step is first or the workbook is reopened: until then the process starts at the next step. A deleted only step leaves a process with no steps. A stage or group left empty stays, so it can be used again or deleted later, and is noted.

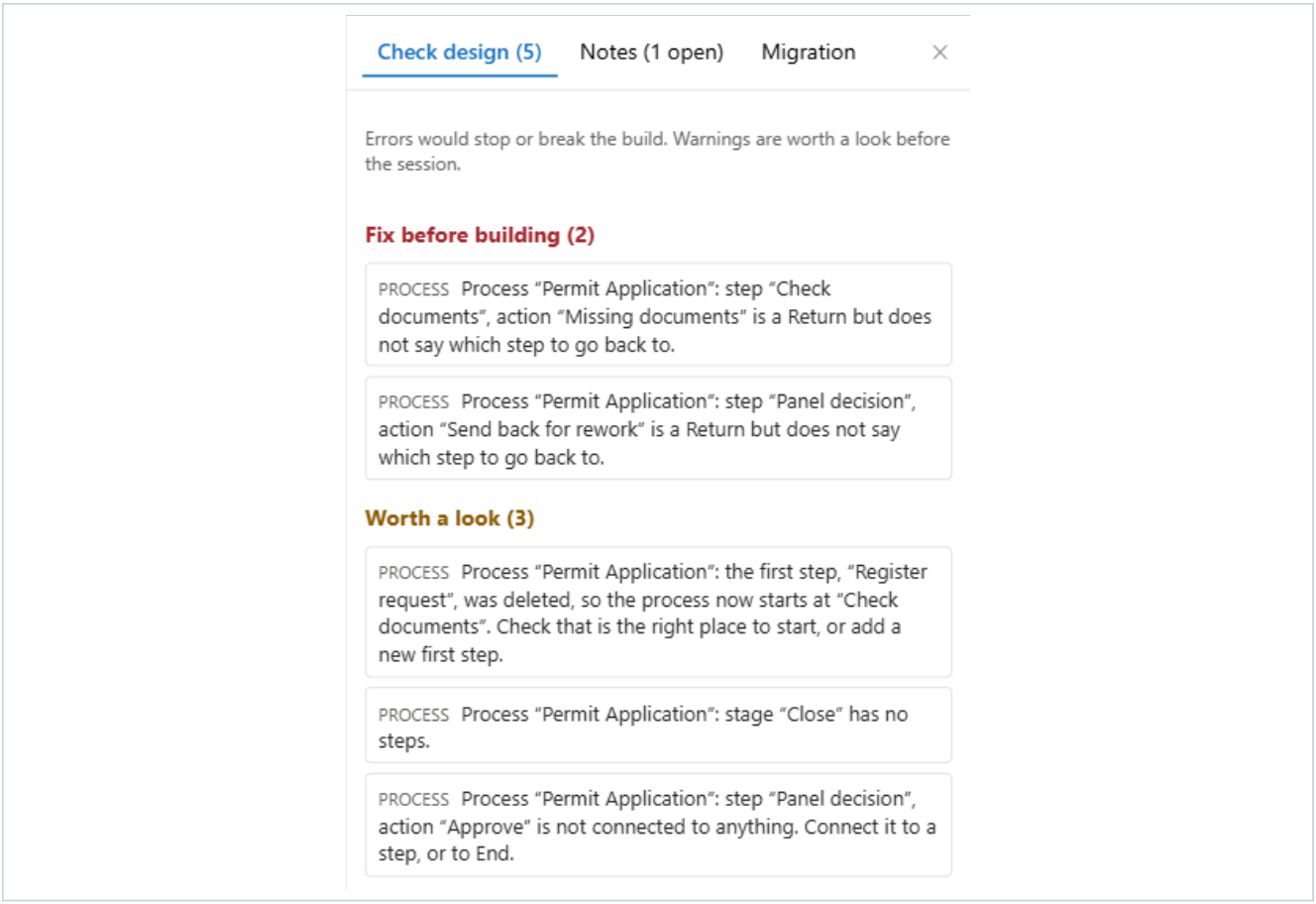


Figure 49. *Check design after deleting the first, a middle and the last step of the sample process.*

Deleting a stage, an action, a document, a rule or an agent

Delete this stage, in the stage panel, works the same way as Delete step. A stage with no steps goes straight away. For a stage that has steps it asks first, and lists the steps, the stage's groups and everything that goes with those steps, by the same rules as deleting a step. It says when the first step of the process is in that stage, or when these are all the steps there are. One Undo brings it all back. With Process design off, a stage that has steps cannot be deleted.

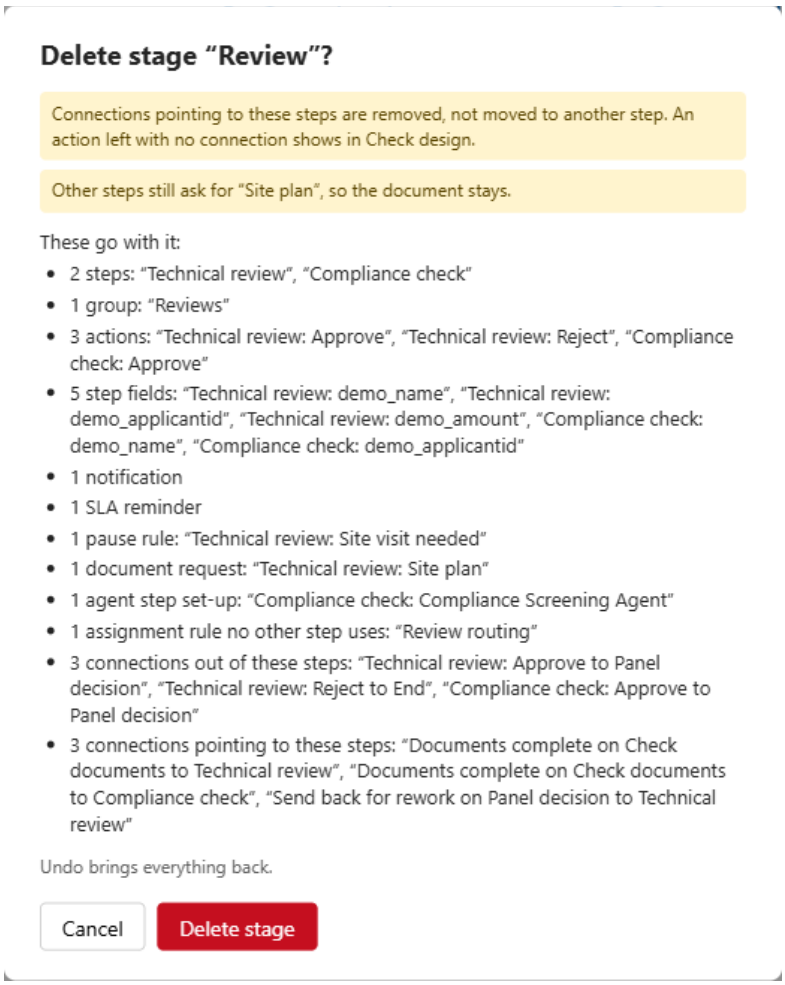


Figure 50. *Deleting a stage that has steps: the steps, the group and everything that goes with them.*

Deleting an action, a document, an assignment rule or an AI agent also asks first, with a short list: an action’s connections and notifications, the steps that stop asking for a document, a rule’s fields and decision table rows. It also names any step left without the rule or the agent. Those steps stay, and Check design reports them. An agent is on one list for the whole workbook, so its list can name steps of other processes.

When a name is already taken

Names must be different: steps, stages, documents and rules within a process, actions on a step, groups within a stage, and processes and agents within the workbook. If you type a name that is taken, a message says so in plain words and the old name stays. A blank name gets the same treatment, and so does calling a step End. Changing only the capitals of a name is allowed.

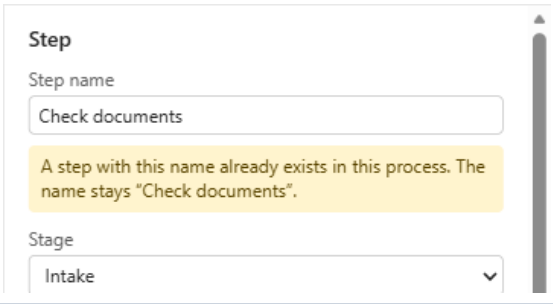


Figure 51. *A refused name: the message says why, and the box goes back to the name that holds.*

16.4 Assignment rules

In real life, who does a step often depends on the case. A rule decides it when the step opens. Rules are defined once for the process and can be created straight from a step with + New rule…. Four types are available.

Rule type	How it decides
By Condition	A decision table you fill in on screen (below). The first row that matches wins.
Built-in	Round robin or Least loaded across a team, queue or list of users.
External API	Calls an outside system. The workbook holds only the name of the connection and a plain-words note of what is sent and what comes back (User, Team or Queue). Addresses and secrets stay in the engine's settings, never in the workbook.
Plugin	Calls your own plug-in logic. Recorded now; built later.

The decision table

For By Condition rules, Open the decision table shows a pop-up in the style of a business rules tool. Pick the table the conditions look at. Add the fields you want to look at with + Add a field…; they become columns. Each row says: when these conditions are all true, assign to this person, team or queue. The last row, Otherwise, always exists and catches everything else.

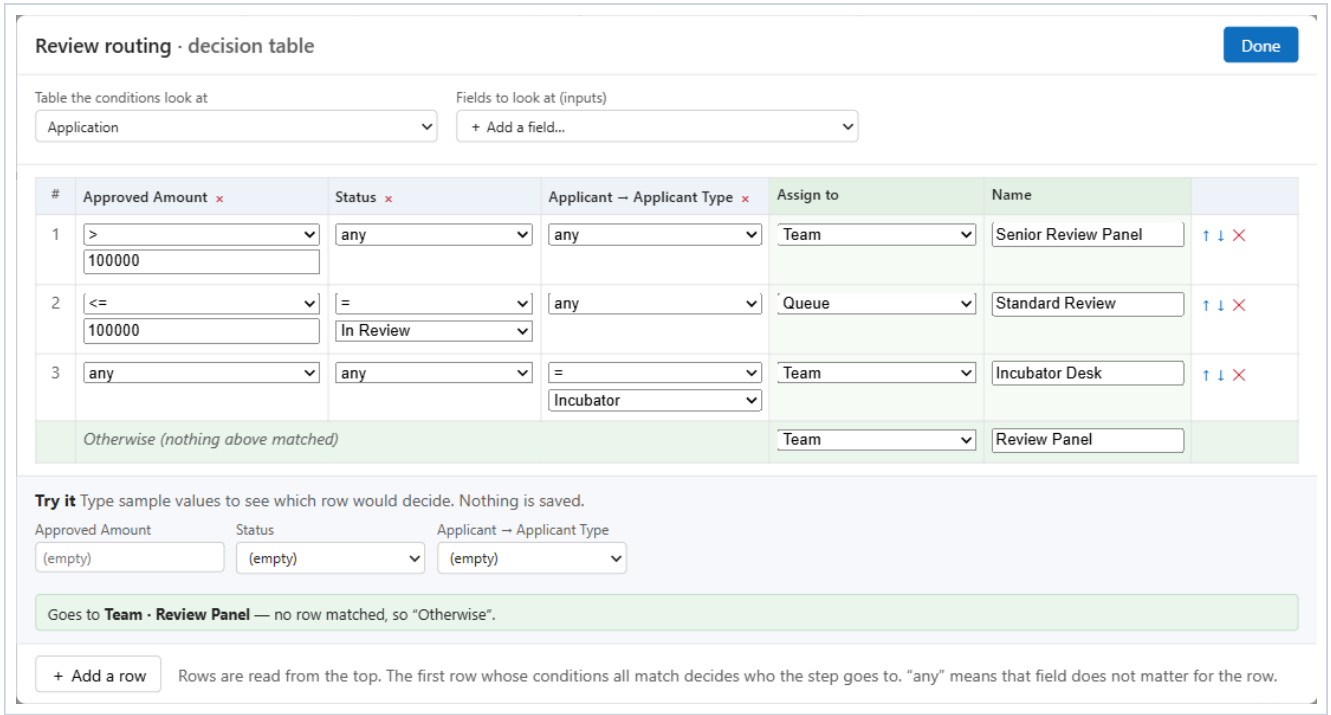


Figure 52. *The decision table for "Review routing". The third column looks through a lookup to the applicant's type. Row 1 is read first.*

- **Fields through a lookup.** The field picker offers the table's own fields and, grouped under each lookup, the fields of the related table (for example Applicant → Applicant Type). Related-table conditions are often the most useful ones.
- **Operators.** Equals, not equals, greater or less than, Contains, Starts with, In, Is empty, Is not empty. Choice fields offer their real option labels.
- **Order matters.** Use the arrows to move a row. Rows are read from the top and the first match decides. "any" means the field does not matter for that row.

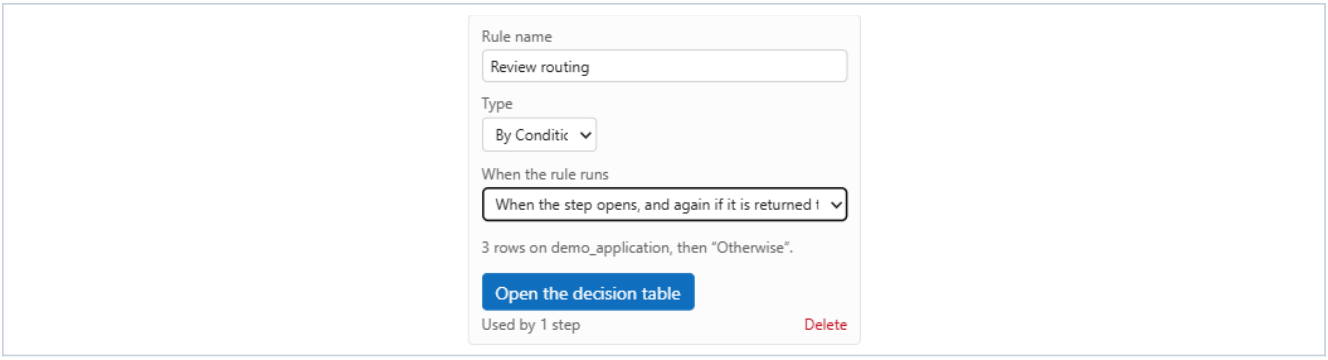


Figure 53. *A rule's settings, including When the rule runs.*

- **When the rule runs.** Once when the step opens (the usual choice); or again if the step is returned to; or again whenever the fields the rule looks at change. Stored in the When it runs column.
- **Try it.** Under the decision table, a Try it panel has one box per field the table looks at. Type values and the matching row lights up with the result, or Otherwise. Use it to check a table with the client without leaving the designer.

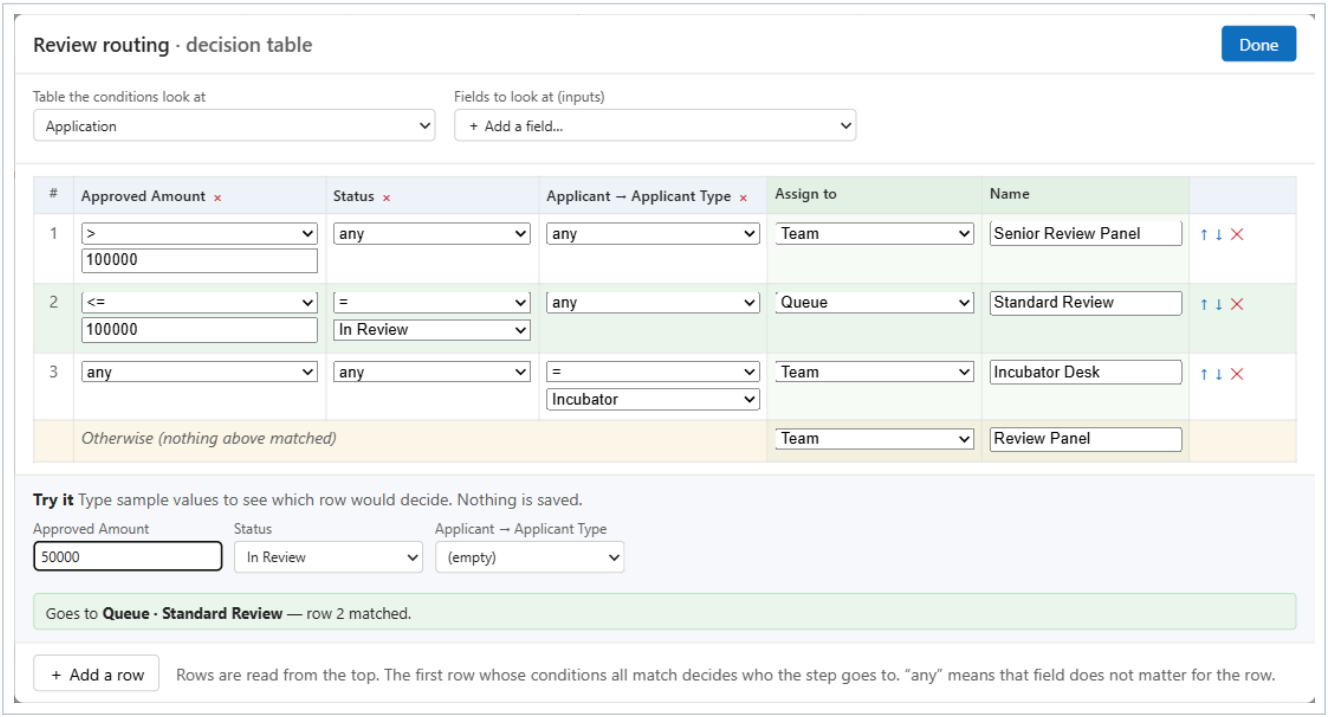


Figure 54. *Try it: values typed on the left; row 2 matches and is highlighted, with who the step would go to.*

16.5 Time: step SLA, whole-process SLA, reminders and pausing

- **Step SLA.** Warn after and Fail after on each step, counted in the process's SLA calendar (the name of a working-hours calendar).
- **Whole-process SLA.** A total time for the request, from submission to the end.
- **Reminders.** As many as needed, on a step or on the whole process: before breach, at breach or after breach (optionally repeating), by email or SMS, to the assignee, owner, manager, a team or a user. For example: 1 day before, at breach, then every day after.
- **Pause reasons.** A list of reasons that may stop the clock (for example Waiting for applicant), for any step or one step. Each has one level of approval, by the manager of the person asking or by a named approval team, and a longest pause after which the clock restarts.

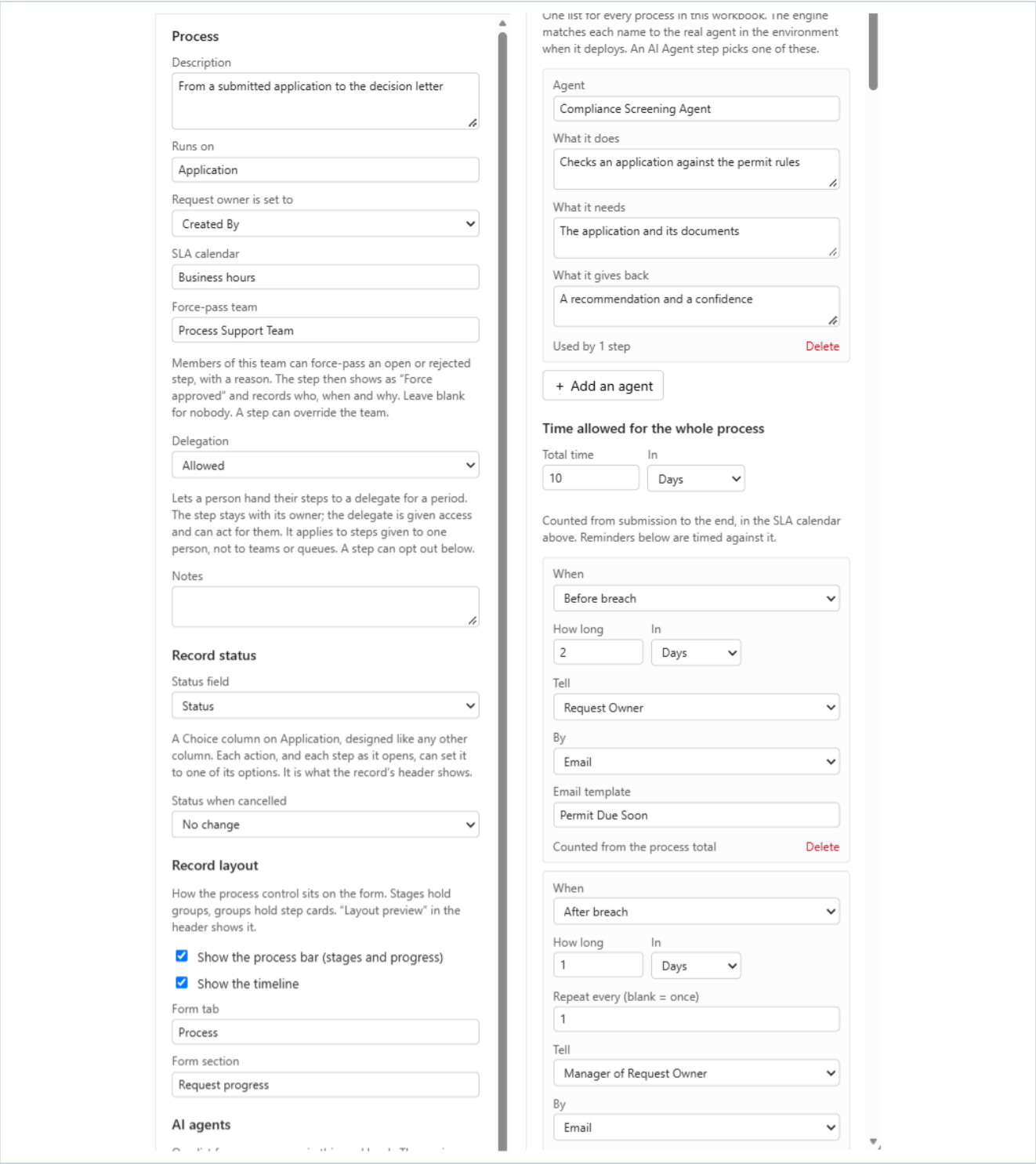


Figure 55. *Process settings. Left: the force-pass team and whole-process time. Right: reminders and pause reasons with their approver.*

16.6 AI Agent steps

An AI Agent Task is a step done by an AI agent. The workbook holds one list of agents, shared by every process; the engine matches each name to the real agent in the environment when it deploys. The step says what the agent is asked to do, what it reads, what it fills in, whether a person confirms its answer (human review), and a confidence limit. The agent's answer is one of the step's actions, and that action's routes decide where the process goes. If the agent fails, or is less sure than the limit, the step goes to the person or queue it is assigned to.

16.7 Rejection, parallel steps and the force-pass team

- **Parallel steps.** When one action leads to several steps with no condition, they open together. A step reached by more than one branch waits until no other open step can still reach it.
- **Reject.** A rejected step does not cancel the steps beside it. They carry on. When they have finished, the process follows the rejection's route: back for rework, or to End (the request then closes as Rejected).
- **Force-pass team.** Set on the process, and optionally overridden on a step. Members of that team can force a step through, an Open or a Rejected one, with a reason that must be filled in. The step is then marked Force approved, and the person, time and reason are recorded. Leave it blank and nobody can.

Status	Meaning
Step: Open / Paused	Waiting for someone; the SLA clock is running or paused.
Step: Completed	The person took a Complete action.
Step: Rejected	The person took a Reject action.
Step: Returned	The person sent it back to an earlier step.
Step: Force approved	The force-pass team passed it, with a reason.
Request: In progress / Completed / Rejected / Cancelled	Cancelled is for a request withdrawn by the owner; a rejection ends the request only when its route goes to End.

16.8 Preview as user

Preview as user runs the process as a pretend, in its own window. The button is on the record and in the designer. It opens the same process screen a record shows by default, so it is a second way to the same thing. Section 16.17 explains the screen. Nothing is saved. Use it with the client to agree what users will see before anything is built.

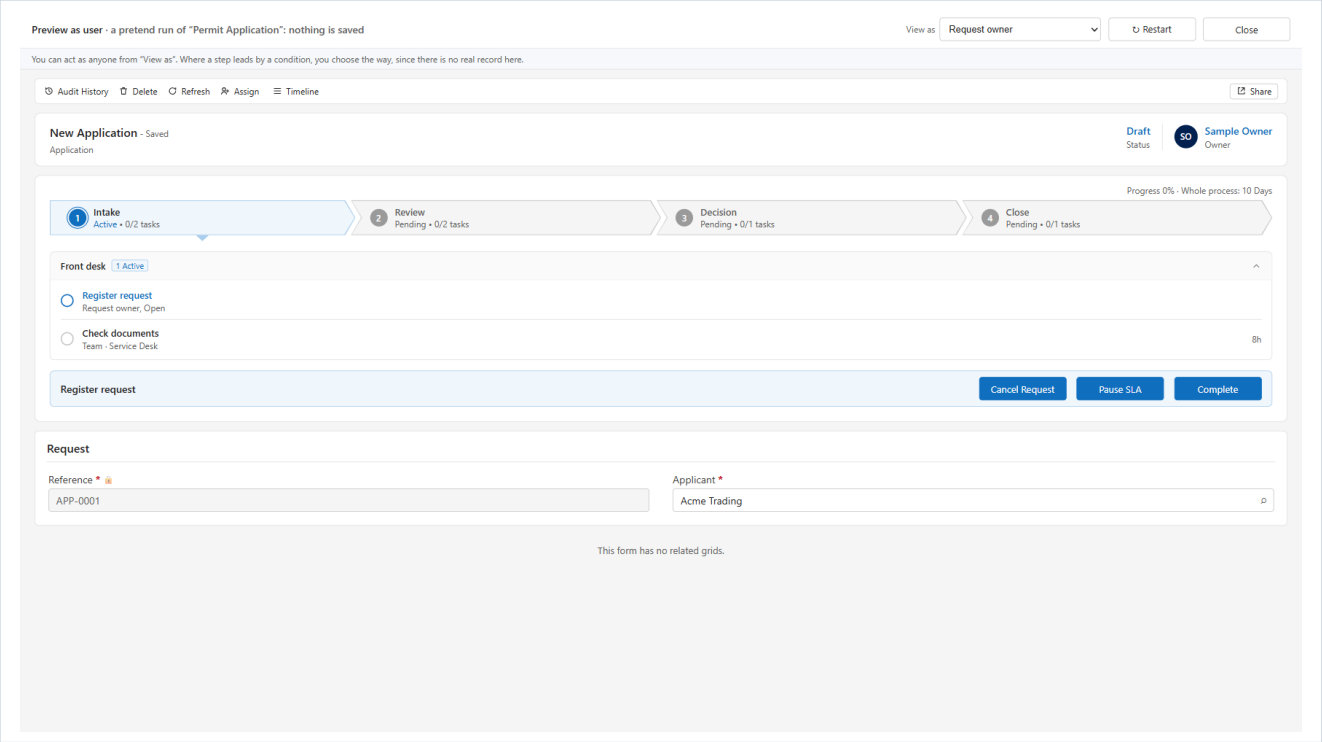


Figure 56. *Preview as the request owner at the start: the stages, the progress %, the step rows, and the bar with the step's name and its blue buttons.*

- **View as.** Choose the request owner, the owner's manager, a team or queue member, someone chosen by a rule, a member of the force-pass team, or someone with read access only. Each sees only what they are allowed to do.
- **Required things block.** A step cannot be completed until its required documents are ticked and its required fields are filled, and actions that need a comment will not run without one. The comment and the documents are asked for in a popup when the button is pressed.
- **Queue steps.** A step given to a queue shows Pick this step first. After a member picks it, others see who has it.
- **Choices.** Where an action leads by conditions, the preview asks Which way?, because there is no real record to decide. In the real process the record's data decides.
- **Delegate.** Choose Delegate, then Acting for. The delegate sees and acts on that person's steps, unless the process or the step does not allow it. Steps given to a team or queue are not available to a delegate, even when the step is marked as one that can be delegated (section 16.10).
- **Rule steps.** A small helper lets you pretend the rule picked a different outcome, to see how each person's view changes.

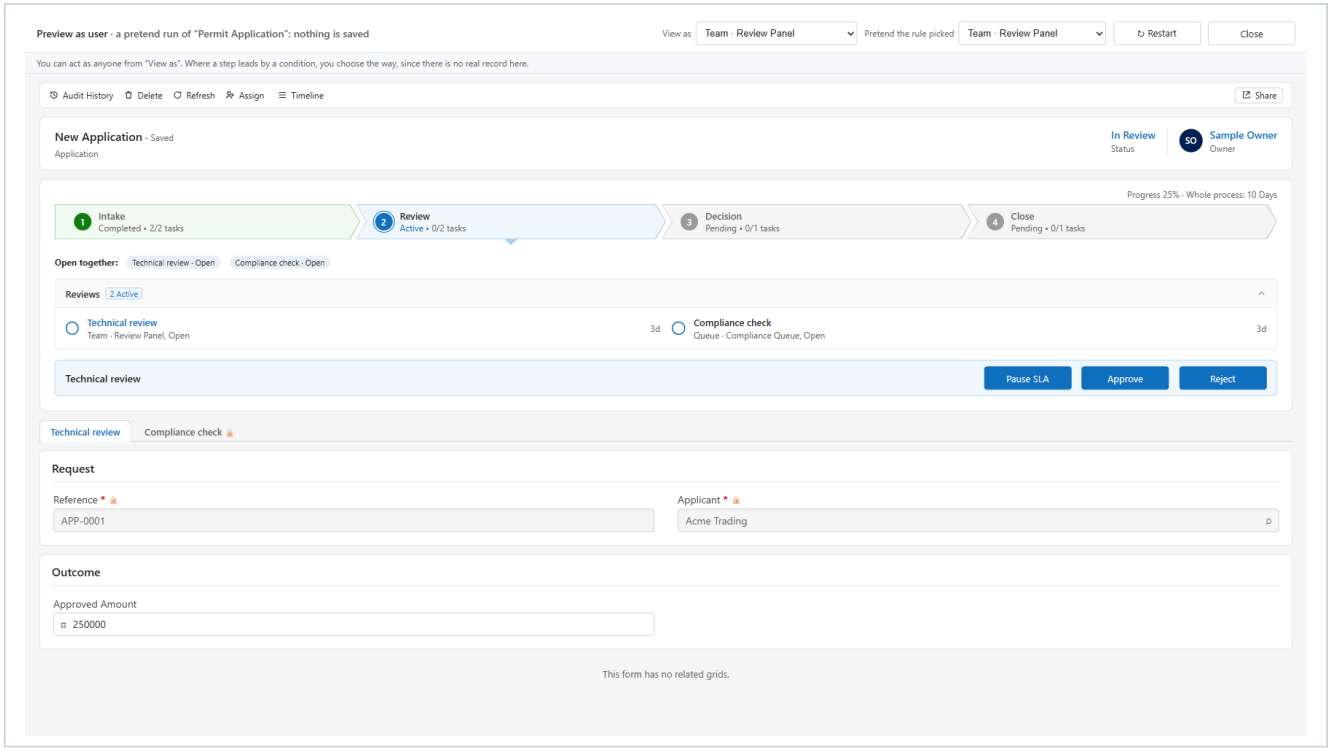


Figure 57. *Both reviews are open together. Viewed as a Review Panel member, only Technical review is theirs; Compliance check shows as someone else's.*

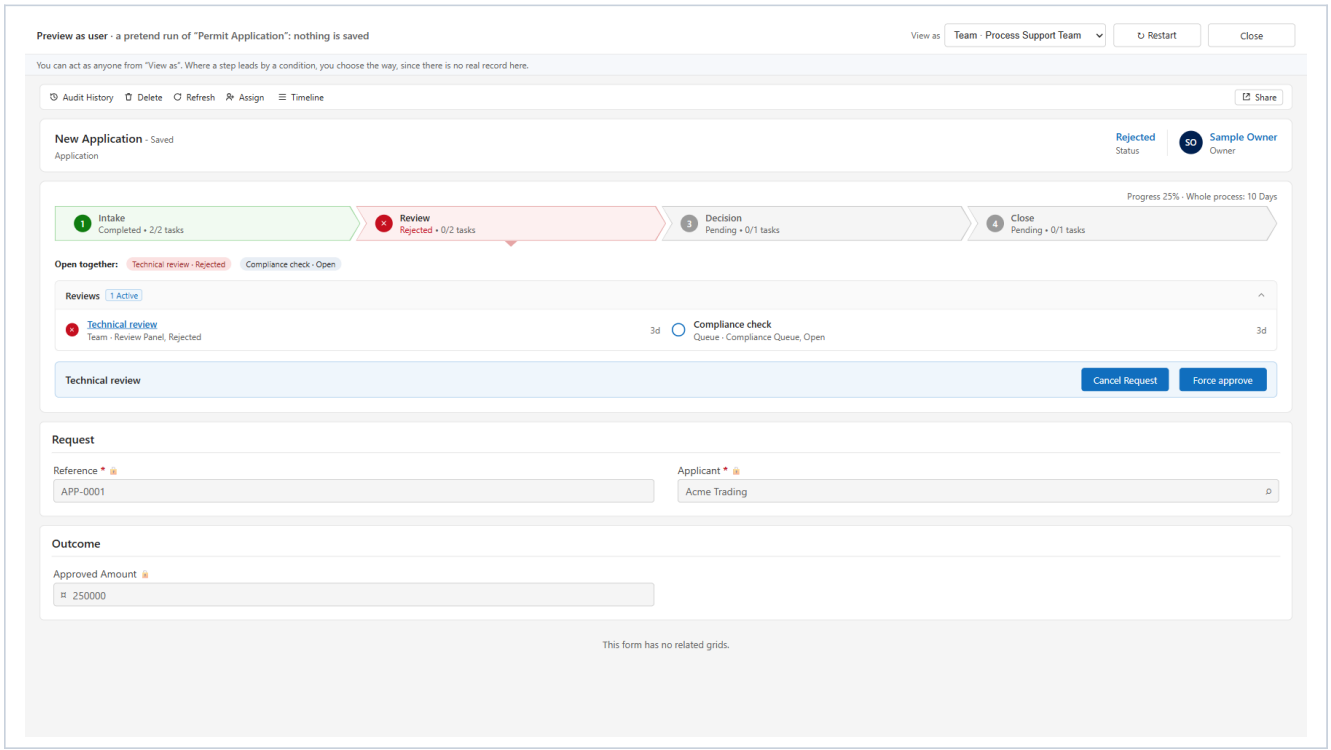


Figure 58. *Technical review was rejected, shown in red. Compliance check carries on. A member of the force-pass team sees a Force approve button on the step's bar.*

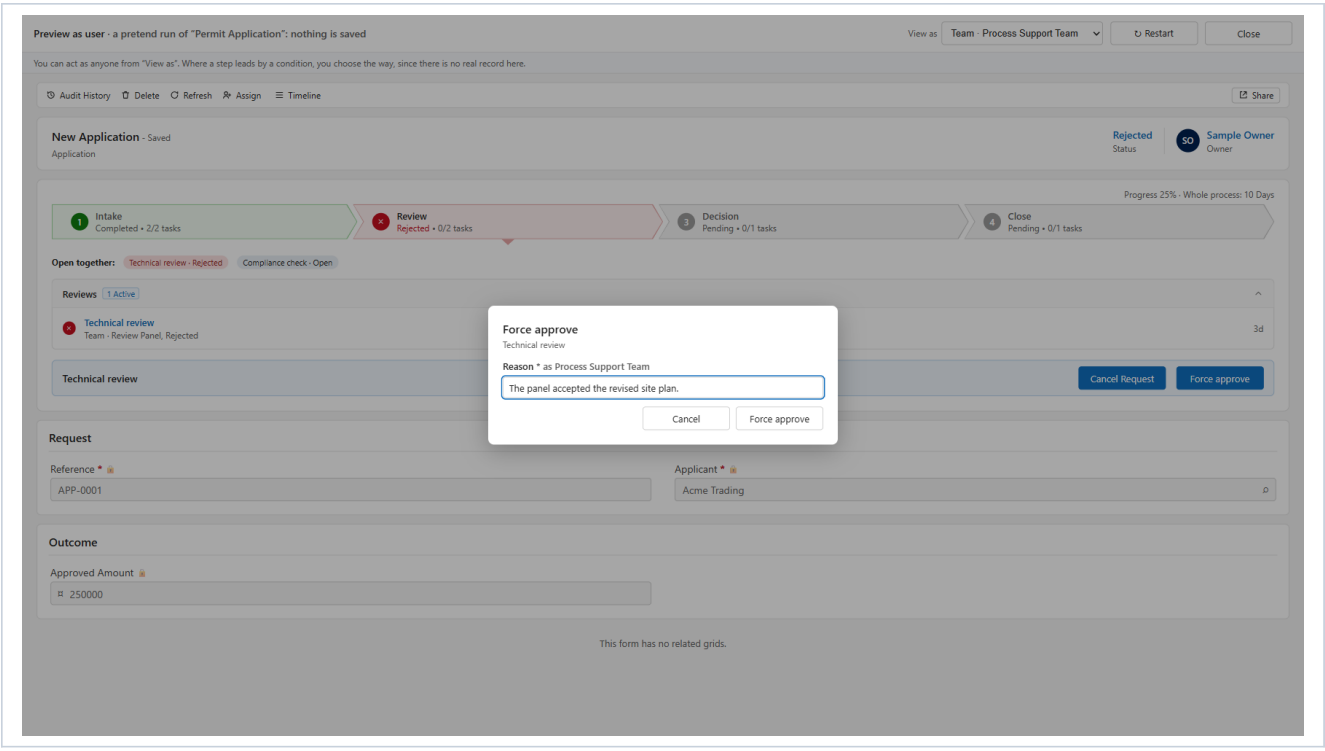
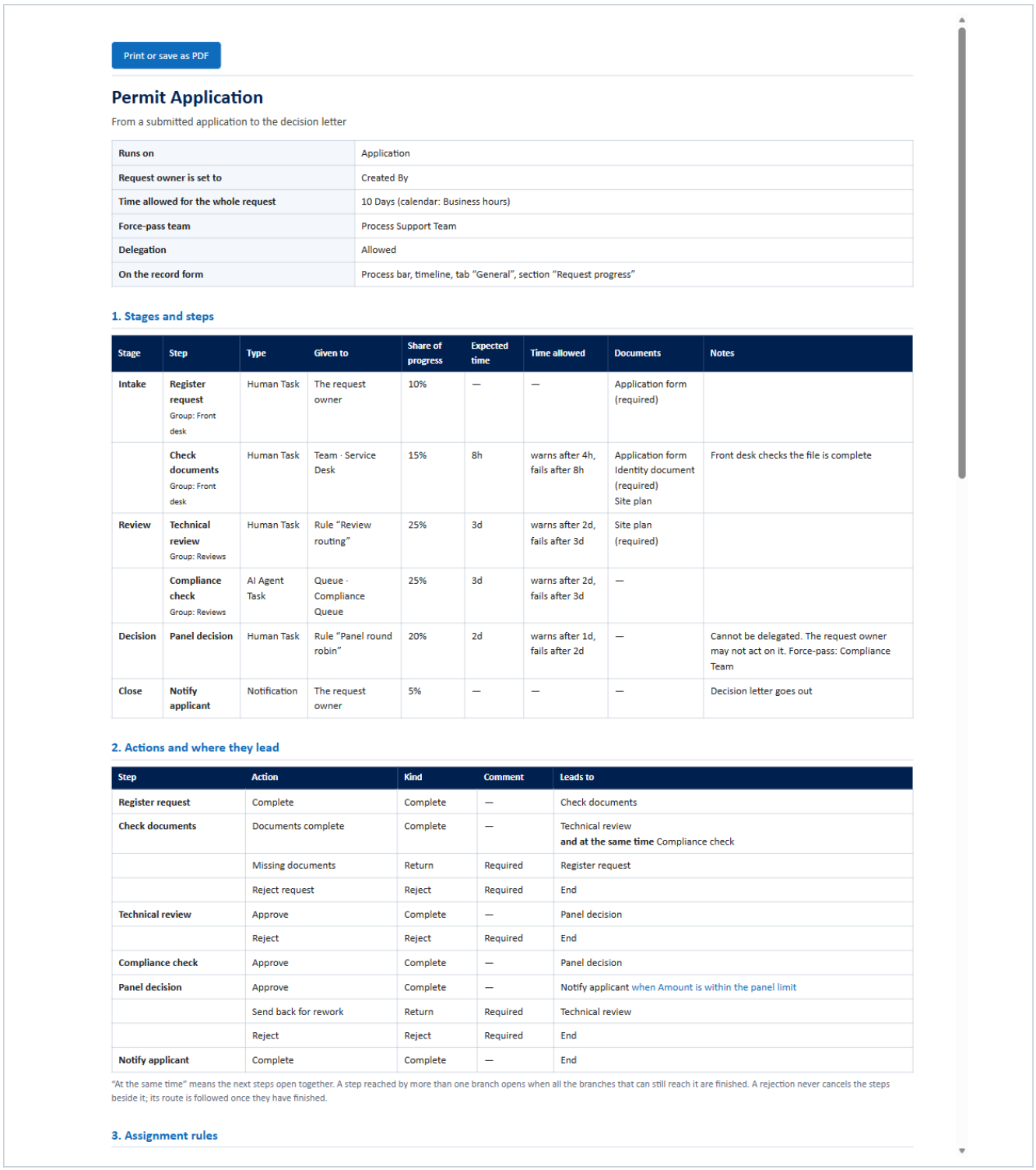


Figure 59. *Force approve asks for a reason. The reason is required and is kept on the step and in the timeline.*

16.9 The process document

The Process document button (printer icon) in the designer opens a clean, printable description of the whole process in a new window: stages and steps, actions and where they lead, assignment rules written as readable tables, messages, reminders, pause reasons, AI steps and documents, then the record status settings and the fields each step shows. The first table gives each step's expected time. Use Print or save as PDF to hand it to the client for sign-off. It is built from the current design, so print it again after any change.



Print or save as PDF

Permit Application

From a submitted application to the decision letter

Runs on	Application
Request owner is set to	Created By
Time allowed for the whole request	10 Days (calendar: Business hours)
Force-pass team	Process Support Team
Delegation	Allowed
On the record form	Process bar, timeline, tab "General", section "Request progress"

1. Stages and steps

Stage	Step	Type	Given to	Share of progress	Expected time	Time allowed	Documents	Notes
Intake	**Register request** Group: Front desk	Human Task	The request owner	10%	—	—	Application form (required)	
	Check documents Group: Front desk	Human Task	Team · Service Desk	15%	8h	warns after 4h, fails after 8h	Application form Identity document (required) Site plan	Front desk checks the file is complete
Review	**Technical review** Group: Reviews	Human Task	Rule "Review routing"	25%	3d	warns after 2d, fails after 3d	Site plan (required)	
	Compliance check Group: Reviews	AI Agent Task	Queue · Compliance Queue	25%	3d	warns after 2d, fails after 3d	—	
Decision	**Panel decision**	Human Task	Rule "Panel round robin"	20%	2d	warns after 1d, fails after 2d	—	Cannot be delegated. The request owner may not act on it. Force-pass: Compliance Team
Close	**Notify applicant**	Notification	The request owner	5%	—	—	—	Decision letter goes out

2. Actions and where they lead

Step	Action	Kind	Comment	Leads to
Register request	Complete	Complete	—	Check documents
Check documents	Documents complete	Complete	—	Technical review **and at the same time** Compliance check
	Missing documents	Return	Required	Register request
	Reject request	Reject	Required	End
Technical review	Approve	Complete	—	Panel decision
	Reject	Reject	Required	End
Compliance check	Approve	Complete	—	Panel decision
Panel decision	Approve	Complete	—	Notify applicant when Amount is within the panel limit
	Send back for rework	Return	Required	Technical review
	Reject	Reject	Required	End
Notify applicant	Complete	Complete	—	End

"At the same time" means the next steps open together. A step reached by more than one branch opens when all the branches that can still reach it are finished. A rejection never cancels the steps beside it; its route is followed once they have finished.

3. Assignment rules

Figure 60. *The first part of a process document.*

16.10 Delegation

Delegation lets a person hand their steps to someone else for a period of time, for example while on leave. It does not change the process design. Only who can act on a step changes.

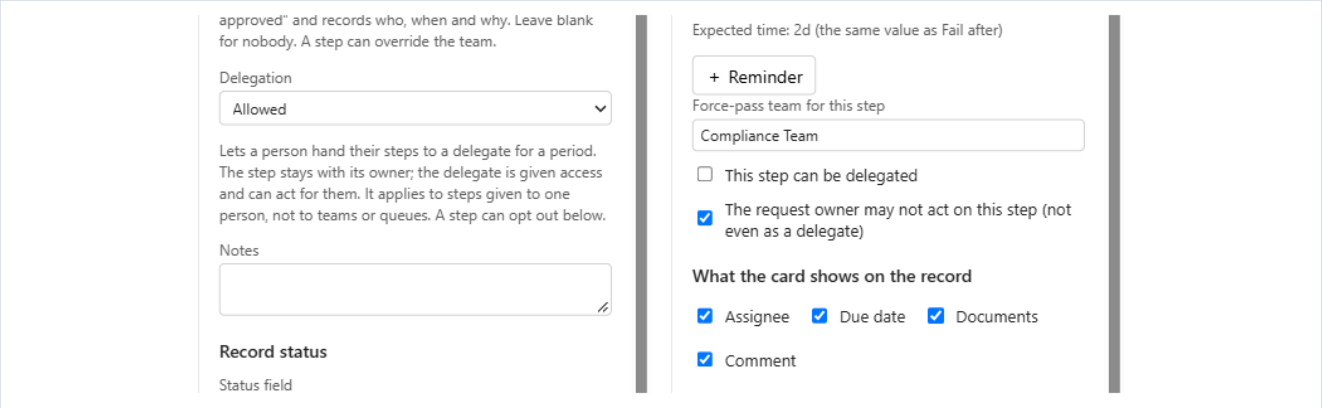


Figure 61. *Left: the Delegation setting on the process. Right: a step with "can be delegated" and "request owner may not act" ticks (here, Panel decision is neither).*

Setting	What it does
Delegation (process)	Allowed, Not allowed, or Needs manager approval. In the Studio's pretend run a blank counts as Allowed. Decides whether people may hand their steps of this process to a delegate at all.
This step can be delegated (step)	Untick it for a step that must always be done by its own assignee, such as a final decision. Such a step is never delegated.
The request owner may not act on this step (step)	Tick it when the person who submitted the request must not decide on it, even by being someone's delegate.

How a delegation works

- **The step stays with its owner.** The delegate is given access and acts for the owner. The owner can still act too, and whoever acts first wins. The history reads "Done by the delegate, on behalf of the owner".
- **Access, not reassignment.** The delegate is given a direct share of the step and of the business record, which gives access to those records without changing the owner.
- **Open steps.** Steps already open when the delegation starts are included, and so are steps that open later. When the delegation ends or is cancelled, the delegate loses access. The owner already holds the steps, so nothing has to be moved back.
- **Single people only.** It applies to steps given to one person: a user, the request owner, the owner's manager, or a person chosen by a rule. Steps given to a team or queue are not delegated, because anyone in them can already pick the step up.
- **One level.** If the delegate is also away, the step goes to the delegator's manager, not to a second delegate.
- **Time.** The SLA keeps running. Leave is not a pause reason.
- **The delegate needs a role.** They must hold a security role that includes the table. A share adds access to records, not table permission.

Delegations themselves (who, to whom, from when to when, all processes or chosen ones) are kept by the engine, not in the workbook. People create their own from a My delegations page, or a manager or admin does it for them. The process sheets only hold the settings above.

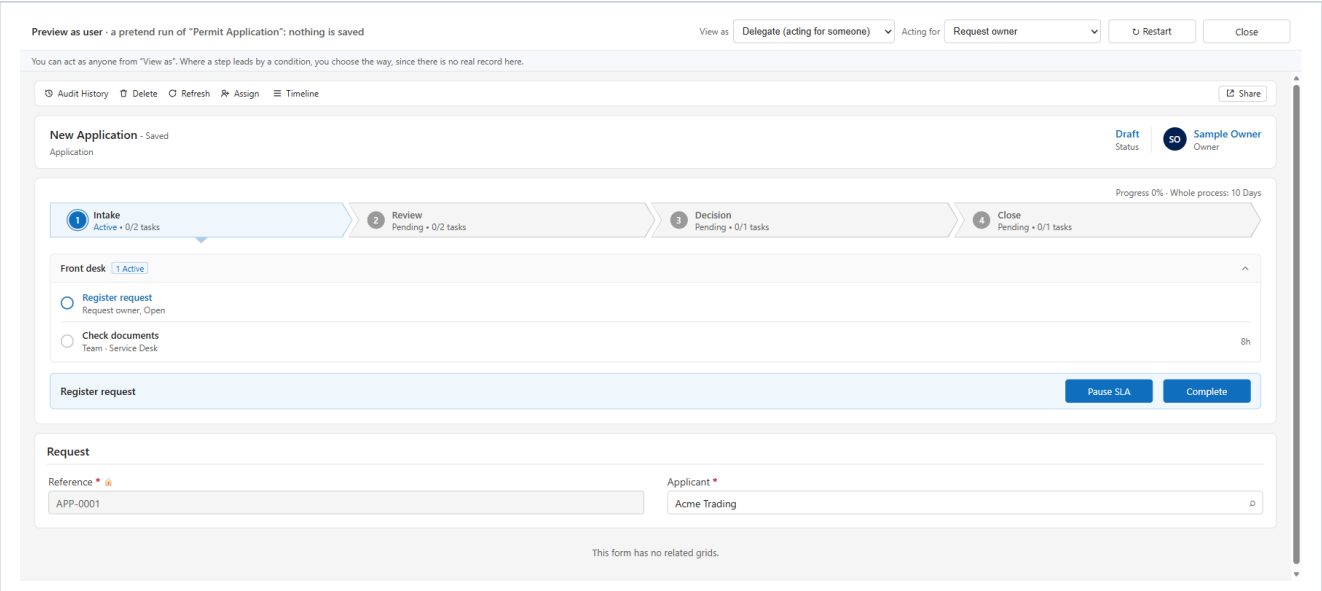


Figure 62. *Preview as a delegate acting for the request owner. The step stays theirs; the timeline will say "on behalf of".*

16.11 How it will run behind the scenes

This is the shape the engine and the screen will follow. It is the agreement the Studio designs to. The one-page version for the engine builder is docs/PROCESS_ENGINE_SPEC.md.

- **The business record stays an ordinary table.** Users work on their own record, for example a Permit Application. It is never turned into an activity.
- **Two tables sit behind it.** Request, one row per run of the process, and Request Step, one row per step that opens.
- **Request Step is an activity table.** Making it an activity gives owner by user or team, queues and the record timeline without extra work. The choice cannot be reversed later, which is why it is made once, here.
- **Assignment, actions, SLA and pause live on the Request Step.** Parallel steps are simply several Request Steps for one request.
- **A control on the main record shows the actions.** A custom control reads the open steps for the record and draws a card for each step the signed-in person may act on, and read-only lines for the rest. It only displays and asks. When the user clicks an action, the engine checks permission again, records it and moves the process on. Hiding a button is not security, so the check is always repeated on the server.
- **Delegates get access, not ownership.** The owner stays the owner of the step. The delegate gets a direct share of it (section 16.10).
- **Two people acting at once.** The first wins; the second is told the step is already done.

16.12 The workbook sheets

Process design is saved in optional sheets. They are created only when used, and a workbook without them is untouched. Details are in docs/EXCEL_FORMAT.md, section 6.17.

Sheet	Content
Process Design	One row per process: table it runs on, request owner rule, SLA calendar, whole-process SLA, force-pass team, delegation, the record layout (process bar, timeline, form tab and section), Status field and Status when cancelled.

Sheet	Content
Process Stages / Process Groups / Process Steps	Stages in order; the groups inside each stage (columns, start collapsed); steps with group, type, assignment, weight, time allowed, force-pass team, delegable, not-by-request-owner, what the card shows and Status when it opens.
Process Actions / Process Routes	The buttons on each step, and one row per connection: from action, to step or End, and an optional condition.
Process Assignment Rules	Rules of each type, with the name of a connection for External API.
Process Rule Inputs / Process Rule Cases	The fields a decision table looks at, and its rows (one row per condition, the outcome repeated).
Process Notifications	Email and SMS sent by steps.
Process SLA Reminders / Process SLA Pause	Reminders for steps and the whole process; pause reasons and their approvers.
Process Agents / Process Agent Steps	The workbook-wide agent list, and each AI step's set-up.
Process Documents / Process Step Documents	Documents the request needs, and which step needs which.
Process Step Fields	Which fields of the main table each step shows, each with its own Required and Read-only. A row means shown (section 16.17).

16.13 What Check design looks at

With Process design on, Check design adds a Process area. It reports, among other things: a Form tab in the process settings that the table does not have, a human step with no actions, a rule with no Otherwise row or one that looks at a field that no longer exists, a rule no step uses, a reminder with nothing to count from, a pause that needs an approval team, an AI step with no agent, a document nobody asks for, an action connected to nothing, a stage or group with no steps, a first or only step that was deleted, or a stage deleted with its steps, and an SLA warning that comes after the failure time. Each item names the step or rule and can be clicked to jump to it.

16.14 Current limits of process design

- The engine reads the process sheets, but not yet the newest settings (Process Step Fields, Status field, Status when it opens, Status when cancelled). Until it learns them the Studio's workbook is the master copy: a workbook downloaded from the engine has lost them.
- Delegation works by a direct share of the step and of its business record, not by access teams and not by reassigning the owner.
- The Plugin rule type is recorded only. External API holds a connection name, never an address or a secret.
- Preview is a pretend run. It does not evaluate route conditions or rule conditions (you choose), and its confidence figure for AI steps is an example.
- Rule conditions look at one table's own fields plus one lookup hop.

16.15 Status: what is finished, and what is left for the engine

On the Studio side, process design is complete: nothing in the screens, the workbook format, the checks or the preview is waiting to be done. What remains is the engine and the control on the record, which are built separately from the spec.

Finished in the Studio	Left for the engine build
Process bar, canvas, step panel, stages, groups, parallel steps, routes	Learn the newest settings: step fields, Status field, Status when it opens, Status when cancelled
Record layout settings and layout preview	Draw the real layout on the form (the control on the record)
Assignment rules, decision table, Try it, When the rule runs	Run rules for real (including External API and Plugin)
SLA, reminders, pause reasons, AI agent steps, documents	Clocks, reminder sending, calling the agent
Reject, join, force-pass team, statuses	Request and Request Step (activity) tables and security roles
Delegation settings and delegate preview	Delegation tables and direct shares
Preview as user, Check design, process document	The control on the record and a My work page
Workbook format (section 16.12) and spec page	Audit trail, and one-at-a-time protection when two people act together

Two small notes. Preview is a pretend run: you choose which way a condition goes, and the AI confidence is an example. And one earlier report, an error message that seemed to stay after being resolved, could not be reproduced; please tell me if you see it again.

16.16 Groups and the record layout

A process has stages, a stage can have groups, and a group holds steps: Process → Stages → Groups → Steps. A group is only a display container. It decides which step cards are shown together, in how many columns, on the record. It never changes the order of steps, the routes, the joins or the SLA, and it is not a permission.

Groups in a stage

Click a stage. The panel lists the groups of that stage, each with its steps. Add a group with + Add a group, give it a name, choose 1, 2 or 3 columns and tick Start collapsed if the user should open it on demand. + step adds a new step straight into the group; the arrows reorder groups; Delete group returns its steps to the stage, ungrouped. Steps that are not in a group sit in a default block at the top.

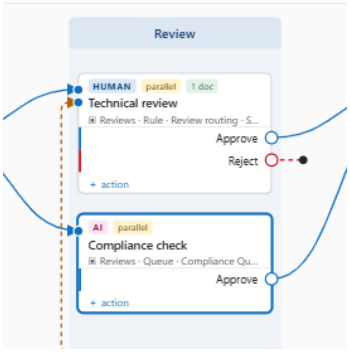


Figure 63. *A stage with one group. "Reviews" shows two columns, so Technical review and Compliance check sit side by side on the record. On the canvas each card names its group.*

Putting a step in a group, and choosing what its card shows

In the step panel, the Group list (under Stage) offers the groups of the step's own stage. Moving a step to another stage clears its group. Lower down, What the card shows on the record has four tick boxes:

Assignee, Due date, Documents and Comment. The step name and status always show. Untick parts to keep a compact card.



Figure 64. *The Group list in the step panel, under Stage.*

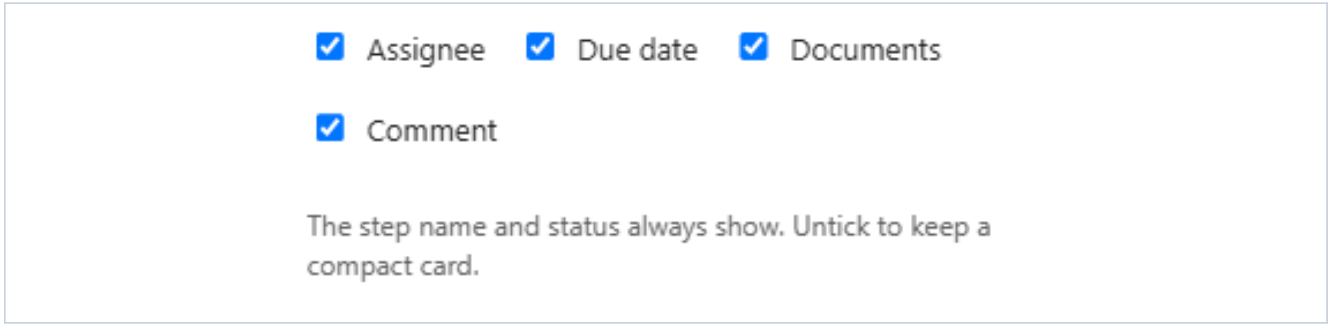


Figure 65. *What the card shows on the record: untick any part for a compact card.*

Record layout

Click the process (the Process settings link in the stage panel) and scroll to Record layout. Show the process bar and Show the timeline switch those parts on or off. Form tab and Form section name where the control sits on the business form, so the engine build can place it. In the Studio, Form tab places the panel for a process with no step fields; with step fields the panel is above the tabs whatever it says (section 16.17). If the table has no such tab, the first tab is used and Check design warns. The ▦ Layout preview button in the designer header draws the result.

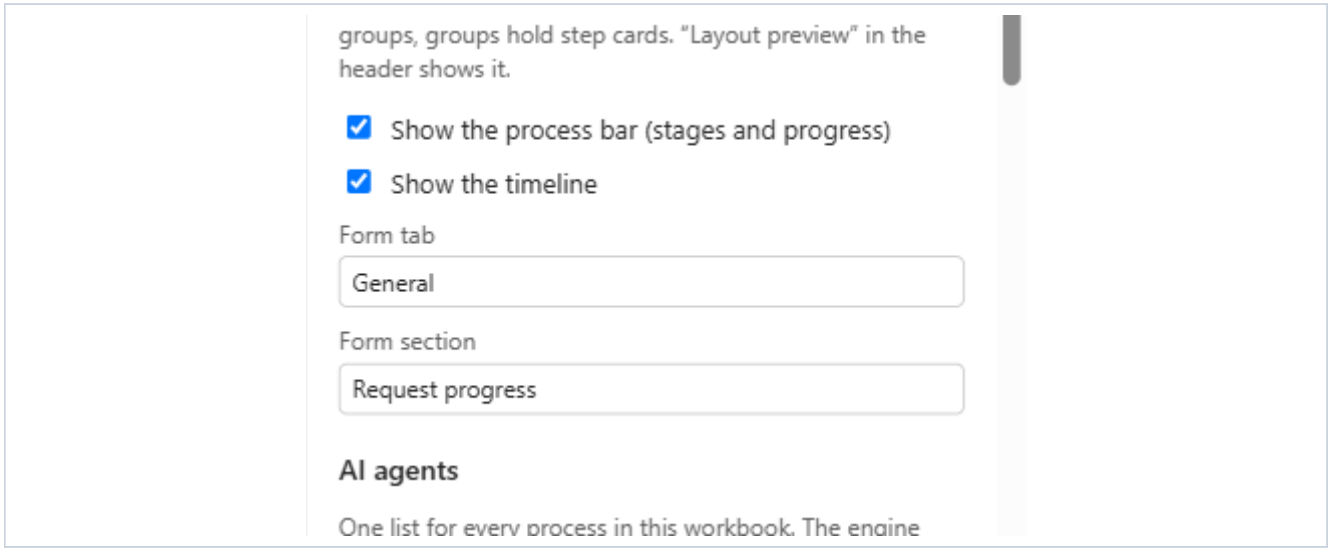


Figure 66. *Record layout in the process settings.*

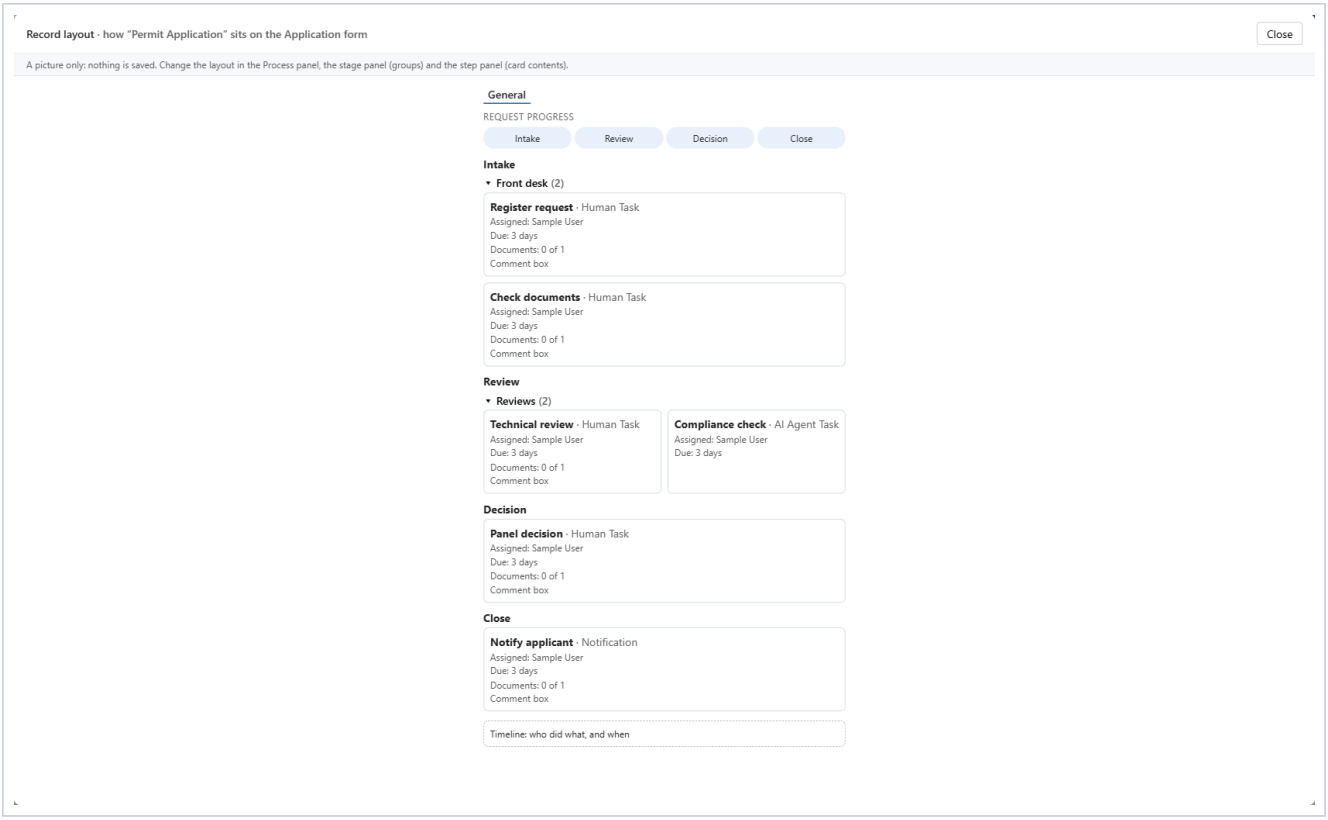


Figure 67. *Layout preview: process bar, then each stage with its groups and step cards. Reviews has two columns; a card with fewer parts ticked is shorter. Click a group heading to collapse it.*

- Preview as user lists the steps under their group heading, and the printable process document shows each step's group and the form layout.
- Check design warns about a group with no steps. A step that names a group the stage does not have is shown in an implicit group, so nothing is lost.
- In the workbook, groups live in an optional Process Groups sheet, plus a Group and a Card shows column on Process Steps and four layout columns on Process Design. Nothing is added to a workbook that does not use them.

> **For the engine build** Process, Stage, Group and Step become separate tables, and each Request Step carries a lookup to its group. The record control draws one block per group, exactly as the layout preview shows.

16.17 The step-driven record screen

When a table's process has step fields, the record opens on the process screen by default. It shows the record the way a person working on it would see it, and only what the step in hand needs. It is a pretend run: nothing typed or done here is saved, and nothing goes into the workbook, so it is safe to use with the client.

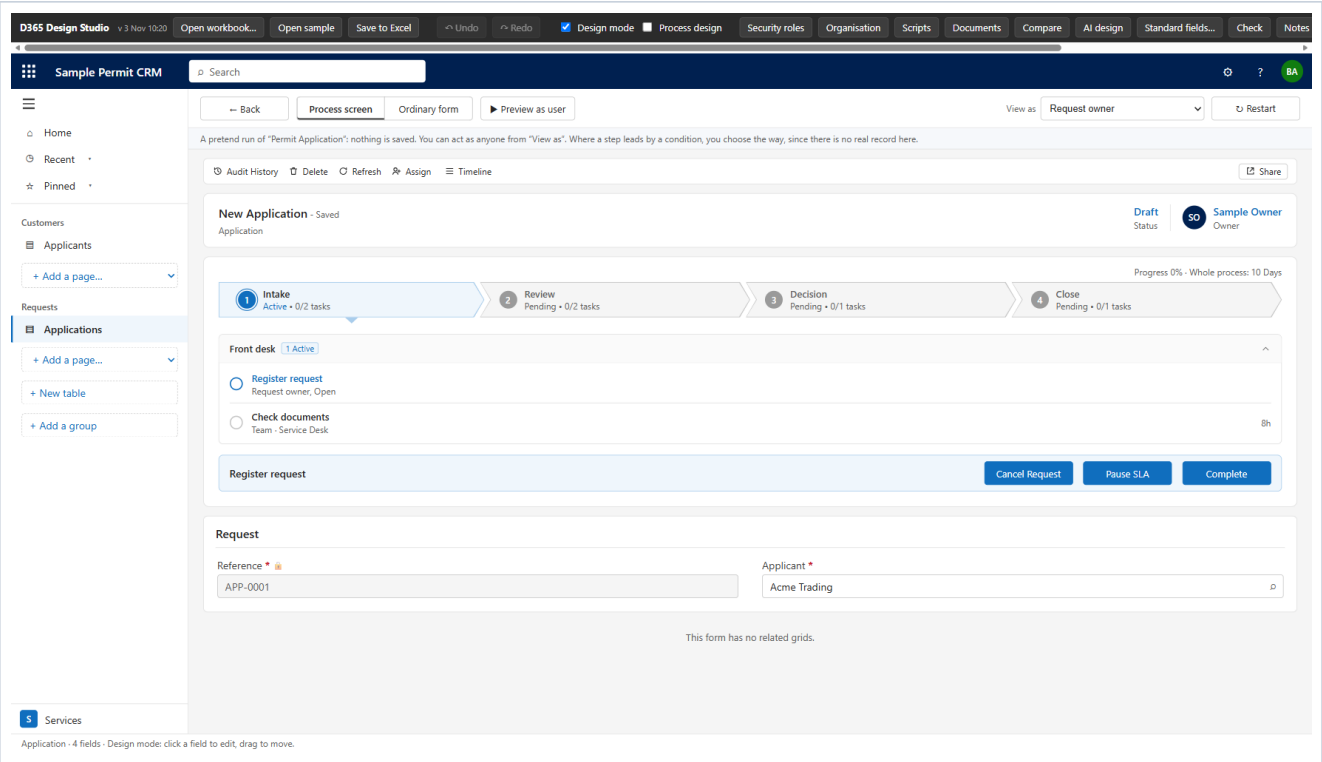


Figure 68. *What a record of the sample shows by default, with Process design unticked: the switch, View as, the stages, the step rows, the bar for the selected step and that step's fields.*

The switch, and what the tick does

The switch at the top of the record flips between Process screen and Ordinary form. The choice lasts until the page is reloaded and is never saved in the workbook. Use Ordinary form to design the form itself: its tabs, sections, fields and command buttons. Clicking a line in Check design also goes to the ordinary form, because that is where the line leads.

The Process design tick only controls the designer tools: the process bar on the ordinary form, Show full process, Fields per step and the Tools menu. With the tick on or off, a record whose process has step fields opens on the process screen. A table with no process, or a process with no step fields, shows the ordinary form and has no switch.

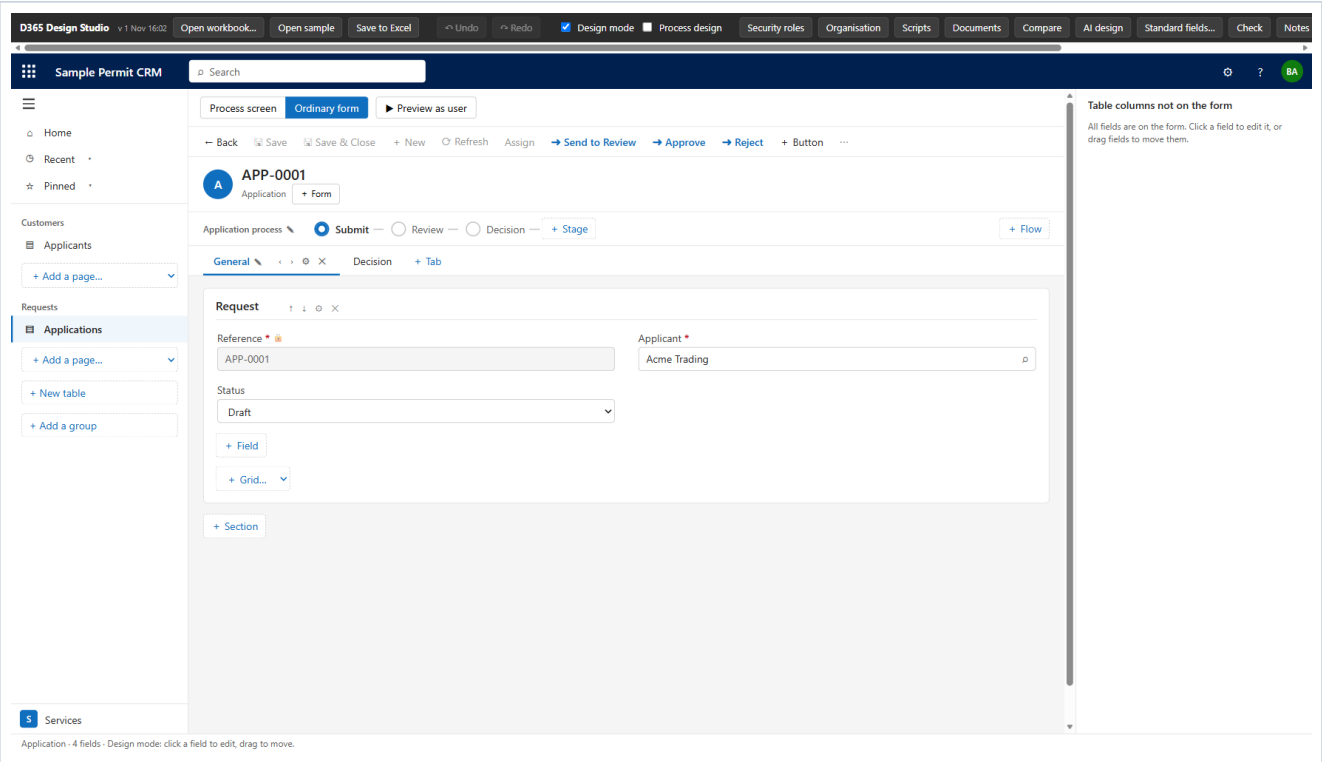


Figure 69. *The same record on the ordinary form. The switch stays at the top, so you can flip back.*

Walking through the process with View as

View as, at the top of the process screen, starts on the request owner. Pick a person and act as them: complete the step, then see the next step open for the right person. Choose that person in View as and carry on, through to the end. A button that needs something, such as a comment or documents, asks for it in a popup. Pick this step is a button on the bar. Where a step leads by a condition, you choose the way, because there is no real record to decide. Restart begins a fresh run. Preview as user is a second way to open the same thing, in its own window.

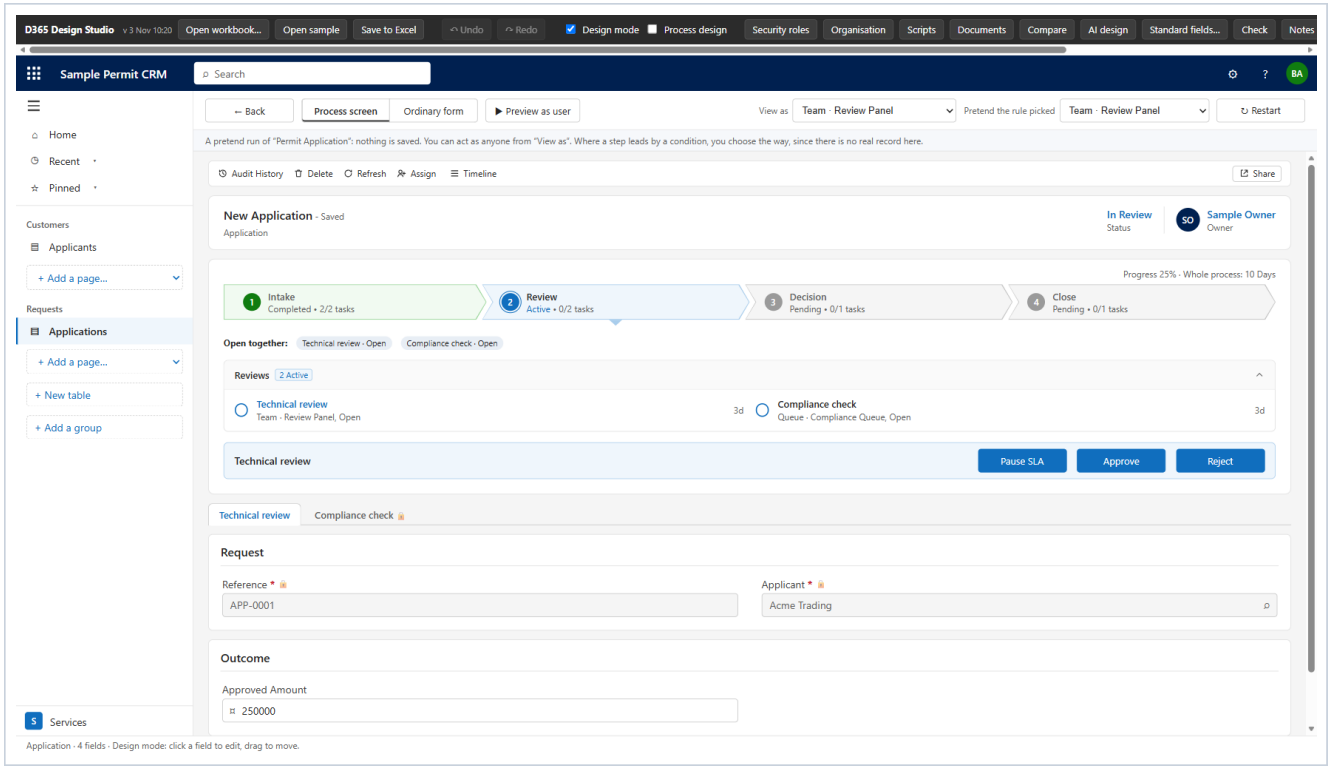


Figure 70. *Part way through, View as set to Team · Review Panel: both reviews are open, Technical review is this person's to act on, and Compliance check's tab has a lock.*

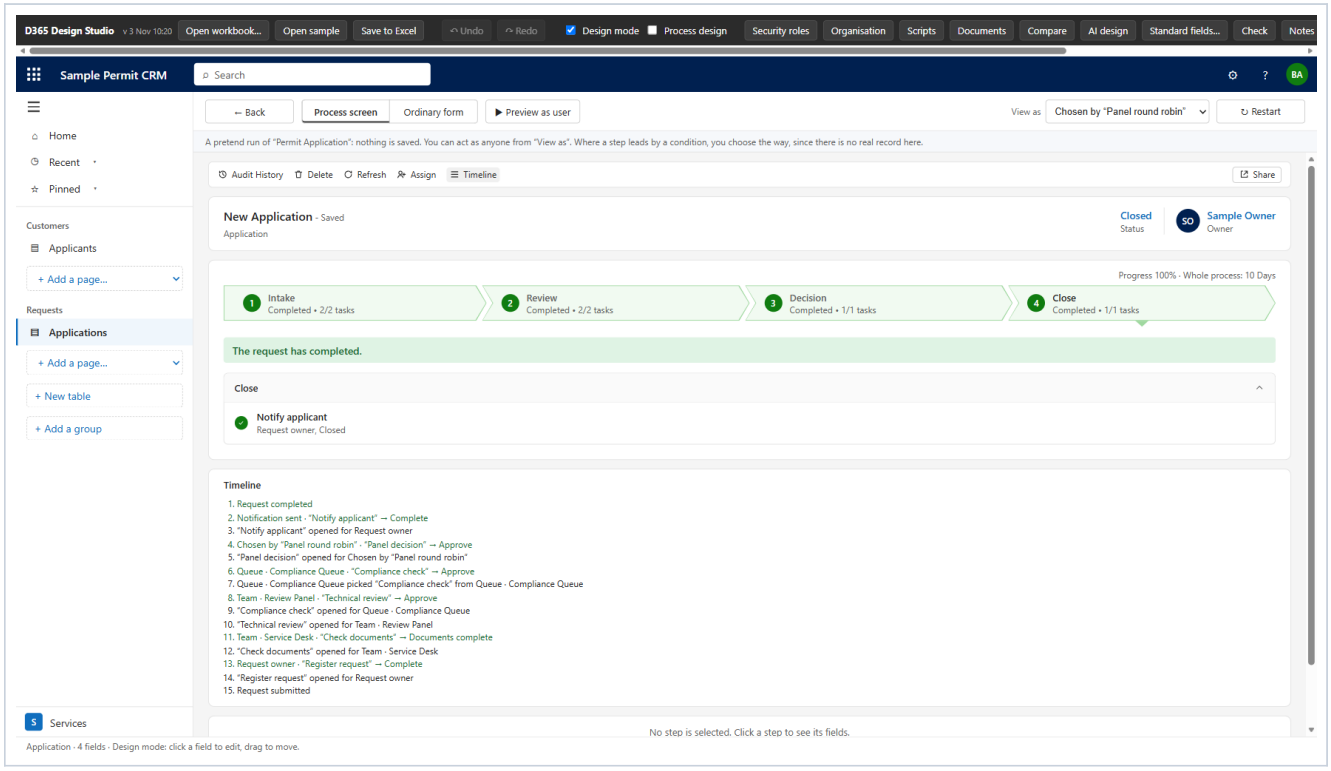


Figure 71. *The end of a walkthrough: the request has completed, the header shows the last status set, and the timeline shows who did what.*

Reading the screen

From the top: the record's header with its one status and its owner; the progress line; the stages as chevrons, each saying how many tasks are done; group cards holding the step rows; a bar for the selected step, with its real action buttons; then the fields that step shows. Click a stage to look at it, or a step to select it. A finished or not-started step can be selected too: its fields show locked and it has no action buttons.

Picking from a queue, pausing and force approve are buttons on the bar when the person may use them. Timeline, on the bar at the top, shows who did what. The stages share the whole width of the screen, however many there are. A long name wraps and then shortens; point at it to read all of it. In a narrow window the stages go onto more rows, and each row still fills the width. Each step's row says who owns it. The selected step's bar has only the step's name and its buttons, solid blue with white text. A greyed one cannot be used by the person chosen in View as; point at it to see why. The other controls on the screen are plain white. The switch shows its chosen side in bold and underlined.

A stage the run has finished shows Completed with no count when its other steps are routes the run did not take, so it does not look as if tasks were left undone. It keeps its count if one of those steps can still be reached going forward. A refusal on the step's bar, such as a required field left empty, is for the person who pressed the button: changing View as clears it.

One area for the fields, and extra tabs

When a process has step fields, everything from the progress line down to the step's bar sits above the tabs and stays in view on every tab. The fields of the selected step show in one area under that bar, in the form's order and with the form's section headings, whichever tabs of the form they are on.

A tab that no step of the process uses stays as an extra tab, as it is on the form, beside a tab called Details. A form tab that already has that name shows as Details (form tab). It is locked for someone who cannot act on the selected step, and a required field on it does not hold the step back. A tab that only other steps use is not an extra tab: its fields show when those steps are selected. With no extra tab there is no tab strip, as in the sample. A process with no step fields keeps the form's tabs in the record header, and the ordinary form does not change.

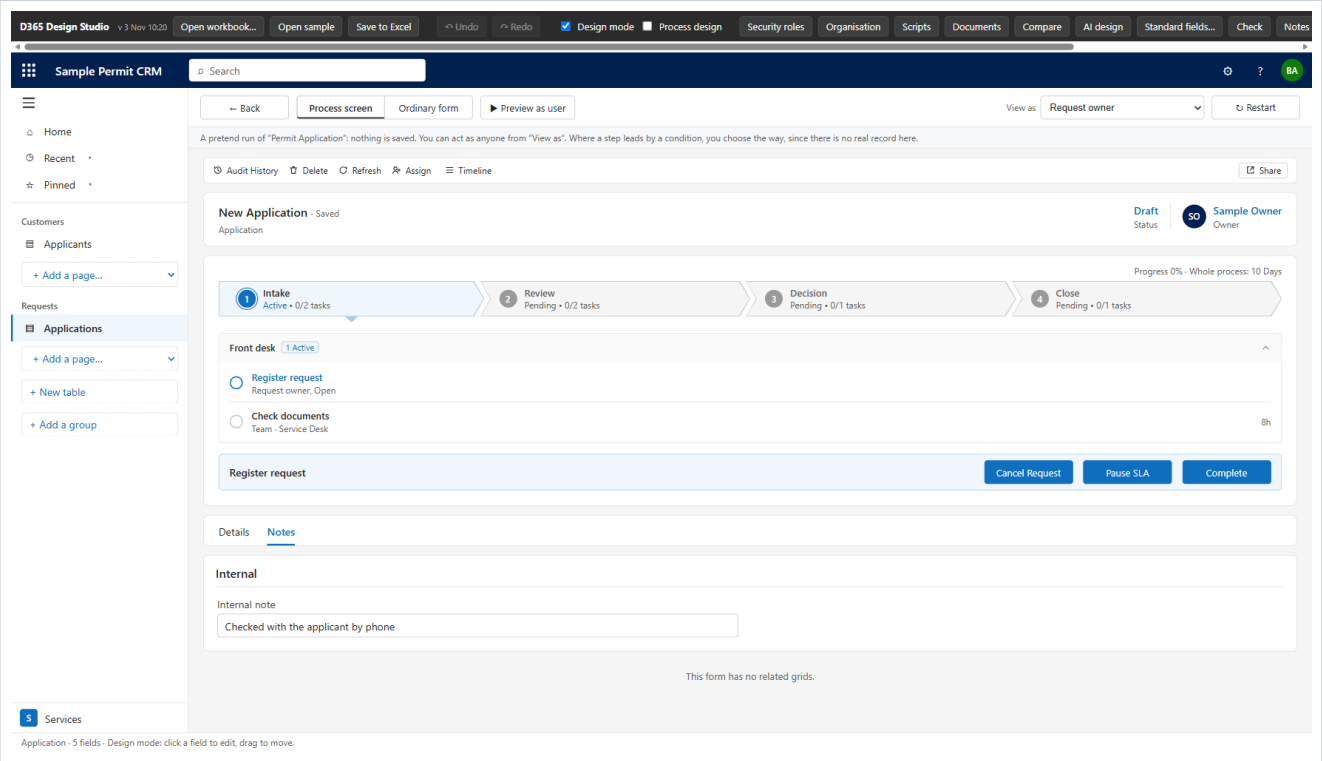


Figure 72. *A Notes tab, added to the sample for this picture, is an extra tab because no step uses it. The stages, the step rows and the step's bar stay in view above the tabs.*

Fields per step

Fields per step, in the designer's top bar or from the Set… link in a step's panel, says which fields of the main table each step shows. Tick a field to show it. For each one, Required and Read-only can follow the table or

be set to Yes or No for this step. Copy from step starts a step from another's choices, and Clear empties it. The left-hand list shows how many fields each step has.

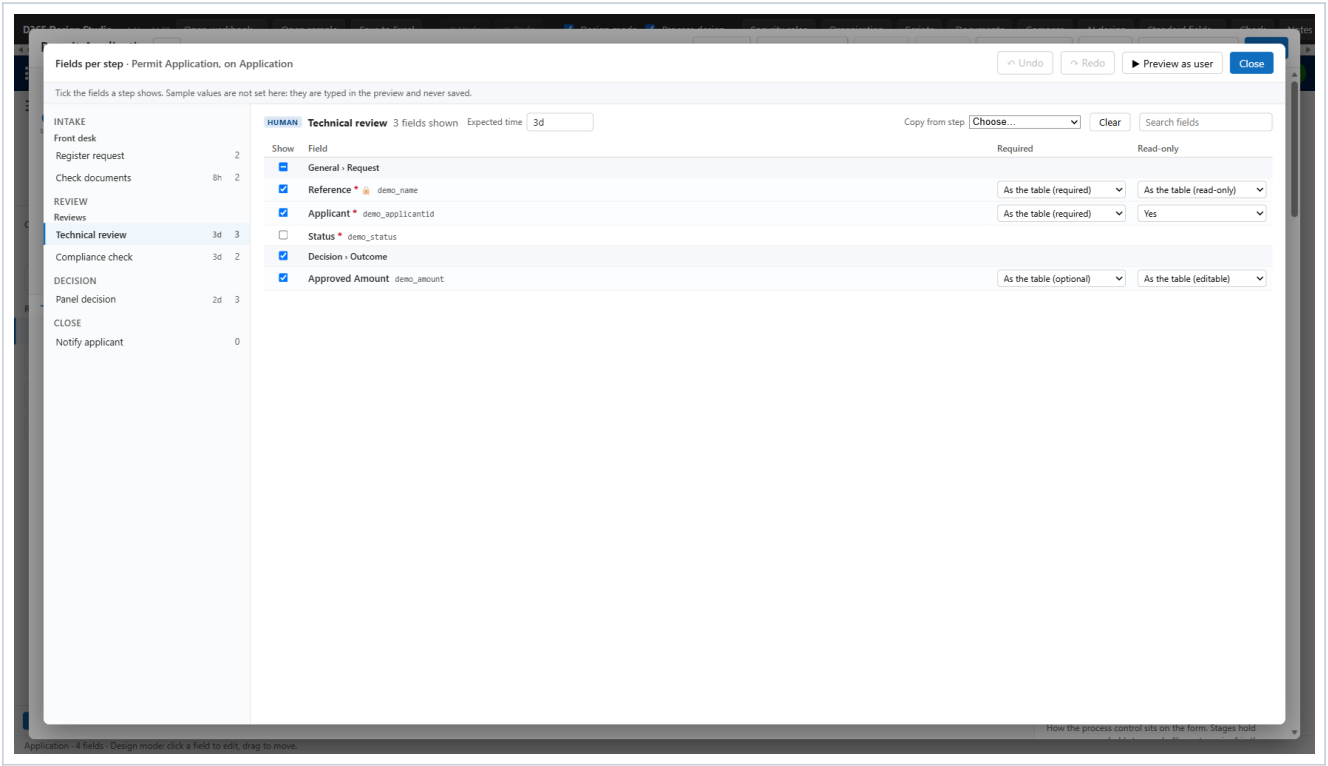


Figure 73. *Fields per step for Technical review. Three fields are ticked, the expected time is at the top, and the left-hand list shows each step's time and field count.*

The rules on a field apply in this order: the role's rules win, then the step's row, then the table sheet for what the row leaves blank. Once a process has any row in this sheet, a step with nothing ticked shows no fields, and Check design warns unless the step runs by itself. Complete refuses an empty required field that the person can fill. A required field that is empty and locked in the step also refuses, and the message says nobody on this step can fill it.

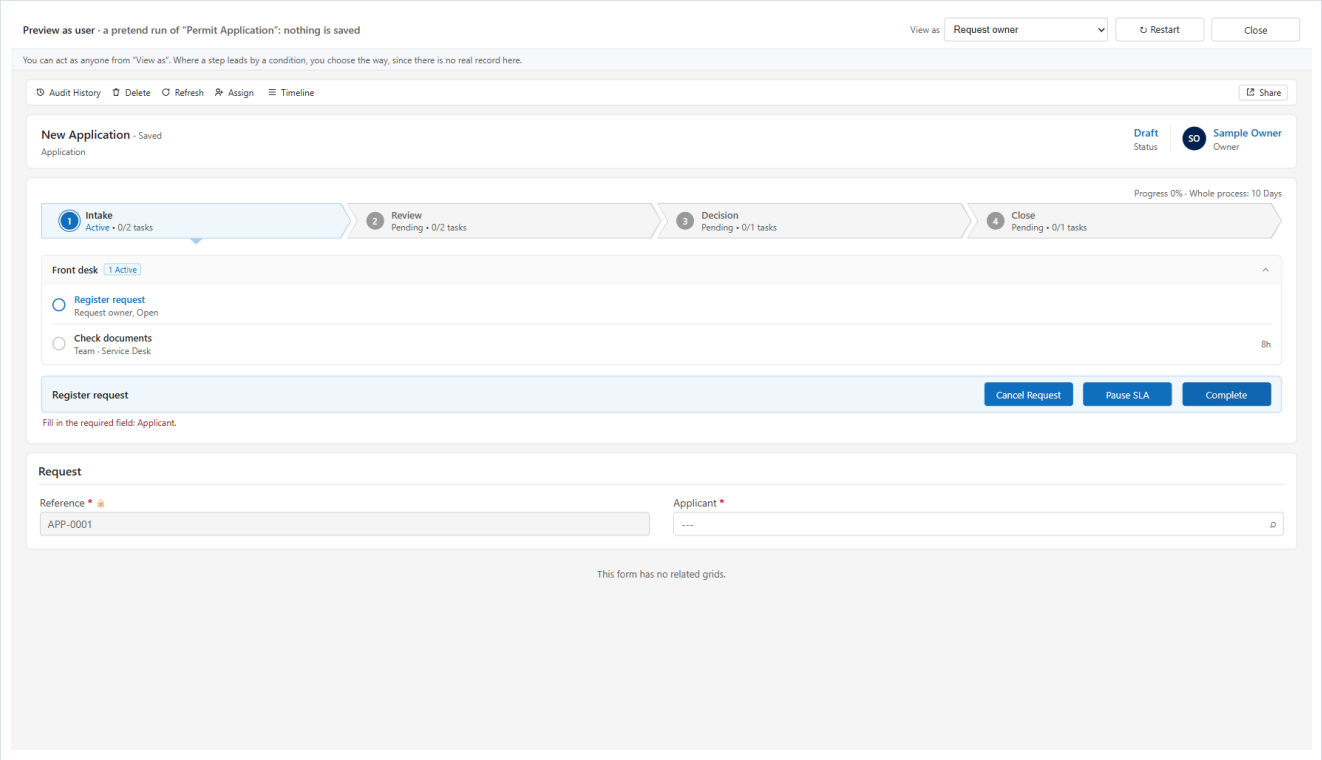


Figure 74. *Complete refuses an empty required field, in plain words.*

A process with no rows in Process Step Fields shows the whole form, on the record and in Preview as user. To get there, press Clear on each step, or use a workbook without that sheet. The sample workbook has rows for both of its processes, Permit Application and Applicant Registration, so records of both Applications and Applicants open on the process screen.

Several open steps

When several steps are open together, the fields area has a tab per open step, as Figure 57 shows. Selecting a tab and selecting a step row are the same thing. A tab has a lock when the chosen person cannot act on that step, and its fields are read-only. Who should see which tab is still an open decision; for now everyone sees all of them.

The record status

There are three statuses. The step status and the request status are the engine's own fixed states and cannot be edited. The record status is a Choice column on the main table, designed like any other column, and it is the one the header shows. Name the column as the process's Status field. Each action's Set the record status to, and each step's Status when it opens, then pick one of its options. If two steps that can be open together both set an opening status, the last one to open wins, and Check design warns.

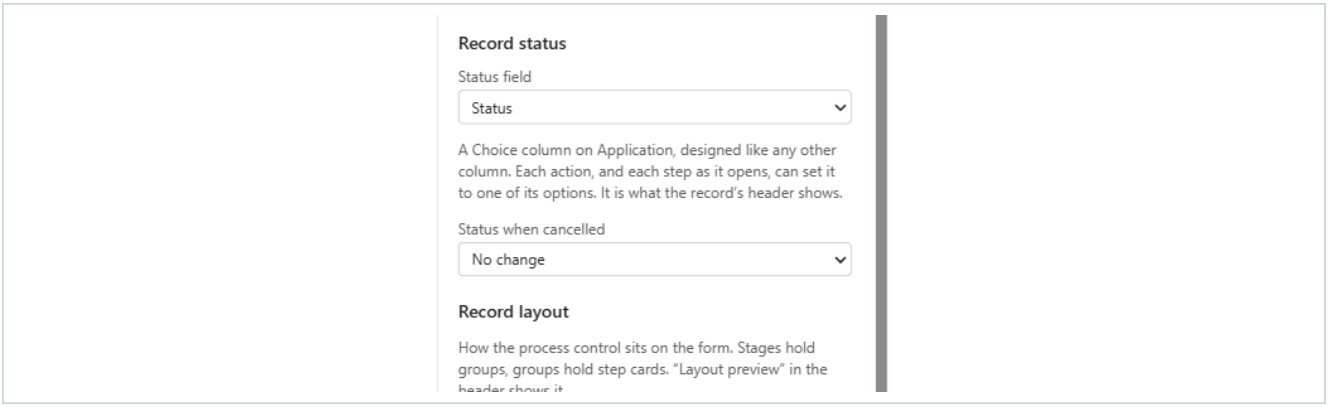


Figure 77. *The Record status block in the process settings.*

Expected time

Expected time is the step's Fail after with its SLA unit, not a separate column. Type it at the top of Fields per step: 1h 20m, 90m, 2d, 1.5h, or a plain number of hours. The unit you type is kept: 2d is saved as Days, and anything with hours or minutes, including 48h, is saved as Hours. Mixing days with hours, such as 1d 4h, is refused, because the length of a working day depends on the SLA calendar. While Warn after is set, a time in the other unit is refused too. The time shows on the step rows. The preview does not count it down.

Cancel Request

Cancel Request is one built-in button on the selected-step bar, on every step of a running request, to the left of the step's own buttons. It is not one of the actions you design. The request owner sees it, and so does an administrator; the Studio has no administrator of its own, so in the preview that means the process's force-pass team. Others do not see it. The real engine asks its security role.

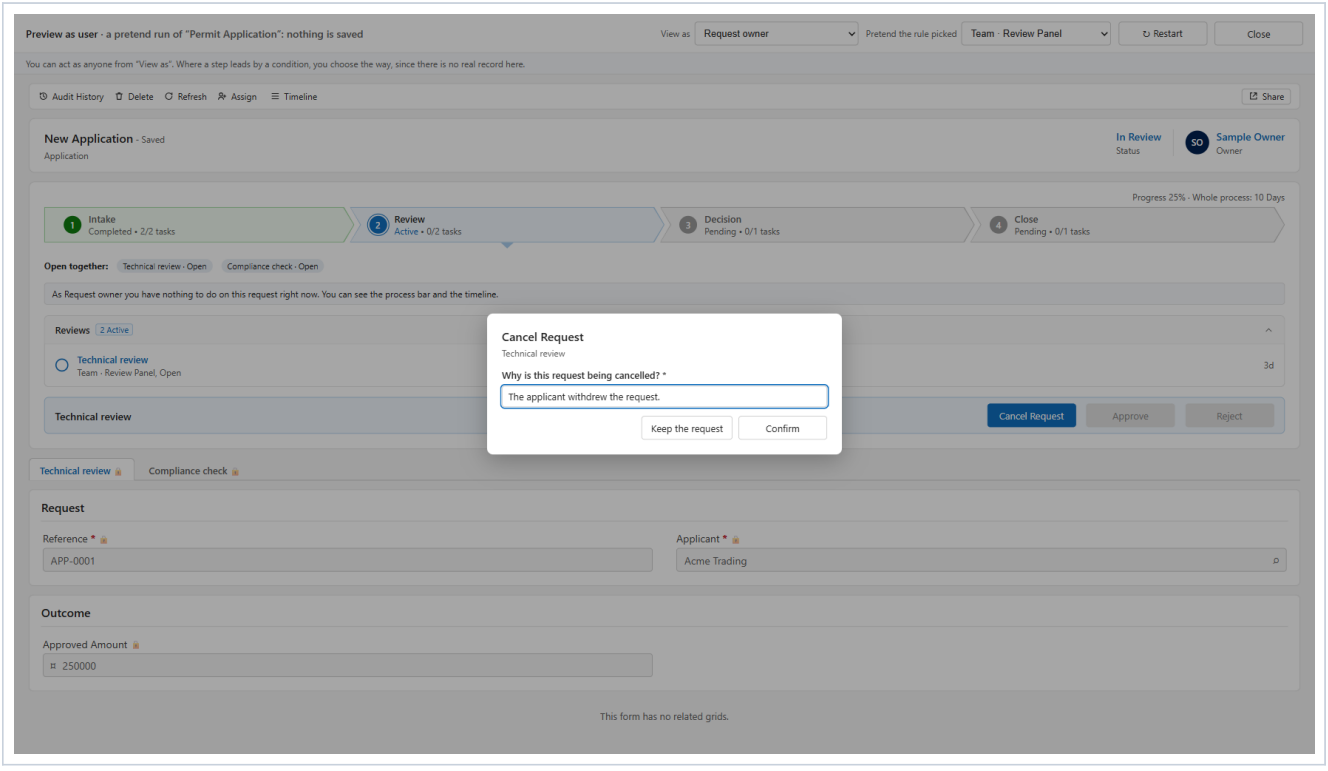


Figure 75. *Pressing Cancel Request opens a popup. A reason is required.*

Confirm closes every open step, ends the request as Cancelled, adds "Cancelled by who: reason" to the timeline and shows a banner with the reason. Required fields and documents never block a cancel, and a

cancelled request cannot be reopened. Restart starts a fresh pretend run. The optional process setting Status when cancelled is the record status it leaves; left blank, the header keeps its last value.

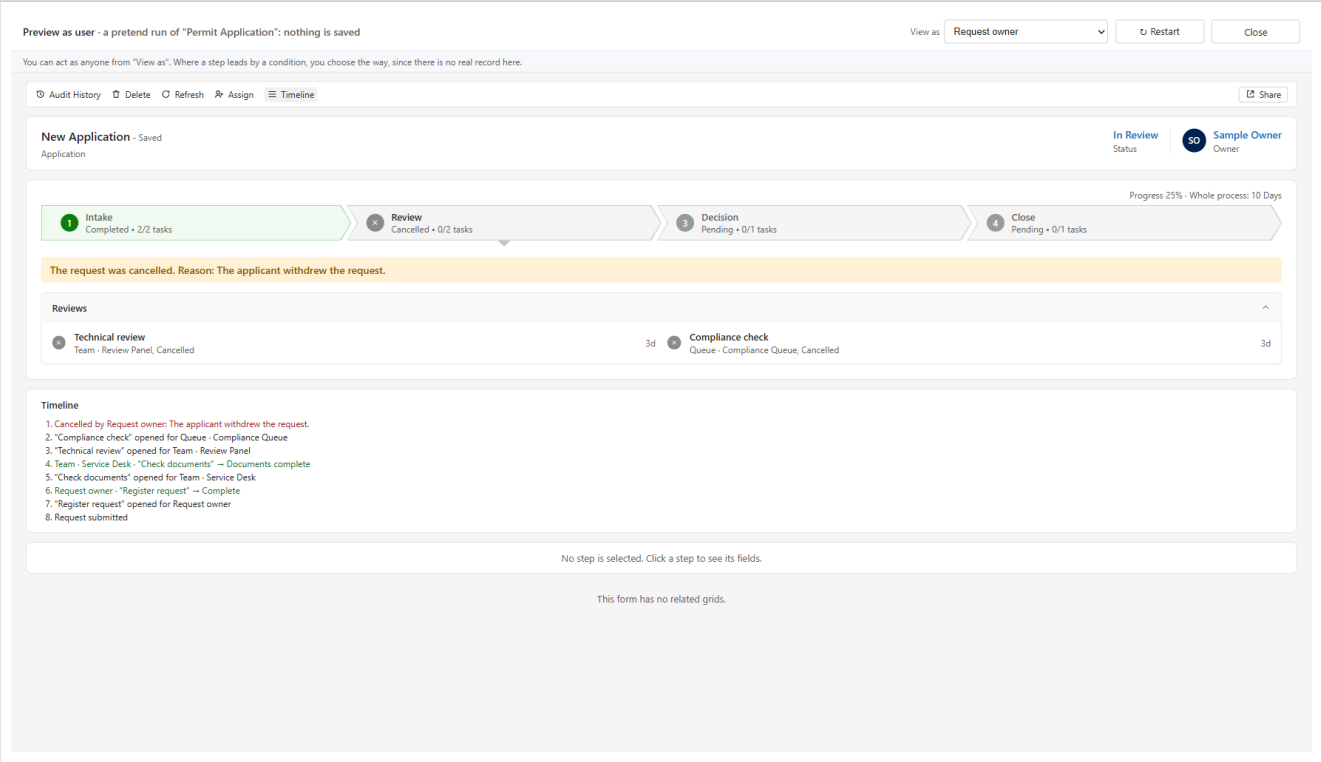


Figure 76. *A cancelled request: the banner with the reason, the closed steps and the timeline.*

Checks, Compare and the process document

When Process design is on and the process has step fields, Check design adds these. Errors: a field required in a step but read-only there; a row whose field is not on the table; a row whose process or step does not exist; a required field that no form shows. Warnings: a step other than a system or notification step that shows no fields; the same field twice on one step; a step whose Reject sets a status while its Complete sets none; a field a step shows that no form places, because the engine needs it on the form. Click a line to go to the step. A field required on the table and read-only in a step is not an error, because an earlier step can have filled it.

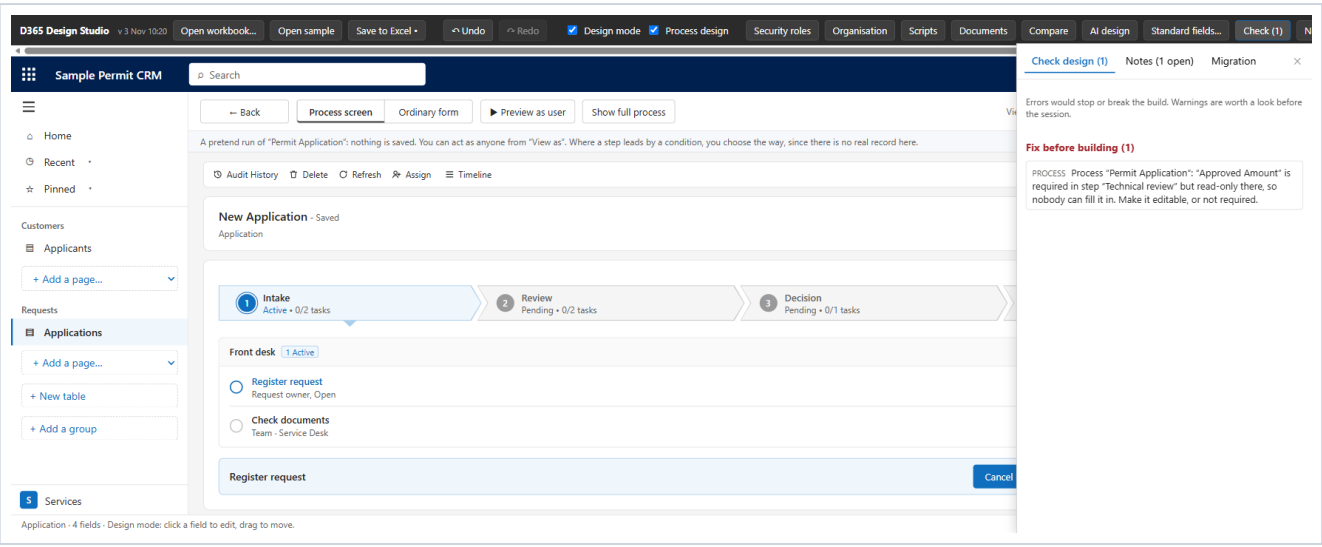


Figure 78. *Check design reports a field required but read-only in a step. Clicking the line opens that step.*

Compare has a Process design area listing differences in the step fields, expected time, Status field, Status when it opens and Status when cancelled, and processes or steps added or removed. The process document lists the status settings, each step's expected time and the fields each step shows.

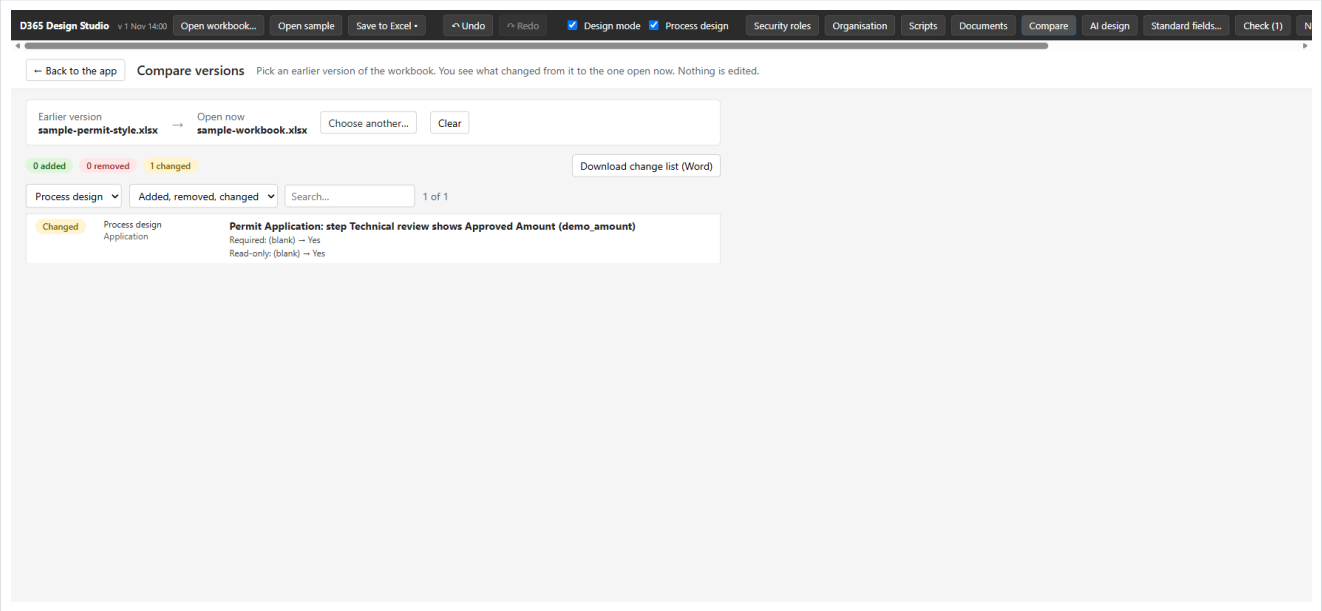


Figure 79. *Compare, filtered to Process design: a step field's settings changed.*

System tasks, popups and Super user

The popup. There is no More button. When a button needs something from the person, a popup opens and asks for it: a required comment, the step's documents with room to add another, which way to go when the action has several conditional connections, what an AI agent recommends, or a reason for force approve, for pausing or for cancelling. Nothing happens until it is confirmed, what is required is checked there and then, and Cancel leaves the step open. A required field on the form is told about under the bar before any popup opens, because the form is where to fill it in. A comment can only be added when a popup opens, so an action that needs nothing takes no comment.

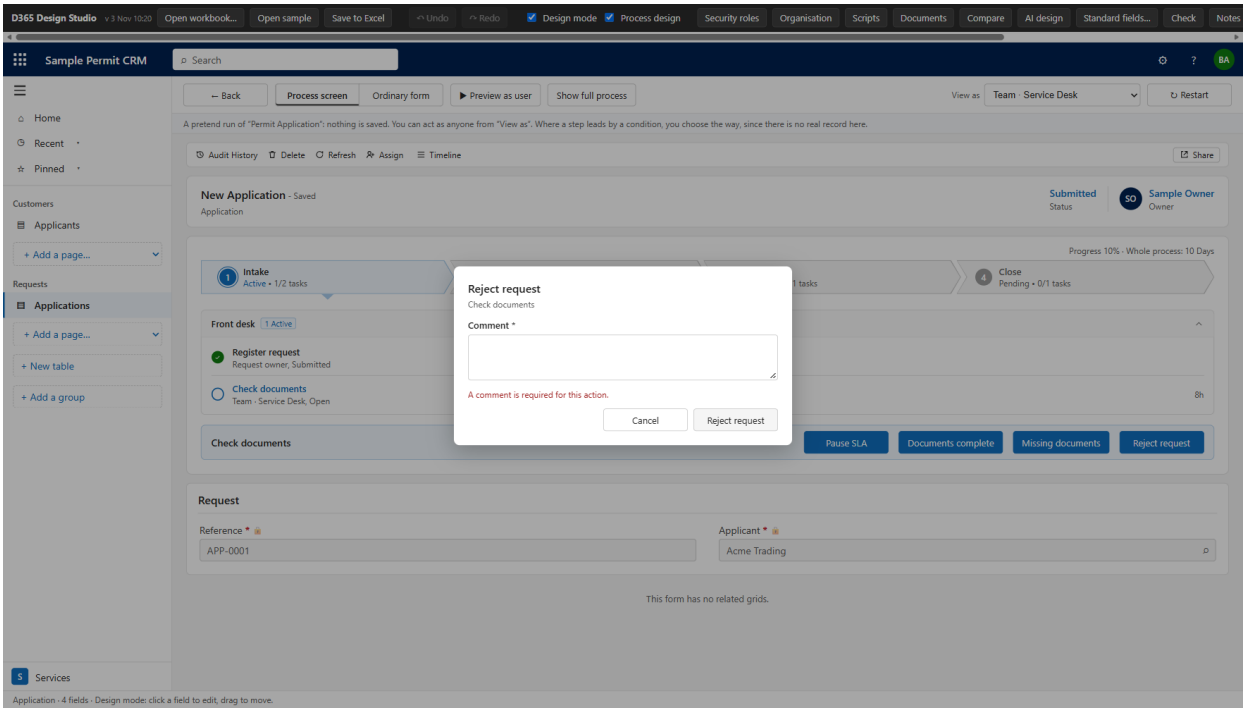


Figure 80. *Reject request asks for its required comment, and will not go on without it.*

The message. When a button needs nothing there is no popup: the action happens at once, and a message comes in from the top with a green tick, such as “Capture details completed”. It goes after about 4 seconds. A message also follows an action confirmed in a popup.

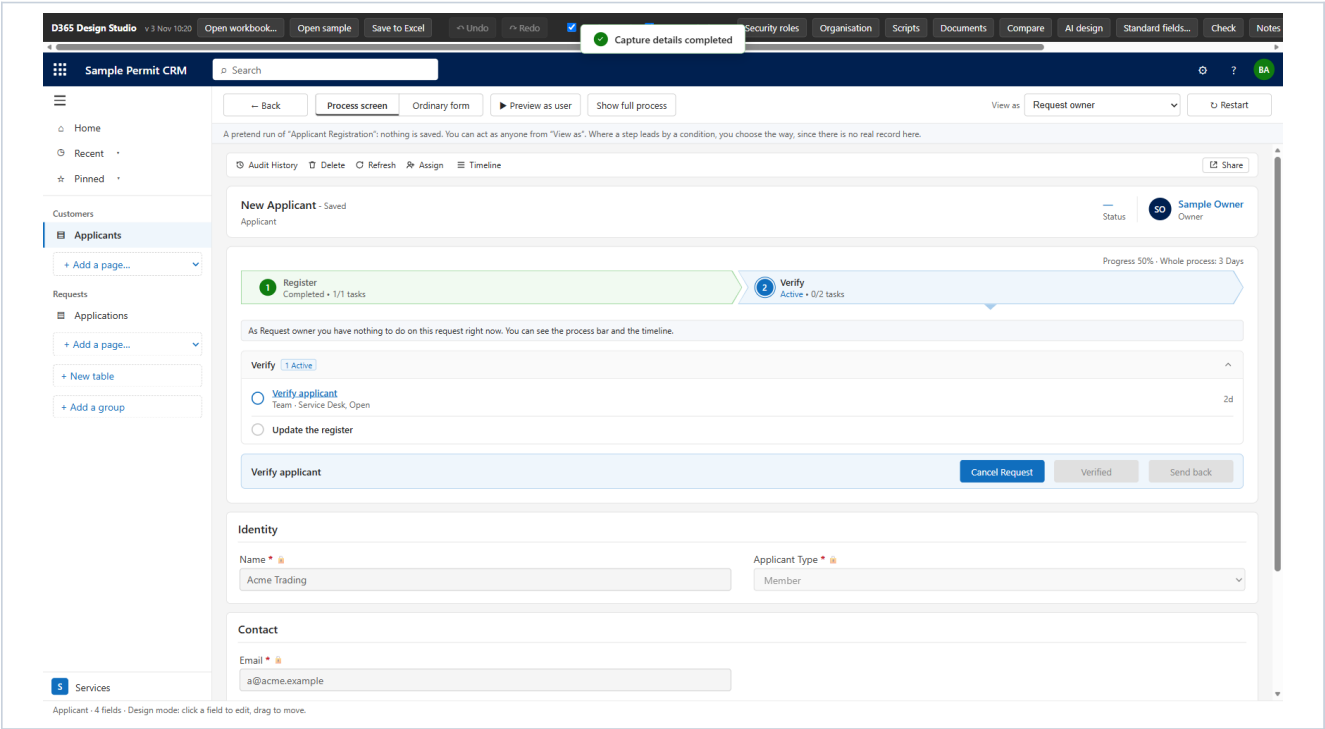


Figure 81. *No popup: the step completes at once and the green message says so.*

Super user. View as has a Super user at the end of its list, for demonstrations. Every action button on every open step works for them, what they do is logged as Super user, and a small “Demo only” label shows. It lasts for the session and is never saved in the workbook. It is the only way to press the buttons of a System Task that waits. The “Pretend the rule picked” list beside View as appears when the selected step’s rule could give it to several people; there is no real record, so you say which.

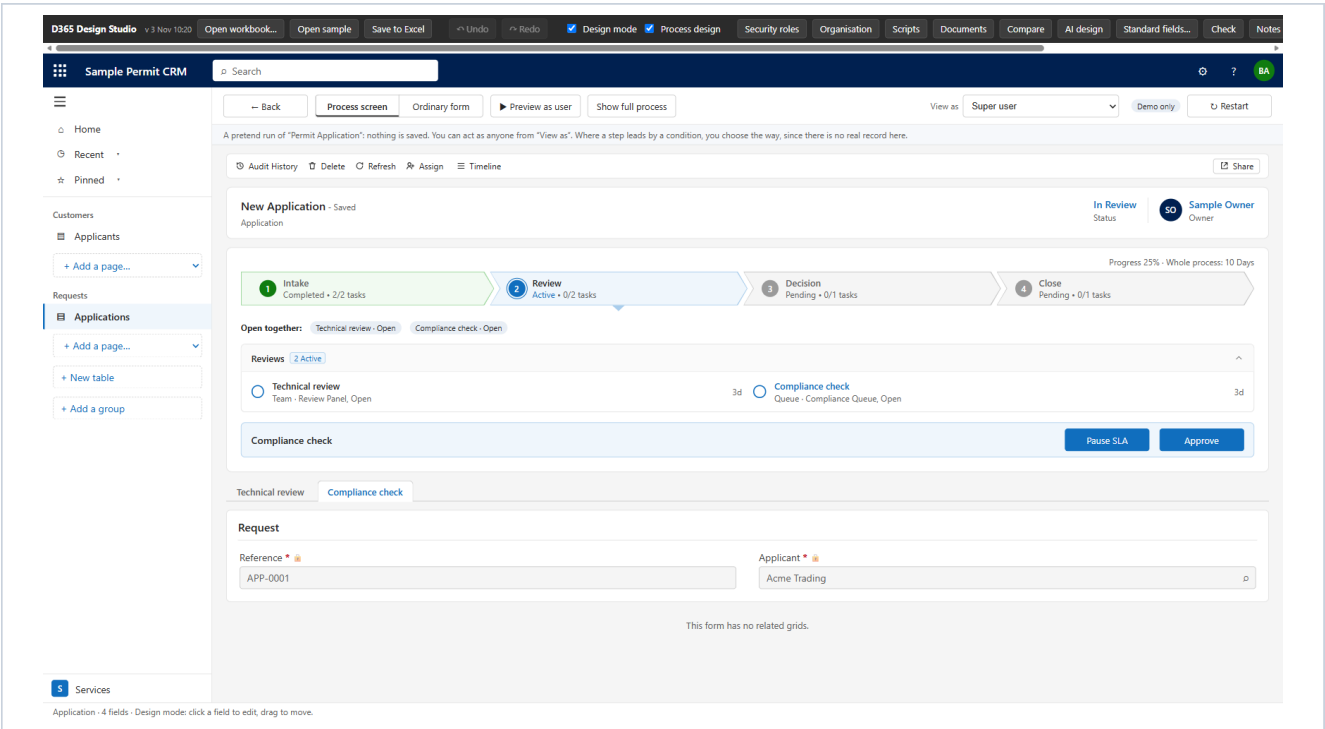


Figure 82. *The Super user, with its Demo only label: the buttons work on both open steps.*

System tasks. A step with the type System Task runs by itself. When one opens, at the start or after any action, it completes at once and the run carries on to the next person's step, or to the end. A Notification on the way runs by itself as it always has. How it chooses: with one action, it takes that action. With several actions and outcome rules, it takes the action of the first case that matches. With several actions and no rules, it waits for an outside event: it stays open and on the screen, and only the Super user can press its buttons. If there are rules and none matches and there is no Otherwise, it stops and stays open, and a message says "System task X could not decide: no outcome rule matched". After 50 automatic steps in one go the run stops and says to check the process for a loop.

Show on screen. A System Task is hidden on the process screen unless its Show on screen is ticked: it is left out of the step rows, the stage's task count, the tabs and the Timeline, though its weight still counts in the progress. Ticked, it is a row saying "Done automatically" or "Waiting for an outside event", with no owner and no buttons for ordinary people. One that waits, or that has stopped, shows even when it is not ticked. It always shows in the designer's diagram, Check design, the process document and Compare. The sample's Applicant Registration has one of each: Duplicate check is hidden and Update the register shows.

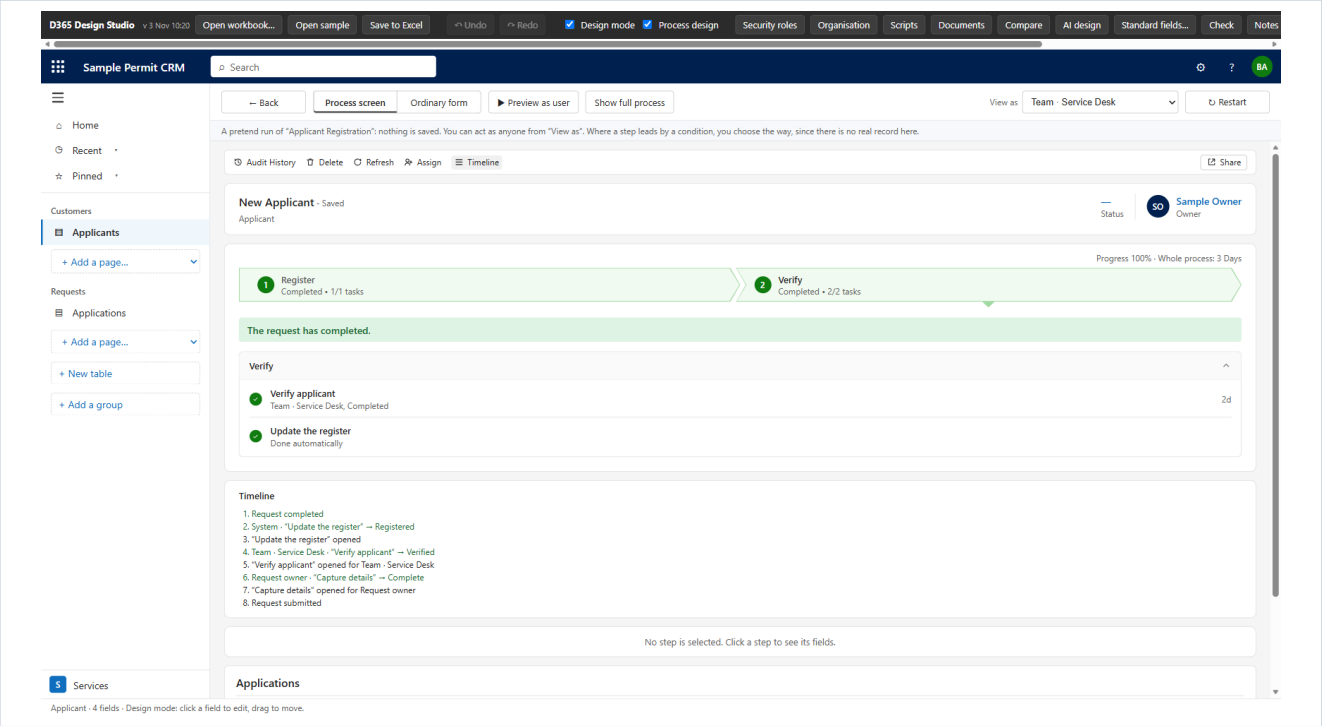


Figure 83. *Applicant Registration at the end: Update the register shows as done automatically. Duplicate check also ran, and is not on the screen.*

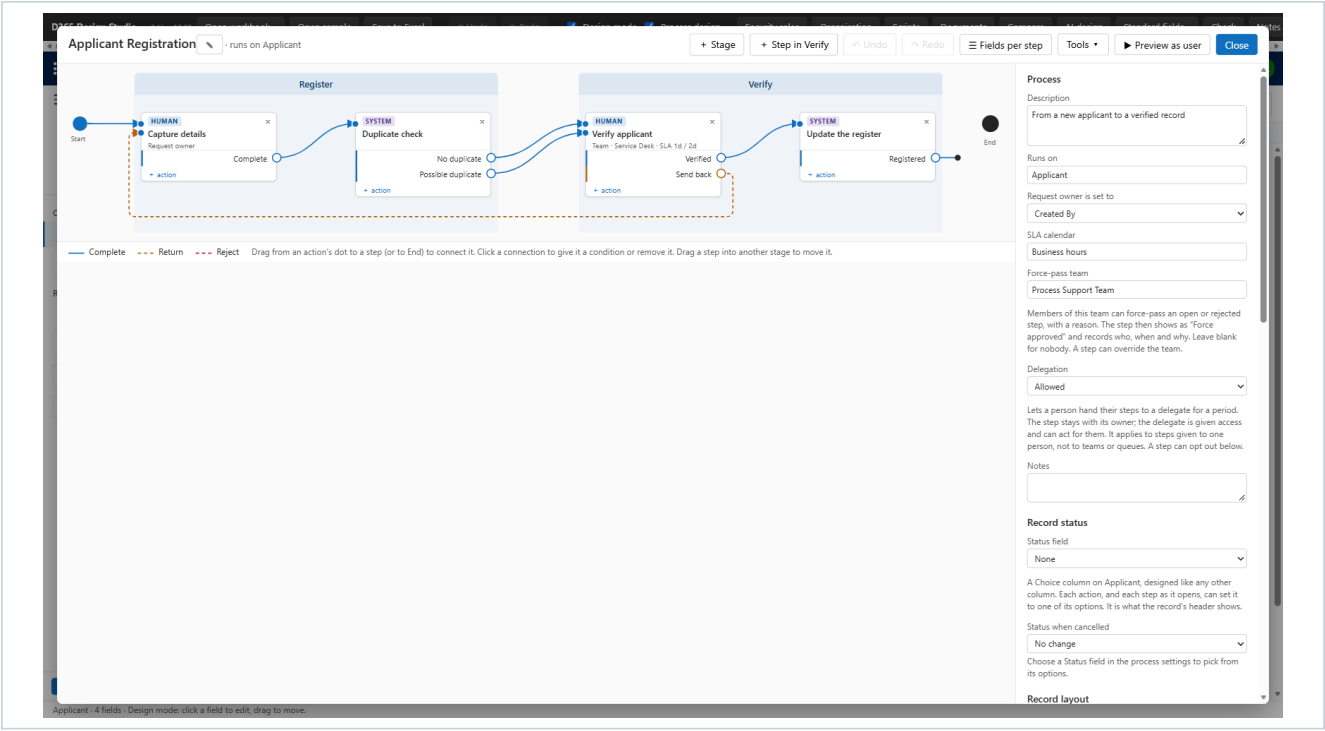


Figure 84. *The designer always shows both System Tasks.*

Outcome rules. Click a System Task on the diagram. Under its type are Show on screen and the Outcome rules: add a case, name it, pick a Field from the table, an Operator and a Value, and the Action to take. + Otherwise adds the case that catches the rest. Cases are read from the top, and Up and Down change the order. The operators are the same as in an assignment rule's decision table, and the rules read what is on the record: what was typed, or otherwise the field's Example value. They are saved in a sheet called Process Outcome Rules, written only when used. Delete step and Delete this stage remove a step's rules and list them first, and deleting an action removes the rules that pick it.

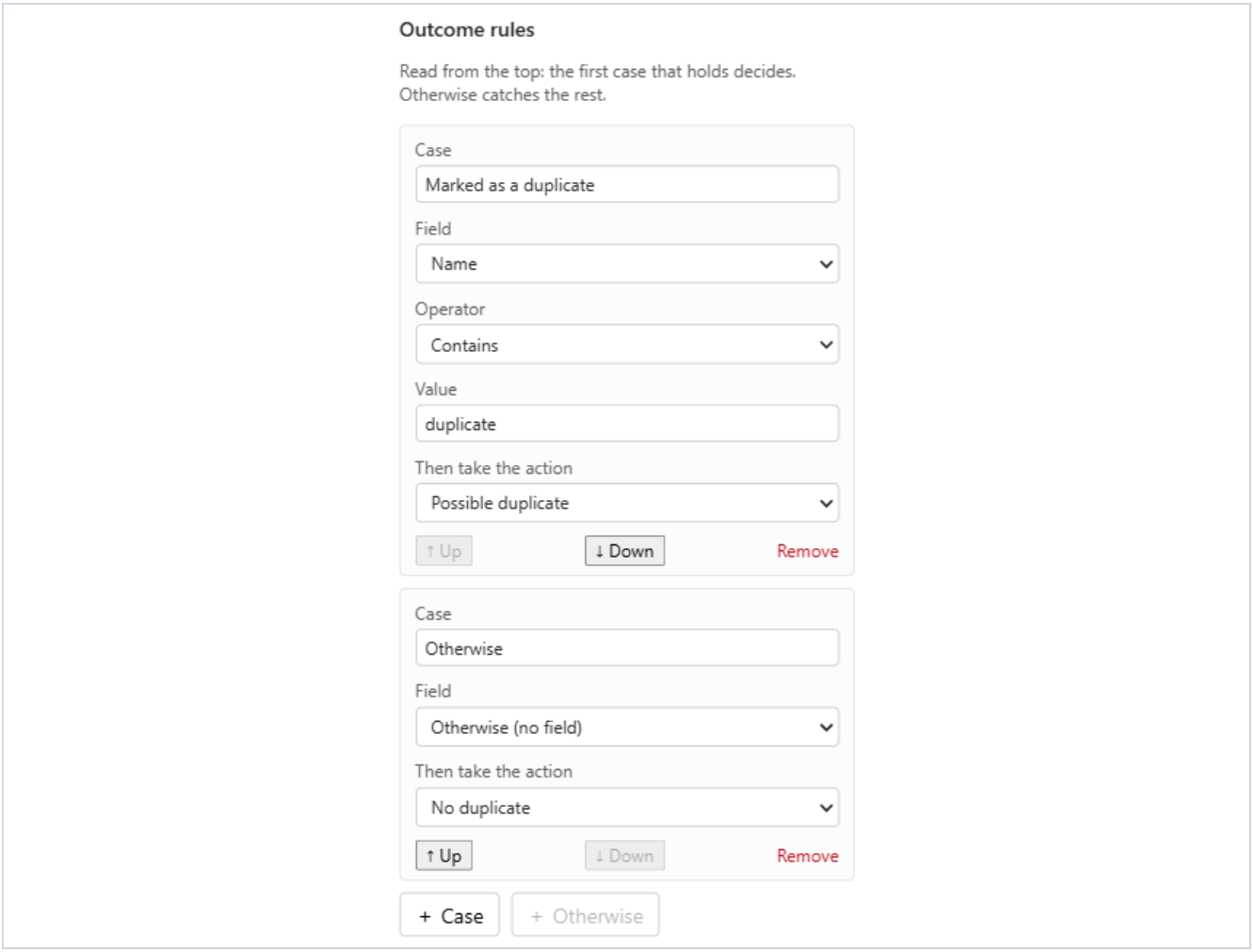


Figure 85. *Outcome rules for Duplicate check: a case on the Name field, then Otherwise.*

Moving actions. Each action in the step panel has Up and Down. The order is the order of the buttons on the screen. One Undo puts a move back.

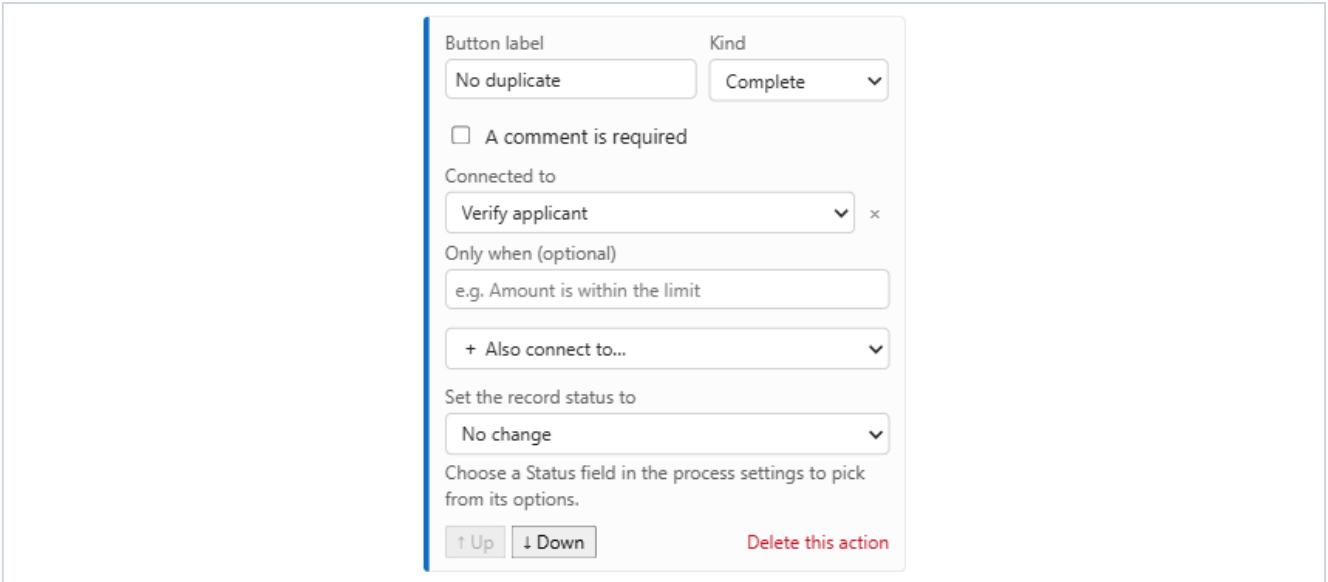


Figure 86. *An action's card, with Up and Down at the foot.*

Check design reports an outcome rule whose action is not one of the step's, or whose field is not on the table, and a System Task with no actions, as errors. It warns about a System Task with several actions and no

rules, because it waits for an outside event and shows on screen, and about rules with several actions and no Otherwise. Clicking a line opens the step.

17. Working with the engine, and current limits

17.1 After the design: building it

1. validate: reads the workbook and reports errors and warnings. Fix errors in the Studio, save, validate again.
2. deploy with --dryrun true: shows what would be created. Then run for real against a non-production environment first.
3. views: creates or updates the views. The second run should report existing views as UPDATED.
4. org: creates business units and teams (dry run is the default).
5. Check in make.powerapps.com and the admin center, then compare against the design.

17.2 Current limits

- Default value and validation rule are recorded for the developer; the build does not apply them yet.
- AI design only adds. It does not edit or remove existing items.
- AI design has been tested with simulated replies. Real replies from Claude or Copilot may need small prompt tuning.
- Table settings such as display name and ownership are not editable yet for existing tables.
- Ownership: the engine reads the Ownership row and creates an Organization-owned table as such. The type cannot be changed once the table exists in Dynamics. The engine warns when an Organization-owned table has an owner field, owner-based views, or privileges below Organization level.
- Form Layout settings (columns, hidden, labels, full width) are read by the engine and checked offline on a real form, but not yet in a live deployment. Moving a section or field to another tab changes the Tab, Section and row order, and takes effect for fields that are not yet on the live form. A field already on a live form is not moved.
- The process sheets (section 16) are read by the process engine, which is built separately, and not by the deployment engine. The newest settings are listed in section 16.14.
- The Studio does not connect to Dynamics. Everything is read from and written to the workbook.
- Sample values typed in Preview are for the demo only. They are not saved anywhere, and Documents previews still use each field's Example value.

Appendix A. Screenshot guide

Use this appendix to retake the screenshots with your own project, for example for a client handover pack. The images in this guide were taken in Chrome at 1760 × 1000 pixels from the built-in sample. A record of the sample opens on the process screen, so click Ordinary form first for the form figures.

Fig.	Screen	How to reproduce it
1	Start screen	Open the HTML file with no workbook.
2	App shell and list	Open the sample. Tables are in the left navigation; the first page opens its default view.
3–5	View designer	Open the view picker for a second view; add a column with + Add a column…; type a filter.
6–10	New table	In design mode click + New table under the navigation. Type a display name; choose Organization to see the note about organization-owned tables; type an existing name to see the error.
11–13	Form and properties	Click a record, then Ordinary form. Click a field to open properties. Click + Field in a section.
15–19	Layout settings	Add a second tab with + Tab. Click the gear icon next to a section or tab. Click a field and look under On the form.
14	New form	Click + Form under the record title.
20	Process flow	Click + Add a business process flow, then + Stage. Select a stage to see the right-hand panel.
21	Command button	Click + Button on the command bar.
25–27	Check, Notes, Migration	Click Check, then the Notes and Migration tabs of the panel.
23	Typed sample values	Click a record, then Ordinary form. Untick Design mode, type in a field, and note Reset sample data on the top bar.
22–24	Preview and View as	Untick Design mode. Choose a role in View as.
44	Process bar	Tick Process design, open an Application record, click Ordinary form.
45–46	Process canvas and step	Open an Application record, click Show full process, then click a step.
48	Decision table	Select Technical review and click Open the decision table.
51–47	Time, pause, AI step	Click the process (or a step) and scroll the panel.
52–55	Preview as user	Click Preview as user. Use View as to switch person; complete steps in order; reject Technical review; view as Team · Process Support Team for Force approve.
59–63	Groups and record layout	Click a stage (Review) for the groups; click Technical review for the Group list and card tick boxes; click the stage and Process settings for Record layout; click ▦ Layout preview.
44	Process bar alone	Tick Process design, open an Application record and click Ordinary form: the older flow bar is hidden.
49–56	Rule timing, Try it, process document	Select a rule step for When the rule runs; open the decision table and use Try it; click Process document.
57–58	Delegation	Click the process for the Delegation setting; click Panel decision for the step ticks; in Preview choose View as Delegate and pick who they act for.

Fig.	Screen	How to reproduce it
68–79	Record screen, step fields, cancel	Open an Application record for the default screen, then use the switch and View as. To finish the walkthrough, act as each person in turn. Tick Process design and click Show full process for Fields per step. To see a refusal, empty Applicant on Register request and press Complete. Cancel Request is on the step bar. For Check design set a step field to Required Yes and Read-only Yes. For Compare choose samples/synthetic/sample-permit-style.xlsx as the earlier version.
28–31	Security roles	Open Security roles. Click each tab.
32–33	Organisation	Open Organisation. Click + Add team.
34	Scripts	Open Scripts.
35	Documents	Open Documents. The preview is on the right.
36–37	Compare	Open Compare. Choose an earlier version of the workbook.
38–43	AI design	Open AI design, type a description, then Propose data model. Paste a reply, apply, then Propose process flow.

Tips for good screenshots

- Use a window about 1700 pixels wide so the whole top bar is visible. Zoom the browser to 100%.
- Use the sample or invented data. Never include real client names, people or amounts in a document that leaves the team.
- Set Design mode on for design screens and off for the preview and View as screens.
- Crop to the part of the screen that matters; keep the caption short and say what to look at.
- Retake a screenshot whenever the Studio changes visibly, so the guide matches the tool.